STIR-L4 · A WORKED EXAMPLE

The complete pipeline

One simulated plan, from the first question to the final report: five sessions, every reply as written

Every STIR-L4 planning effort ends with a final report: the plan as the team certified it, followed by a protocol written in the standard template for systematic review protocols (PRISMA-P 2015). Part 1 is that report for this run, as the assistant produced it, tidied for readers (see Fidelity below). Part 2 is the conversation that led to it, so that readers can see how each decision was reached. The user, a simulated solo reviewer, plans a review of what kind of weight is lost with SGLT2 inhibitors.

About this run

- **User:** a second AI playing Uday, a terse solo reviewer, from a script the first author approved (eval/personas/P1_full_pipeline.md).
- **Assistant:** Claude Opus 5.5 (Anthropic), given only the STIR-L4 prompt and the files a team would upload: the prompt, the helper kit, the model file and the core-properties file. The sessions were run while the prompt was being finalized, on development versions that differ from the released prompt (version 1.0, Supplementary File 1) in how the assistant talks with the team and writes the report, not in the STIR-L4 model.
- **Searches:** real. 169 logged retrievals from PubMed (including its MeSH index and PubMed Central), OpenAlex and ClinicalTrials.gov, run live on 25 October 2026, each saved; every reference in the protocol was found in a saved search result and checked again against PubMed and Crossref.
- **What the report leaves out:** STIR-L4 in a chat app plans a review; it does not screen studies or extract data. The pilot screening and the commitment in Sessions 3 and 4 followed the prompt's steps at the time and tested the criteria; they stay in the log. Paper as an Object, Literature as a Tensor and Registering Evidence Dynamically were designed at Step 7 but are not built in the prompt version; they are being developed for a dedicated planning tool.
- **Fidelity of this document:** every message is reproduced word for word, so drafts in the conversation carry working version numbers. The report in Part 1 is the file the helper kit wrote, with editorial changes only, each listed in docs/example/clean_report.py: working version labels and draft history removed, every PMID and DOI linked, and the plan's Methodology entry showing the methods as certified rather than the protocol's certification line. No content was changed. Only the layout and the lettered notes are added.

Contents

- **Part 1, the STIR-L4 report:** 1. the plan, as certified; 2. the protocol (PRISMA-P items, what planning found, how STIR-L4 and AI were used, gaps, references).
- **Part 2, the conversation:** Session 1 (steps 1 to 3), Session 2 (3 to 6), Session 3 (6 to 9), Session 4 (10 to 12), Session 5 (the protocol and the final report).

Margin notes

R Resuming a session from saved files

S The soundness check of the question

F Feedback between decisions (reopening)

C Certifications and progress

L Before the search: extraction form, registry, atlas

V Verification and safeguards

- **P** Pilot screening by the person (a test of the criteria)
- **W** Writing the protocol and the final report
- **H** Honesty about what was not done

Session 1 keeps the numbered notes of the first example (1 to 9).

Part 1. The STIR-L4 report

From the assistant's planning folder (reports/2026-10-25_final_report.md, written by the helper kit from the saved plan and the certified protocol), with the editorial changes listed on the cover. Every PMID and DOI is a link.

STIR-L4 planning report: which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

Planning team: Uday. Planning record: 783 entries, 2026-10-25 to 2026-10-25. STIR-L4 prompt and helper kit: development versions, before release 1.0; AI model: Anthropic Claude Opus 5.5 claude-opus-5-5 (as recorded: interface, read by Uday).

1. The plan, as certified

Topic: the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

Research focus: which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

Individual Motivation

Certified by Uday on 2026-10-25 at 03:21 UTC (L000058); reason, as given: "testing the workflow"

Uday (summarized): interest — which kind of weight is lost: fat or muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional curiosity; academic activity

Team's Purpose

Certified by Uday on 2026-10-25 at 06:59 UTC (L000251); reason, as given: "broadened for feasibility"

Uday (summarized): focus — which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup; kinds of purpose — research gaps; finding the right question; the review — a systematic review on this topic; with the results — decide whether a full review is worth it; maybe publish

Need for a Review

Certified by Uday on 2026-10-25 at 07:26 UTC (L000292); reason, as given: "broadened for feasibility"

Uday (summarized): case — new review: no review examined so far answers the focus, which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup; grounds — previous reviews; knowledge gaps; clinical justification (used off label for weight in Asia; muscle or water loss matters for older and lean patients; it matters for any adult without diabetes on these drugs); impact analysis (tell clinicians whether the weight lost is fat; show where trials are missing); uncertainties — whether one answer fits every reason for treatment; the drug's own effect, when trials give it with another drug; whether body scans can measure each part of a small loss; not searched — PROSPERO, OSF Registries, Cochrane, Epistemonikos; OpenAlex reviews past the first 100; full texts

Review Objective

Certified by Uday on 2026-10-25 at 07:34 UTC (L000304); reason, as given: "broadened for feasibility"

Uday (summarized): objectives — understand a topic; identify new areas for research; wording — to understand which kind of weight is lost with SGLT2 inhibitors (fat, muscle or body water) within all adults without diabetes, whatever the drug is given for, any country (Asian people as a subgroup), and to identify new areas for research: where trials are missing

Feasibility

Certified by Uday on 2026-10-25 at 06:47 UTC (L000232); reason, as given: "broaden is right"

Uday (summarized): decision — broaden: all adults without diabetes, any country; Asian people as a subgroup; kinds of feasibility — literature availability; FMEA (very few trials)

Research Question

Certified by Uday on 2026-10-25 at 08:13 UTC (L000370); reason, as given: "this is the question"

Uday (summarized): question — which kind of weight is lost with SGLT2 inhibitors, compared with placebo, no treatment or usual care: fat, muscle or body water, in adults without diabetes (Asian people as a subgroup); framework — PICO; context — any country, whatever the drug is given for; Asian people as a subgroup; population — adults without diabetes; people with prediabetes count as without diabetes; intervention — any SGLT2 inhibitor, no minimum treatment time; comparison — placebo, no treatment or usual care; outcomes — fat, muscle and body water; muscle as lean or fat-free mass, muscle itself, and also strength or physical performance; fat, lean mass, muscle itself, body water and strength each kept apart; the measuring method recorded for each study; inclusion and exclusion criteria — randomized controlled trials and controlled cohort studies; another treatment only when both groups get the same one; studies that also enrol people with diabetes only if they report those without diabetes apart; a pair of groups within a larger trial counts if it meets these criteria

Paper as an Object, Literature as a Tensor and Registering Evidence Dynamically: building them is not available in this prompt version (it is being developed for a dedicated planning tool), so they may not behave as intended here; any designs drafted at Step 7 are kept in the planning folder but were not applied to studies.

Search Strategy

Certified by Uday on 2026-10-25 at 10:08 UTC (L000489); reason, as given: "finds both known studies"

PubMed — the search shown in protocol item 10, records created up to 2026-10-25: SGLT2 inhibitors, by subject heading and each drug's name, AND fat, lean or muscle, body water, strength or performance, their measuring methods or body weight, AND randomized trials or cohort studies, animal-only records left out; ClinicalTrials.gov — the search shown in protocol item 10, any status: the drug names among treatments AND the parts of the weight among outcomes; OpenAlex — the planned search shown in protocol item 10, in titles and abstracts only, with the four body-weight phrases added, published up to the run date, run in the next session; WHO trial portal — the drug names of the ClinicalTrials.gov search, searched by Uday by hand in the next session; not searched — Embase and CINAHL, which Uday cannot search, named in the protocol with the reason, and other paid databases, out of reach; filters — no language, start-date or population limit, with adults, the comparison, diabetes status and Asian people judged at screening; known studies — PubMed 29235586, 33879970, 39361103 and 42104798, checked in PubMed and OpenAlex; stopping rule — the strings stop changing when every known study is found and none of the records added by the last change meets the question, as Uday screens them; peer review — none, as there is no second searcher; the protocol says the strings were not peer reviewed

Methodology

Certified by Uday on 2026-10-25 at 11:51 UTC (L000655); reason, as given: "fits a solo review"

Uday (summarized): type — a scoping review of every study that meets the question, then a systematic review of each part of the weight that reaches 3 or more similar randomized trials; sources — PubMed, ClinicalTrials.gov, OpenAlex and the WHO trial portal, as in the approved search, with Embase and CINAHL named as not searched; selecting studies — Uday alone screens titles and abstracts and reads full texts, with the assistant's suggested decision beside each record, and about 2 weeks later screens again a random tenth of all his decisions, drawn by code and without seeing his first decision, reporting how often the two passes agree; collecting data — Uday alone fills the approved extraction form, with the assistant's suggested values and their passages beside each; judging each study, systematic part only — risk of bias for each outcome, trials with RoB 2 and cohort studies with ROBINS-I, cohort studies judged and described there; pooling — a meta-analysis for a part only if 3 or more similar randomized trials report it, cohort studies not counted, otherwise results tabled and described part by part and reported with SWiM; how 'similar' is judged and how to pool settled if a part reaches 3; trust — GRADE for each part; Asian people — results described apart; reporting — PRISMA-ScR for the scoping part and PRISMA 2020 for the systematic part; limitations — one person screens, reads and judges

Certified again together with the protocol (section 2) on 2026-10-25 at 15:17 UTC (L000780); reason, as given: "follows the standard template"

2. Protocol

Which kind of weight is lost with SGLT2 inhibitors in adults without diabetes: fat, muscle or body water? A scoping review, with a systematic review of the parts with enough trials: protocol

Status: certified by Uday on 2026-10-25 with the reason “follows the standard template” (L000780). The protocol writes every decision Uday certified in full sentences, item by item in the order of the PRISMA-P 2015 checklist [34], with the searches reported as PRISMA-S asks [14], a numbered reference for each methodological choice, and three closing sections: what planning found, how STIR-L4 and AI were used, and gaps. '(my suggestion)' marks wording the assistant drafted that came from neither Uday nor a retrieved fact; Uday accepted it when he certified the protocol. The word GAP marks what is missing and who can supply it; every gap is listed again at the end. Planning-record ids (L followed by six digits) are given for audit; numbered references are listed at the end.

Administrative information

Item 1a. Identification

This document is the protocol of a scoping review, with a systematic review of each part of the weight that has enough trials. Title: *Which kind of weight is lost with SGLT2 inhibitors in adults without diabetes: fat, muscle or body water? A scoping review, with a systematic review of the parts with enough trials: protocol* (my suggestion, from Uday's focus and methods).

Item 1b. Update

This is a new review, not an update of an earlier review: Uday certified the need for a new review, because no review examined during planning answers the focus (record: L000292).

Item 2. Registration

The protocol is to be registered with OSF Registries (Uday's choice; record: L000679). GAP: registration number and date. None yet: on 2026-10-25 Uday chose not to submit the protocol, as planning was a test (record: L000708). If he submits it, he adds the number and date here. Facts shown when Uday chose the register: most of the scoping review protocols examined in a recent study of high-impact journals were registered on OSF [1]; PROSPERO is restricted to systematic reviews of the effects of interventions with a health-related outcome [2]; and much of what PRISMA-P recommends was not stated in PROSPERO registrations [3], so this full protocol is kept beside the register entry (my suggestion).

Item 3a. Authors and contact

Author: Uday, the name as he gave it, the only author. Institutional affiliation: left blank at Uday's request (record: L000679). GAP: e-mail address and postal address of the corresponding author. Uday adds them on the register's form when he submits; none is kept in the planning files.

Item 3b. Contributions

Uday planned the review and will screen the records, extract the data, judge each study and write the report; he is the guarantor of the review. The AI assistant is not an author (my suggestion).

Item 4. Amendments

This protocol is not an amendment; it has not been registered. After registration, a change is made only if Uday agrees to reopen the decision concerned; the changed decision is certified again with its reason, dated in the planning record and submitted to OSF Registries as an amendment.

Item 5a–5c. Support

5a. The review has no funding (Uday); no other support is named. 5b. There is no funder or sponsor (my suggestion, as there is no funding). 5c. No funder, sponsor or institution had a role in developing the protocol (my suggestion).

Introduction

Item 6. Rationale

Uday's topic is the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, in any country, with Asian people as a subgroup; the agreed focus is which kind of weight is lost: fat, muscle or body water (records: L000250, L000243). Why it matters (Uday's clinical justification, summarized): these drugs are used off label for weight in Asia; losing muscle or water matters for older and lean patients; and it matters for any adult without diabetes who takes these drugs. What the review would change (Uday's expected impact, summarized): it would tell clinicians whether the weight lost is fat, and show where trials are missing (record: L000292). What is already known (FACT): of the 9 reviews found during planning that may bear on the focus, 7 name body fat, 0 name muscle or lean mass and 1 names body water, in their titles and abstracts. 3 of them are reviews of weight in people without diabetes or with and without it [4, 5, 6]; all 3 name body fat and none names muscle, lean mass or body water. Their search end, as their abstracts give it: May 2021 for PMID 34897070; not stated in the abstract (abstract only: unknown) for PMID 34542222; not stated in the abstract (abstract only: unknown) for PMID 42339604. The latest search end stated among the 9 is August 2025, in a review that the rule placed with the focus because it names body composition and without diabetes [7]. The nearest evidence, in Uday's words, is from reviews in people with type 2 diabetes (FACT, from their abstracts; no full text was read). By a declared rule over the review records examined during planning (type 2 diabetes named as the population, muscle or lean mass named, and a systematic review or meta-analysis by its title; record: L000771), 6 reviews were found; none names adults without diabetes. In a meta-analysis of 8 randomized trials with 541 participants (its title names older adults) [8] (2026), fat mass fell (standardized mean difference −0.53); muscle mass fell less (standardized mean difference −0.35), which the authors call substantially smaller than the fat loss. In a network meta-analysis of 33 randomized trials with 5,399 participants [9] (2026), fat mass fell consistently (for example dapagliflozin −1.75 kg); lean mass fell modestly with empagliflozin (−1.22 kg), ipragliflozin (−0.99 kg) and canagliflozin (−0.80 kg) but not with dapagliflozin or tofogliflozin, and no SGLT2 inhibitor significantly reduced skeletal muscle mass compared with placebo; strength and physical performance were rarely reported. In a meta-analysis of 5 randomized trials of at least 12 months with 695 participants [10] (2026), fat mass fell (−1.26 kg); fat-free mass did not differ significantly (−0.53 kg; 95% confidence interval −1.62 to 0.56). In a meta-analysis of 19 randomized trials with 1,482 participants [11] (2023), fat is not reported in the abstract; lean body mass fell by 0.66 and skeletal muscle mass by 0.35 (units not given); the quality of the evidence was low. In a meta-analysis of 25 randomized trials with 2,286 participants [12] (2023), fat mass fell (−1.49); lean mass (−0.80), skeletal muscle mass (−0.38) and skeletal muscle index (−0.12) fell as well; body water fell (−0.96); units are not given in the abstract. In a meta-analysis of 18 studies with 1,430 participants [13] (2022), fat mass fell (−1.16 kg); lean mass changed by −0.76 kg (95% confidence interval −1.53 to 0.01) and skeletal muscle mass fell (−1.01 kg). Their search end, as their abstracts give it: not stated in the abstract (abstract only: unknown) [8]; not stated in the abstract (abstract only: unknown) [9]; April 22, 2024 [10]; 1 May 2023 [11]; February 2023 [12]; not stated in the abstract (abstract only: unknown) [13]. The abstract of [10] names type 2 diabetes only as the condition in which these drugs are used; the people in its trials are not described there. Reviews were searched in PubMed and OpenAlex only (see What planning found). Uncertainties Uday recorded with the need (summarized): whether one answer fits every reason for treatment; the drug's own effect, when trials give it with another drug; and whether body scans can measure each part of a small loss.

Item 7. Objectives

The objectives (certified; Uday, summarized) are to understand which kind of weight is lost with SGLT2 inhibitors (fat, muscle or body water) within all adults without diabetes, whatever the drug is given for, in any country (Asian people as a subgroup), and to identify new areas for research: where trials are missing (record: L000304). The question (certified; framework PICO) is: which kind of weight is lost with SGLT2 inhibitors, compared with placebo, no treatment or usual care: fat, muscle or body water, in adults without diabetes (Asian people as a subgroup) (record: L000370). Participants: adults without diabetes; people with prediabetes count as without diabetes. Intervention: any SGLT2 inhibitor, with no minimum treatment time. Comparators: placebo, no treatment or usual care. Outcomes: fat, muscle and body water; muscle as lean or fat-free mass and muscle itself, and also strength or physical performance; each kept apart.

Methods

Item 8. Eligibility criteria

Study characteristics (certified; record: L000370). Participants: adults without diabetes, whatever the drug is given for and in any country; people with prediabetes count as without diabetes; a study that also enrols people with diabetes is eligible only if it reports those without diabetes apart. Intervention and comparators: any SGLT2 inhibitor, with no minimum treatment time, compared with placebo, no treatment or usual care; another treatment is allowed only when both groups get the same one, and a pair of groups within a larger trial counts if it meets these criteria. Outcomes: whole-body fat, lean or fat-free mass, muscle itself, body water, and strength or physical performance, each kept apart, with the measuring method recorded for each study. Designs: randomized controlled trials and controlled cohort studies. Report characteristics: no language limit and no start-date limit (certified search strategy; record: L000489); records up to the search date; any publication status, including results posted in trial registers (my suggestion).

Item 9. Information sources

The review searches the following sources (certified search strategy; record: L000489). PubMed, through the NCBI E-utilities, records created from 1800/01/01 to the search date; searched on 2026-10-25. ClinicalTrials.gov, through its API version 2.0.5, studies of any status; searched on 2026-10-25, with data as of 2026-10-23T09:00:04 as the register stated. OpenAlex, through its works endpoint, in titles and abstracts, works published up to the run date: planned, not yet run; it was refused on 2026-10-25 because the free daily allowance of the network was used up and no access code is set (records: L000457, L000593). The WHO International Clinical Trials Registry Platform (the WHO trial portal), searched by Uday by hand with the drug names of the ClinicalTrials.gov search: planned, not yet run. Not searched: Embase and CINAHL, which Uday cannot search (no access), and other paid databases, which are out of reach. The review will not contact study authors or investigators and will not check reference lists (my suggestion). Dates of coverage: from the earliest record in each source to the search date. GAP: run dates and counts for OpenAlex and the WHO trial portal. Not run when planning ended; Uday supplies them when he runs these searches (OpenAlex reads past its first 100 results only with a free access code).

Item 10. Search strategy

The searches are reported as PRISMA-S asks [14]: each source and platform, the date searched, the full string, limits and filters, the known-study check and peer review.

PubMed (NCBI E-utilities, ESearch; searched 2026-10-25; records created 1800/01/01 to 2026/10/25, the bound checked in PubMed's query translation; no language, publication-date or population limit; adults, the comparison, diabetes status and Asian people are judged at screening): 2,026 records (record: L000459). Blocks: 1. SGLT2 inhibitors, by subject heading and each drug's name; 2. fat, lean or muscle, body water, strength or performance, their measuring methods or body weight; 3. randomized trials or cohort studies; records about animals only are left out (certified).

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(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab]
OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab] OR enavogliflozin[tiab] OR
ganagliflozin[tiab] OR remogliflozin[tiab] OR sergliflozin[tiab] OR janagliflozin[tiab] OR
licogliflozin[tiab]) AND ("Body Composition"[Mesh] OR "Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR
"Sarcopenia"[Mesh] OR "Body Water"[Mesh] OR "Muscle Strength"[Mesh] OR "Physical Functional Performance"
[Mesh] OR fat[tiab] OR adipos*[tiab] OR muscle*[tiab] OR muscular[tiab] OR "body composition"[tiab] OR "lean
mass"[tiab] OR "lean body mass"[tiab] OR "lean tissue"[tiab] OR "fat-free mass"[tiab] OR "fat free mass"
[tiab] OR sarcopeni*[tiab] OR "body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"
[tiab] OR "extracellular fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR
"water loss"[tiab] OR "fluid loss"[tiab] OR "grip strength"[tiab] OR handgrip[tiab] OR "hand grip"[tiab] OR
dynamomet*[tiab] OR "physical performance"[tiab] OR "physical function"[tiab] OR "walk test"[tiab] OR "minute
walk"[tiab] OR "gait speed"[tiab] OR "walking speed"[tiab] OR "chair stand"[tiab] OR DXA[tiab] OR DEXA[tiab]
OR "dual-energy x-ray"[tiab] OR bioimpedance[tiab] OR "bioelectrical impedance"[tiab] OR "body weight"[tiab]
OR "weight loss"[tiab] OR "weight reduction"[tiab] OR "weight change"[tiab]) AND ("Randomized Controlled
Trial"[pt] OR "Controlled Clinical Trial"[pt] OR "Clinical Trial"[pt] OR randomized[tiab] OR randomised[tiab]
OR randomly[tiab] OR placebo[tiab] OR trial[tiab] OR crossover[tiab] OR "cross-over"[tiab] OR "Cohort
Studies"[Mesh] OR cohort*[tiab] OR "Observational Study"[pt] OR propensity[tiab] OR retrospective*[tiab] OR
prospective*[tiab])) NOT ("Animals"[Mesh] NOT "Humans"[Mesh])
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ClinicalTrials.gov (API version 2.0.5, studies endpoint; searched 2026-10-25; any status; the drug names among interventions AND the parts of the weight among outcome measures): 540 registered studies (record: L000468).

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Interventions: "SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR
"sodium-glucose co-transporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR
canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR
henagliflozin OR bexagliflozin OR enavogliflozin OR ganagliflozin OR remogliflozin OR sergliflozin OR
janagliflozin OR licogliflozin
Outcome measures: fat OR adipose OR adiposity OR muscle OR muscles OR muscular OR "body composition" OR "lean
mass" OR "lean body mass" OR "lean tissue" OR "fat-free mass" OR "fat free mass" OR sarcopenia OR "body
water" OR "extracellular water" OR "intracellular water" OR "extracellular fluid" OR "fluid volume" OR
"plasma volume" OR hydration OR "water loss" OR "fluid loss" OR "grip strength" OR handgrip OR "hand grip" OR
"physical performance" OR "physical function" OR "walk test" OR "minute walk" OR "gait speed" OR "walking
speed" OR "chair stand" OR DXA OR DEXA OR "dual-energy x-ray" OR bioimpedance OR "bioelectrical impedance"
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OpenAlex (works endpoint; titles and abstracts; works published up to the run date; corpus core): planned string, not yet run.

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR "sodium-
glucose co-transporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR
ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR
bexagliflozin OR enavogliflozin OR ganagliflozin OR remogliflozin OR sergliflozin OR janagliflozin OR
licogliflozin) AND (fat OR adipose OR adiposity OR muscle OR muscles OR muscular OR "body composition" OR
"lean mass" OR "lean body mass" OR "lean tissue" OR "fat-free mass" OR "fat free mass" OR sarcopenia OR "body
water" OR "extracellular water" OR "intracellular water" OR "extracellular fluid" OR "fluid volume" OR
"plasma volume" OR hydration OR "water loss" OR "fluid loss" OR "grip strength" OR handgrip OR "hand grip" OR
"physical performance" OR "physical function" OR "walk test" OR "minute walk" OR "gait speed" OR "walking
speed" OR "chair stand" OR DXA OR DEXA OR "dual-energy x-ray" OR bioimpedance OR "bioelectrical impedance" OR
"body weight" OR "weight loss" OR "weight reduction" OR "weight change") AND (randomized OR randomised OR
randomly OR placebo OR trial OR crossover OR "cross-over" OR cohort OR "observational study" OR propensity OR
retrospective OR prospective)
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WHO trial portal: GAP: the exact string for the WHO trial portal, built from the drug names of the ClinicalTrials.gov search when Uday runs it by hand.

Search development (FACT): the first PubMed search returned 739 records (record: L000424) and missed a 2026 trial that Uday keeps as a known study [15]; adding four phrases for body weight gave the search above, which returned 1,287 more records and lost none of the first search's records (record: L000461). Known-study check (FACT): the PubMed search retrieves 4 of the 4 studies Uday confirmed as known studies [16, 17, 15, 18] (record: L000460); the same check is planned for OpenAlex. Peer review: none; Uday has no second searcher, so the strings were not peer reviewed (no PRESS check). Stopping rule (certified): the strings stop changing when every known study is found and none of the records added by the last change meets the question, as Uday screens them. Records and duplicates: every page of every response is saved unchanged with its sha256 check code; duplicates were removed by code (PubMed identifier, then DOI, then normalized title and year, with title-and-year matches confirmed by Uday): 2,566 records before and 2,565 after (record: L000517).

Item 11a. Data management

Every search result, screening decision, extracted value and certification is kept in the planning folder: saved results with their sha256 check codes, a record of every step that is never edited, and dated versions under version control; licensed full texts are kept out of any shared copy (certified; record: L000703).

Item 11b. Selection process

Uday alone screens titles and abstracts and reads full texts, with the assistant's suggested decision and its supporting passage beside each record; only Uday admits or excludes a record. About two weeks later he screens again a random tenth of all his decisions, drawn by code and without seeing his first decision, and reports how often the two passes agree (certified; record: L000655). Uday's reason, as he gave it when he certified the methods: 'fits a solo review'; he plans the review alone, with about 5 hours a week (team details, summarized). Published evidence bearing on this choice: single-reviewer screening of titles and abstracts misses more relevant studies than dual screening [19, 20]; false exclusions under single-reviewer screening fall most often on nonrandomized studies of interventions [21]; one reviewer's full-text screening is less sensitive than two reviewers' [22]; and evidence comparing AI tools as second screeners is limited [23].

Item 11c. Data collection process

Uday alone fills the certified data-extraction form, with the assistant's suggested values and the exact passage for each beside it (certified; record: L000655). Data will not be sought or confirmed from investigators (my suggestion). GAP: whether the extraction form is piloted before use, and on how many studies. Not decided; Uday decides.

Item 12. Data items

For each study the certified data-extraction form (record: L000380), records: PubMed and DOI identifiers; one study key linking every report of the same study; design: randomized controlled trial (parallel or crossover) or controlled cohort; how groups were formed; who was blinded; trial registration number; funding source; country where the study was done (also one of the two marks of the Asian subgroup, kept apart from race or ethnicity); number enrolled and analysed in each group; age; percent women; diabetes status: share with prediabetes; if people with diabetes were also enrolled, whether results for those without diabetes are given apart; Asian subgroup: race or ethnicity as the study reports it, kept apart from the country where the study was done; any results given for Asian participants apart; what the drug was given for (the condition); starting weight and body mass index; which SGLT2 inhibitor, dose and treatment time; other treatment both groups get (for example usual medicines or a diet); placebo, no treatment or usual care, as the study describes it; total weight change; whole-body fat: fat mass, percent body fat; fat in particular places: not recorded - whole-body fat only (fat under the skin, around the belly organs or inside organs such as the liver or heart is not recorded); lean or fat-free mass; muscle itself: muscle mass, area or volume; body water: total; outside or inside cells and plasma volume when reported; strength; physical performance, each as the study tests it; the share of the weight lost that is fat, lean or water, when the study gives it; measuring method; every time point; for each group the number analysed, the starting value and the change or end value with its spread; the difference between groups with its confidence interval; how missing data were handled. A value that a study does not report is recorded as not reported, never guessed; each value keeps its exact passage, where it is in the paper, and whether it is reported, not reported or unclear.

Item 13. Outcomes and prioritization

Main outcomes (from the certified question): whole-body fat (fat mass, percent body fat), lean or fat-free mass, muscle itself (mass, area or volume) and body water, each kept apart. Additional outcomes: strength or physical performance, and total weight change (my suggestion: the question names strength with 'also', and the form records total weight). Rationale: the focus asks which kind of weight is lost. For every outcome the measuring method and every time point are recorded.

Item 14. Risk of bias in individual studies

For the systematic part only, risk of bias is judged for each outcome: randomized trials with RoB 2 [24] and cohort studies with ROBINS-I [25]; cohort studies are judged and described there (certified; record: L000655). The judgments feed the GRADE rating of each part of the weight (item 17). Uday makes these judgments alone; agreement between raters using RoB 2 was low in one study [26].

Item 15a. Criteria for quantitative synthesis

A part of the weight is pooled in a meta-analysis only if 3 or more similar randomized trials report it; cohort studies are not counted toward the 3 (certified; record: L000655). Published guidance bearing on this rule: heterogeneity cannot be reliably estimated when very few studies are available [27]; with very few studies, refraining from a quantitative meta-analysis may be the best solution [28]; and whether to pool cohort studies with randomized trials affects the certainty of evidence and needs guidance [29].

Item 15b. Summary measures and methods of combining

GAP: summary measures, handling of data, the method of combining studies, the measure of consistency, and how 'similar' trials are judged. Not decided; if a part reaches 3 similar randomized trials, Uday settles them and registers them as an amendment before any pooling (certified rule; registering them as an amendment was my suggestion).

Item 15c. Additional analyses

Results for Asian participants are described apart (certified). No other subgroup or sensitivity analyses are planned (my suggestion).

Item 15d. If quantitative synthesis is not appropriate

The scoping part covers every study that meets the question. Where a part of the weight is not pooled, results are tabled and described part by part and reported with SWiM [30] (certified; record: L000655).

Item 16. Meta-bias(es)

For each trial, the outcomes in its register entry are compared with those it reports (my suggestion). No statistical test for publication bias is planned, as few trials are expected (my suggestion).

Item 17. Confidence in cumulative evidence

The certainty of the evidence for each part of the weight is rated with GRADE (certified), following the Core GRADE guidance on risk of bias and publication bias [31].

Reporting and limitations (after item 17)

The scoping part will be reported with PRISMA-ScR [32] and the systematic part with PRISMA 2020 [33] (certified). This protocol follows PRISMA-P 2015 [34], whose authors encourage its use, as applicable, for any type of review protocol, and its searches follow PRISMA-S [14]. Limitations (certified): one person screens, reads and judges; the assistant's suggestions are not checked by a second person; and the author of the prompt and kit uses them and judges their suggestions alone.

What planning found

Every statement in this section is a FACT read by code from the planning record and the saved search results.

Existing reviews (need for a review). Reviews were searched in PubMed (with the systematic-review filter) and OpenAlex (works typed as reviews); each search is listed below with its count (records, not studies). After duplicates were removed by code, 241 review records remained (record: L000259); by a declared rule over titles and abstracts, 9 may bear on the focus, 37 cover the population with other outcomes, 44 cover the focus in other populations, 38 name fat or muscle only in another sense (for example liver fat), and 113 neither. What the 9 that may bear on the focus cover is summarized under item 6, with what 6 of those on the focus in other populations, reviews in people with type 2 diabetes, found on fat and on lean mass or muscle. Not searched for reviews: PROSPERO, OSF Registries, Cochrane and Epistemonikos; OpenAlex results past the first 100 of each search were not read, and no full text was read.

Source	What it looked for	Run	Records	Read
PubMed	Topic as first given: all four topic concepts	2026-10-25	1	1
PubMed	Focus as first agreed: SGLT2 inhibitors and fat or muscle	2026-10-25	70	70
OpenAlex	Topic as first given: all four topic concepts	2026-10-25	1,108	100
OpenAlex	Focus as first agreed: SGLT2 inhibitors and fat or muscle	2026-10-25	2,203	100
PubMed	Focus as first agreed, with body water added	2026-10-25	73	73
PubMed	Published protocols for the focus as first agreed, with body water added	2026-10-25	1	1
OpenAlex	Focus as first agreed, with body water added	2026-10-25	2,343	100
PubMed	Topic as broadened: SGLT2 inhibitors, weight lost, without diabetes	2026-10-25	19	19
OpenAlex	Topic as broadened: SGLT2 inhibitors, weight lost, without diabetes	2026-10-25	1,279	100

Soundness of the question. Of the 13 PubMed records tagged as randomized controlled trials in people without diabetes (any region; record: L000262), the reasons for treatment named in their titles are: overweight or obesity 4; fatty liver 3; none

of these 1; prediabetes or insulin resistance 2; kidney disease 1; heart failure 2; polycystic ovary syndrome 1. None names fat, muscle and body water together (record: L000279).

Feasibility. PubMed counts on 2026-10-25 (records): the objective as then certified (fat, muscle or body water in Asian people without diabetes): 5; the same in any region: 154; trial records only, of the latter: 51. ClinicalTrials.gov on 2026-10-25: 473 studies naming the drug terms among interventions, with a status of recruiting, not yet recruiting, enrolling by invitation or active, not recruiting (record: L000198); of these, 36 name an SGLT2 inhibitor among interventions and fat, muscle or body water among outcomes, and 23 of those do not name diabetes among their conditions (columns of rule step5-rule-2; record: L000205). Time (computed from Uday's answers; record: L000215): 92 days, 13.1 weeks, about 66 hours at 5 hours a week. Uday certified broadening to all adults without diabetes in any country, with Asian people as a subgroup (record: L000232).

Searches for the review. PubMed: 2,026 records; ClinicalTrials.gov: 540 registered studies (completed 225; recruiting 103; unknown 83; not yet recruiting 51; active not recruiting 40; terminated 22; withdrawn 13; enrolling by invitation 3; with results posted: 90). Known studies: 4 of 4 retrieved by the PubMed search. Records to screen: 2,566 before duplicates and 2,565 after. OpenAlex and the WHO trial portal: not run.

Pilot screening (a test of the criteria). To test the eligibility criteria, Uday screened 26 records by title and abstract: 4 met the criteria, 12 were unclear and 10 did not (record: L000772).

How STIR-L4 and AI were used

This section follows the reporting items of the 2025 position statement on AI use in evidence synthesis [35], which supports the RAISE recommendations [36].

Framework, prompt and kit. Planning used STIR-L4 [37], which represents planning as ten elements joined by 27 directed flows, with each decision certified by a named person. The assistant followed the STIR-L4 planning prompt and its helper kit, in development versions made before release 1.0; each file is saved unchanged with its sha256 check code (records: L000003, L000107, L000735, L000006, L000736).

AI model. Anthropic Claude Opus 5.5 (model id claude-opus-5-5), as Uday read it from the run settings, in sessions 1 to 5 (records: L000738, L000739, L000740, L000741, L000742, L000743). Settings such as temperature: not available. Dates: 2026-10-25 (session 1 03:01 to 03:58 UTC; session 2 05:01 to 07:45 UTC; session 3 07:47 to 11:00 UTC; session 4 11:05 to 12:51 UTC; session 5 from 14:28 UTC).

What the AI did, by step. Topic, interests, focus and purpose: recorded Uday's answers and reflected them back in his words, and drafted the kinds of motivation and purpose for him to choose. Need for a review: ran and logged searches for existing reviews in PubMed and OpenAlex, grouped the records by a declared rule, read search end dates from abstracts, appraised the soundness of the question from logged records, and drafted the case. Objective: drafted its wording from Uday's answers. Feasibility: counted records for candidate searches, listed ongoing studies in ClinicalTrials.gov and computed the time available. Question: looked up Medical Subject Headings and counted what trial records name, and drafted the question. Before the search: drafted the extraction form, the features to count and the evidence map. Search strategy: ran the PubMed and ClinicalTrials.gov searches, checked the known studies and drafted the stopping rule. Pilot screening: removed duplicates by code, ordered the records and suggested a decision with its supporting passage for each record Uday screened. Methods and protocol: retrieved published guidance, estimated the hours the methods need, and drafted the methods and this protocol. Across planning the record holds 169 logged retrievals, 192 computations and 89 drafts.

What people decided. Uday certified each of the 11 decisions (16 certification records, some after a reopening, each with a reason in his own words): the motivations, the team's purpose, the need for a review, the objective, feasibility, the question, the extraction form, the features to count, the evidence map, the search strategy and the methods with the protocol. He confirmed the known studies and the duplicate match, made all 26 screening decisions of the pilot screening and chose not to submit the protocol (a test). The assistant admitted or excluded no record and certified nothing.

How the AI's work was checked. Every count in this protocol is read from the planning record, where it was computed by code from the saved response; the saved results were re-counted by code against the record at the end of sessions, with the results kept in the planning record. Every passage quoted to Uday was checked by code against the saved title, abstract or full text. Uday read each draft before he certified it. No second person checked the assistant's suggestions.

Use during the review (certified): the assistant will run and log searches, count records by code, suggest screening decisions and extracted values with their exact passages, and draft wording; Uday makes and certifies every decision, with his reason recorded.

Interests and sharing (certified): Uday wrote the STIR-L4 planning prompt and its helper kit used here (summarized); no other interest is declared (my suggestion). The prompt, kit, record of every step and code are shared with the review (my suggestion); a 2026 study of reviews that disclose AI use found that sharing of prompts and human-validation procedures was insufficient [38].

Limitations. The assistant's suggestions are not checked by a second person; the author of the prompt and kit uses them and judges their suggestions alone; evidence on AI tools as second screeners is limited [23].

Gaps

- Item 2: registration number and date. None yet: on 2026-10-25 Uday chose not to submit the protocol, as planning was a test (record: L000708). If he submits it, he adds the number and date here.
- Item 3a: e-mail address and postal address of the corresponding author. Uday adds them on the register's form when he submits; none is kept in the planning files.
- Item 9: run dates and counts for OpenAlex and the WHO trial portal. Not run when planning ended; Uday supplies them when he runs these searches (OpenAlex reads past its first 100 results only with a free access code).
- Item 10: the exact string for the WHO trial portal, built from the drug names of the ClinicalTrials.gov search when Uday runs it by hand.
- Item 11c: whether the extraction form is piloted before use, and on how many studies. Not decided; Uday decides.
- Item 15b: summary measures, handling of data, the method of combining studies, the measure of consistency, and how 'similar' trials are judged. Not decided; if a part reaches 3 similar randomized trials, Uday settles them and registers them as an amendment before any pooling (certified rule; registering them as an amendment was my suggestion).

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Part 2. Session 1

Session 1 (from the first example). The topic, each interest, the focus and the purpose are set; Uday asks to "just start", so Step 3's searches run early; the session ends with a pause.

UDAY (SIMULATED USER) session 1, message 5

passed 16/16. begin

STIR-L4 ASSISTANT session 1, message 6

Step 1 of 12: Topic | **Certified 0 of 8** | **Waiting on you: 1** | **Focus: not yet stated**

Done: Recorded your quiz result (passed, 16 of 16). Planning starts now, with your topic.

Next: I will repeat your topic back in one sentence, then ask each of you which part of it interests you most.

Your turn, team: What topic would you like to plan a systematic review on? Describe it in your own words, as fully or as briefly as you like; we will sharpen it together.
For example: "Exercise programs for older adults after a hip fracture"
Or just a few words: "sleep problems after stroke"

1. **The status line.** Every reply opens with it: the step (1 of 12), how many of the eight decisions are certified, what the team owes the assistant, and the research focus once stated.

2. **The same shape every time.** Done (what was recorded), Next (what comes after your answer) and Your turn, always last, with one plain question and example replies. The topic is asked first, on its own.

UDAY (SIMULATED USER) session 1, message 7

role of SGLT2i in weight management in non diabeteic asian populations

STIR-L4 ASSISTANT session 1, message 8

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 3** | **Focus: not yet stated**

Done: Recorded your topic, in my words for now: *the role of SGLT2 inhibitors in weight management in Asian populations without diabetes.* We will sharpen it step by step into your research focus (Step 2), objective (Step 4) and question (Step 6).

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	now	each member present
3. Need for a Review	whether a new review or an update is needed	next	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—

Step	Settles	Status	Waiting on
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

The top line shows the step, approvals (of 8), what awaits you and your focus.

Quick commands (any wording) and labels	What they mean
"where are we"	the plan map and what comes next
"next"	moves on; never accepts or certifies anything
"why"	how this answer affects your other decisions
"show <item>"	shows that item in full
"skip"	the question stays open
"back"	reopens an approved decision only if the team agrees
"pause"	a summary, then stop
FACT · JUDGMENT · DRAFT	found in a search · what one of you said · my wording for you to edit
"Added by me: nothing"	every word came from you

Next: I will reflect your answers back to confirm.

Your turn, each member present: please give the name I should record (add "solo" if you are planning alone or trying out this prompt). Then, in your own words:

1. Within this topic, which part or question interests you most, and why?
2. What do you hope to get from this review yourself? You may say "prefer not to say".
3. For the team: what should the review achieve, and which decision or work should its results inform?

Say if I may keep your exact words (topic included), or only summaries you confirm. Approvals (certify) and reasons are always kept as given.
Example: "Sam (solo). 1. ... 2. ... 3. ... Summaries are fine."

3 **Plan map and quick commands, once.** The twelve steps, what each settles and who it waits on, then the commands the team may type in any wording and what the labels FACT, JUDGMENT and DRAFT mean.

4 **The research focus is asked for directly.** Before any logistics: which part of the topic interests each member and why, what each wants from the review, and what the review should achieve and inform. Consent is asked once.

UDAY (SIMULATED USER) session 1, message 9

uday. solo. all of it i guess. learning. do a systematic review on this topic

STIR-L4 ASSISTANT session 1, message 10

Step 2 of 12: Interests, focus and purpose	Certified 0 of 8	Waiting on you: 2 (Uday)	Focus: not yet stated

Done: Recorded you, Uday, as planning solo, so you certify every decision; the topic is now under your name. You didn't say whether I may keep your exact words, so I'll keep summaries that you confirm.

JUDGMENT (Uday, summarized):

- Interest: the whole topic, for now; no single part chosen yet.
- For yourself: to learn.
- For the review: a systematic review on this topic; no decision or work for its results named yet.

Next: I will reflect your answers back in one short summary for you to confirm, with the kinds of purpose and motivation to choose from.

Your turn, Uday: two short follow-ups, because your focus and purpose steer every search and draft that follows.

1. If the review could answer only one question about this topic, which would you pick?
2. Once the review is done, what would you do with its results?

For example: "1. <one question, in your words>. 2. <what you would use the results for>."
Or: "1. Keep the whole topic as it is. 2. I would use them to <...>."

4 **Vague answers are probed, never filled in.** "All of it" and "do a systematic review" get two short follow-up questions. The assistant adds no aim of its own.

UDAY (SIMULATED USER) session 1, message 11

what kind of weight is lost, fat or muscle. decide if a full review is worth it, maybe publish

STIR-L4 ASSISTANT session 1, message 12

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat or muscle (to confirm)

Done: Recorded your one question and your plans for the results. Your focus narrows the topic to one question; the topic stays as recorded.

JUDGMENT (Uday, summarized), reflected back:

What	Your answer, summarized
Your interest, and so the research focus	which kind of weight is lost: fat or muscle
What you seek	to learn
The review should achieve	a systematic review on this topic
Its results should inform	whether a full review is worth it; maybe a publication

Added by me: nothing. Your topic names Asian people without diabetes; the focus names no population (question 1).

Kinds of purpose (what the review is for, for the team), none chosen for you:

#	Kind of purpose	What it means
1	Summary of current knowledge	what is known now
2	Research gaps	what is not yet known
3	Finding the right question	sharpening what to ask
4	Direction for future work	what research should come next
5	Clinical guideline	a recommendation for practice
6	Resolving conflicts	settling studies that disagree
7	Policy development	a decision for an organization or health system
8	Quantitative analysis	pooling results in a meta-analysis
9	Other	say what, in your words

DRAFT: kinds of motivation (what you get from the review) your answers seem to fit:

Kind of motivation	Why it seems to fit
Learn the process (how reviews are done), personal or institutional curiosity (about this topic), or both	you want to learn; question 3 settles which
Academic activity (coursework, a thesis or a publication)	you may publish

Next: I will record your choices, finish setting up the search tools, then ask you to certify your motivations, then the purpose.

Your turn, Uday: one reply settles the focus and purpose that every later search follows.

1. Is everything above right? Reply "yes" or say what to change, and keep the focus a) within Asian people without diabetes, or b) with no population limit.
2. Which kinds of purpose apply? Give their numbers.
3. Learning means: a) how reviews are done; b) this topic; c) both.

For example: "1. yes, <a or b>. 2. <the numbers>. 3. <a, b or c>."

4 **The focus is reflected back, with nothing added.** The status line now carries the focus, marked "(to confirm)". The kinds of purpose are shown with short explanations, none chosen for the user; the motivation is a DRAFT to confirm.

UDAY (SIMULATED USER) session 1, message 13

yes. 2 3. yes

STIR-L4 ASSISTANT session 1, message 14

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle

Done: Recorded that you confirmed the focus wording, and recorded your kinds of purpose and kinds of motivation. Because you are planning solo, I skipped the questions about other members, databases, deadlines and rules. They are recorded as not given.

JUDGMENT (Uday, summarized):

What	Recorded
Research focus	which kind of weight is lost: fat or muscle
Population for the focus	within Asian people without diabetes, or no limit (to confirm)
Kinds of purpose	research gaps (what is not yet known); finding the right question (sharpening what to ask)
Kinds of motivation	learn the process (how reviews are done); personal or institutional curiosity; academic activity (coursework, a thesis or a publication)

Your "yes" did not settle the population or what learning means. Both kinds of learning (the first two motivations) are included; say which to drop if one is wrong.

The search tools work: PubMed (an index of medical journals), OpenAlex (an open index of research papers) and ClinicalTrials.gov (a registry of clinical trials).

When convenient: If you like, type a contact email for PubMed searches (PubMed asks for one); I keep it private.

Next: I will ask you once more about the population, then show your purpose for you to certify.

```
CERTIFY Individual Motivation | Uday (summarized): interest — which kind of weight is lost: fat or
muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional
curiosity; academic activity | reason: ___ | Uday, solo
```

Your turn, Uday: your motivations come first, and the purpose follows. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

5 **Solo planning skips the team questions.** Set-up is reported in one plain sentence. The certification line is filled in except for the reason; replying "certify, because ..." is enough.

UDAY (SIMULATED USER) session 1, message 15

Certify

STIR-L4 ASSISTANT session 1, message 16

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle

Done: Recorded your certify; it needs your reason.

Your turn, Uday: What is your reason, in a few words?

5 **A bare "Certify" gets one question.** Only the reason is asked for, with no rule numbers and nothing to retype.

UDAY (SIMULATED USER) session 1, message 17

testing the workflow

STIR-L4 ASSISTANT session 1, message 18

Step 2 of 12: Interests, focus and purpose | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle

Done: Recorded your certification of your motivations, with your reason as given. Nothing you have approved is affected.

```
CERTIFY Individual Motivation | Uday (summarized): interest — which kind of weight is lost: fat or muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional curiosity; academic activity | reason: testing the workflow | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	now	Uday
3. Need for a Review	whether a new review or an update is needed	next	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—

Step	Settles	Status	Waiting on
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member. Your motivations are certified; the team's purpose is next.

Next: I will show your purpose, naming your focus, for you to certify.

Your turn, Uday: one choice. I am asking a second and last time, because the purpose names your focus and Step 3 will look for existing reviews of it. Your topic names Asian people without diabetes; the focus names no population. Should the focus be:

a) within Asian people without diabetes, or
b) with no population limit?

For example: "a"
Or: "b"

5 **Certified 1 of 8.** The completed line is shown back with the reason as given, and the plan map is shown because a decision was certified. The open population choice is asked a second and last time.

UDAY (SIMULATED USER) session 1, message 19

a

STIR-L4 ASSISTANT session 1, message 20

Step 2 of 12: Interests, focus and purpose | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your choice: the focus now covers only Asian people without diabetes, which narrows it.
This may change the Individual Motivation you approved, because the agreed purpose shapes what each member seeks from the review; nothing changes unless you agree.

JUDGMENT (Uday, summarized): the purpose below uses only answers you confirmed: the narrowed focus, the kinds of purpose you chose and what you would do with the results. You chose those kinds of purpose before the focus narrowed, so check that they still hold. Added by me: nothing.

Next: once you certify, I will say what Step 3 will check for your focus, then search for existing reviews of your topic and of your focus.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose names the focus that every later search follows. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

4 **The focus is carried into the purpose.** The purpose line names the narrowed focus and is built only from answers the user confirmed ("Added by me: nothing").

UDAY (SIMULATED USER) session 1, message 21

just start the review process

STIR-L4 ASSISTANT session 1, message 22

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Moved to Step 3 (need for a review), as you asked; your purpose stays open. Searched PubMed and OpenAlex for existing reviews. Nothing you have approved is affected.

Steps: 1 done · 2 now · 3 started early · 4–12 open

FACT: PubMed's review filter finds 1 record for your topic [5] and 70 for SGLT2 inhibitors with fat or muscle [8]; OpenAlex's review label, 1,108 [13] and 2,203 [16].
FACT: Of 206 records examined, 9 may bear on your focus: 7 name body fat, none muscle or lean mass [18]; the latest stated search end is August 2025 [17].

Still unknown: what full texts and unsearched sources show for your focus.

How to read the strings: `[Mesh]` a MeSH subject heading, including narrower headings; `[tiab]` words in the title or abstract; `[sb]` a PubMed subset, such as systematic reviews; `*` any word ending (truncation).
Also: MeSH means Medical Subject Headings, PubMed's subject index; `"` an exact phrase; `AND` every block must match; `OR` any one term in the block may match; `(` parentheses group the terms of a block.

Search 1: Your topic: all four topic concepts (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Weight"[Mesh] OR
weight[tiab] OR obes*[tiab] OR overweight[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR asian*
[tiab] OR asia[tiab] OR chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*[tiab] OR
korea[tiab] OR indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab] OR
malaysia*[tiab] OR vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR
bangladesh*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	subject headings, title/abstract words, with truncation	23,338 [1]
2	weight management	topic	subject headings, title/abstract words, with truncation	1,650,180 [2]
3	Asian populations	topic	subject headings, title/abstract words, with truncation	2,263,145 [3]
4	without diabetes	topic	title/abstract words	66,022 [4]
All	combined	—	blocks joined as shown	**1** [5]

Block counts are before the filter; the combined count is after it.

Search 2: Your focus: SGLT2 inhibitors and fat or muscle (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	subject headings, title/abstract words, with truncation	23,338 [6]
2	fat or muscle (which kind of weight is lost)	focus	subject headings, title/abstract words, with truncation	1,524,751 [7]

Block	Covers	From	How it searches	Records
All	combined	—	blocks joined as shown	**70** [8]

Block counts are before the filter; the combined count is after it.

Search 3: Your topic: all four topic concepts (OpenAlex; published any to 2026-10-25; filter: type: review)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin) AND (weight OR obesity OR overweight) AND (Asian OR Asians OR Asia OR Chinese OR China OR Japanese OR Japan OR Korean OR Korea OR Indian OR India OR Taiwan OR Thailand OR Singapore OR Malaysia OR Vietnam OR Indonesia OR Philippines OR Pakistan OR Bangladesh) AND ("non-diabetic" OR nondiabetic OR "without diabetes" OR "without type 2 diabetes" OR "absence of diabetes" OR normoglycemic)
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	words in title, abstract or text	122,281 [9]
2	weight management	topic	words in title, abstract or text	15,691,147 [10]
3	Asian populations	topic	words in title, abstract or text	33,895,133 [11]
4	without diabetes	topic	words in title, abstract or text	384,375 [12]
All	combined	—	blocks joined as shown	**1,108** [13]

Block counts are before the filter; the combined count is after it.

Search 4: Your focus: SGLT2 inhibitors and fat or muscle (OpenAlex; published any to 2026-10-25; filter: type: review)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin) AND (fat OR muscle OR muscles OR adipose OR adiposity OR "body composition" OR "lean mass" OR "lean body mass" OR "fat-free mass" OR sarcopenia)
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	words in title, abstract or text	122,281 [14]
2	fat or muscle (which kind of weight is lost)	focus	words in title, abstract or text	8,528,973 [15]
All	combined	—	blocks joined as shown	**2,203** [16]

Block counts are before the filter; the combined count is after it.

Left out on purpose in searches 2 and 4: weight management and your population, so reviews that don't name them are not missed; the rule below checks the population. PubMed could not find “absence of diabetes” and searched without it [4].

Not searched or not yet read	Who can do it
PROSPERO and OSF Registries (registers of planned reviews)	you, Uday, with strings I will prepare
Cochrane and Epistemonikos (libraries of systematic reviews)	you, Uday
Review protocols published in journals, through PubMed	me, after you certify
OpenAlex past the first 100 works of searches 3 and 4: 1,008 and 2,103 more [18]	me, with a free access code I will ask about later

DRAFT rule (my terms, yours to edit): a record may bear on your focus if its title or abstract names fat or muscle in your sense (body fat, not liver fat) and names Asian people or people without diabetes.

FACT, from [17]: Reviews that may bear on your focus

#	Title	Year	Says (quoted from the abstract)	PMID
1	Selective SGLT2 inhibitors in MASLD/MASH: an outcome-specific systematic review and meta-…	2026	“SGLT2 inhibitors also improved body weight, body mass index, visceral adiposity and glycaemic control in T2DM-associated MASLD.”	42517896
2	Effectiveness of sodium-glucose cotransporter 2 inhibitors in patients with metabolic dys…	2026	“In addition, reductions were noted in body weight, visceral adipose tissue, waist circumference, glycated hemoglobin, and homeostatic model assessment of insulin resistance.”	42215421
3	Emerging Effects of Glucagon-Like Peptide-1 Receptor Agonists and Sodium-Glucose Cotransp…	2026	“Both classes of medications showed an overall improvement of body composition and metabolic profile, resulting in positive erectile function outcomes.”	42011503
4	Comparison of efficacy and safety of pioglitazone and SGLT2 inhibitors in treating Asian …	2025	“Improvement in body composition, metabolic parameters, glucose parameters, and incidence of adverse effects as a secondary outcome”…	40043473
5	Comparison of sodium-glucose cotransporter-2 inhibitors and thiazolidinediones for manage…	2023	…“with SGLT2 inhibitors (n = 156) showed significant decrease in visceral fat area (VFA;”	36931466
6	GLP-1 RAs and SGLT-2 Inhibitors for Insulin Resistance in Nonalcoholic Fatty Liver Diseas…	2022	“-1.573[-2.523 to -0.495]), visceral fat (-0.637[-0.992 to -0.284]), weight (-2.394[-4.625 to -0.164]), fasting blood sugar (-0.662[-1.377 to -0.021])”…	35909522
7	Effect of Sodium-Glucose Cotransporter 2 Inhibitors on Weight Reduction in Overweight and…	2021	“There were no significant changes in fat mass, blood pressure, low-density lipoprotein cholesterol or high-density lipoprotein cholesterol with SGLT2 inhibitor treatment.”	34897070
8	Sodium-glucose co-transporter 2 inhibitors on weight change and cardiometabolic profiles …	2021	…“were observed, but not for the changes in waist circumference, fat mass, blood pressure, and lipid profile compared with placebo.”	34542222
9	Sodium-Glucose Co-Transporter 2 Inhibitors for Non-Alcoholic Fatty Liver Disease in Asian…	2020	“compared with Control: -0.468 (-0.685 to -0.251), p < 0.001), body composition (BMI: after treatment: -0.225 (-0.456 to 0.005), p = 0.055;”	33643221

FACT, from [17]: search end dates in the same abstracts

#	Search end date, as the abstract states it
1	unknown (abstract only; no date stated)
2	unknown (abstract only; no date stated)
3	August 2025: …“systematic search of Medline was conducted from 1969 to August 2025, according”…
4	December 2024: …“Technology Journal Database (VIP) were searched from inception to December 2024.”
5	unknown (abstract only; no date stated)
6	March 2021: …“databases (Medline, Embase, PubMed) were searched from inception until March 2021.”
7	May 2021: …“Cochrane Central Register of Controlled Trials were searched through May 2021.”
8	unknown (abstract only; no date stated)
9	July 2020: …“on treatments of NAFLD in type 2 diabetes in July 2020, and”…

DRAFT	Parts of your focus, in the 9 abstracts
Covered	body fat
Not covered	muscle or lean mass; Asian people and people without diabetes, named together
Unknown	what the full texts say, and the records in the table of what is not searched or read
Examined	70 PubMed and 200 OpenAlex records, 206 after removing duplicates [18]
Related	40 more reviews name fat or muscle but neither Asian people nor people without diabetes [18]

Say "show broader" (33 reviews: same drug and population, other outcomes [18]) or "show all"; all are also beside the chat.

Sources: [1] L000068 PubMed, run 2026-10-25; [2] L000069 PubMed, run 2026-10-25; [3] L000070 PubMed, run 2026-10-25; [4] L000071 PubMed, run 2026-10-25; [5] L000072 PubMed, run 2026-10-25; [6] L000073 PubMed, run 2026-10-25; [7] L000074 PubMed, run 2026-10-25; [8] L000075 PubMed, run 2026-10-25; [9] L000076 OpenAlex, run 2026-10-25; [10] L000077 OpenAlex, run 2026-10-25; [11] L000078 OpenAlex, run 2026-10-25; [12] L000079 OpenAlex, run 2026-10-25; [13] L000080 OpenAlex, run 2026-10-25; [14] L000081 OpenAlex, run 2026-10-25; [15] L000082 OpenAlex, run 2026-10-25; [16] L000083 OpenAlex, run 2026-10-25; [17] L000084 PubMed, run 2026-10-25; [18] L000085 computation from [8], [13], [16] and [17], run 2026-10-25

Next: after you certify, I will ask whether these reviews change how you see your focus.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose is the last item before Step 3. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."

Opened beside the chat: stir-l4-workspace/data/derived/step3_review_selection.csv

6 **"Just start" is honored without skipping anything.** Step 3 is started early and marked in the status line; the purpose stays open and is still asked for, last.

7 **Every search is shown in full.** Lead FACT lines, a key to the field tags, then each search with its exact string and a count for every block, the date bound checked in PubMed's own translation, what was left out on purpose and why, what has not been searched yet, the reviews found with quoted passages, and one Sources line tying every number to a logged query. (This reply is long; see the last page.)

UDAY (SIMULATED USER) session 1, message 23

what was the search strategy?

STIR-L4 ASSISTANT session 1, message 24

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your question. Here is the short form of the four searches from my last reply. Nothing you have approved is affected.

These searches looked only for existing reviews (Step 3). The search for your own review comes at Step 8, once your question is set.
In PubMed and OpenAlex I ran two searches each: all four topic concepts joined (SGLT2 inhibitors, weight, Asian people, without diabetes), and SGLT2 inhibitors joined with fat or muscle, your focus.
Code then picked the reviews that may bear on your focus, using the rule shown last time.

FACT:

#	Source	Looks for	Filters and dates	Records
1	PubMed	Your topic: all four topic concepts	systematic-review filter; created 1800/01/01 to 2026/10/25 (checked)	1 [1]
2	PubMed	Your focus: SGLT2 inhibitors and fat or muscle	systematic-review filter; created 1800/01/01 to 2026/10/25 (checked)	70 [2]
3	OpenAlex	Your topic: all four topic concepts	type: review; published any to 2026-10-25	1,108 [3]
4	OpenAlex	Your focus: SGLT2 inhibitors and fat or muscle	type: review; published any to 2026-10-25	2,203 [4]

Say "show strings" to see each search string again.

Sources: [1] L000072 PubMed, run 2026-10-25; [2] L000075 PubMed, run 2026-10-25; [3] L000080 OpenAlex, run 2026-10-25; [4] L000083 OpenAlex, run 2026-10-25

Still open from my last reply:

- Your certification of the team's purpose (the line below).

Next: after you certify, I will ask whether these reviews change how you see your focus.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose is the last item before Step 3. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "show strings"

8 **Asking again gets a compact recap.** One row per search, with an offer to show the strings again; nothing is sent to a file.

UDAY (SIMULATED USER) session 1, message 25

what's the next step?

STIR-L4 ASSISTANT session 1, message 26

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your question. Nothing you have approved is affected.

The next step is yours: certify your purpose (the line below), finishing Step 2. Step 3 (need for a review) then continues, one question per reply:

1. whether these reviews change how you see your focus;
2. why it matters for patients or practice;
3. the impact you expect.

Then I prepare strings for your PROSPERO and OSF Registries searches, ask about the free OpenAlex access code, and draft the case for a new review, an update or no review, for you to certify.

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	now	Uday
3. Need for a Review	whether a new review or an update is needed	started early	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—

Step	Settles	Status	Waiting on
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

When convenient: If you like, type a contact email for PubMed searches (PubMed asks for one); I keep it private.

Next: after you certify, I will ask the first Step 3 question above.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose is all that Step 2 still needs. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

8 **"What's the next step?" gets the plan map.** Plus the next three questions in order, and the one item that blocks Step 2.

UDAY (SIMULATED USER) session 1, message 27

certify, focus is right

STIR-L4 ASSISTANT session 1, message 28

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your certification; searched PubMed for published review protocols.
This may change the motivations you approved, because the purpose shapes them.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: focus is right | Uday, solo
```

Step 3 checks whether existing or planned reviews already answer your focus (summarized): which kind of weight is lost: fat or muscle, within Asian people without diabetes.

Steps: 1 done · 2 done · 3 now · 4–12 open

FACT: PubMed's review filter finds no published protocols for your topic [2] or focus [3].

Still unknown: what full texts and unsearched sources show.

How to read the strings: `[ti]` words in the title.

Search 1: Your topic, protocols only: all four topic concepts and protocol (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
```

```
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Weight"[Mesh] OR
weight[tiab] OR obes*[tiab] OR overweight[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR asian*
[tiab] OR asia[tiab] OR chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*[tiab] OR
korea[tiab] OR indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab] OR
malaysia*[tiab] OR vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR
bangladesh*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab]) AND (protocol[ti])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	review protocols (the word protocol in the title)	added	words in the title	114,264 [1]
All	combined	—	blocks joined as shown	**0** [2]

Block counts are before the filter; the combined count is after it.

Search 2: Your focus, protocols only: SGLT2 inhibitors, fat or muscle and protocol (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab]) AND (protocol[ti])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	review protocols (the word protocol in the title)	added	words in the title	114,264 [1]
All	combined	—	blocks joined as shown	**0** [3]

Block counts are before the filter; the combined count is after it.

Not searched yet, or left out on purpose	Who, or why
PROSPERO and OSF Registries; Cochrane and Epistemonikos	you, Uday, later in this step
OpenAlex past its first 100 works per search	me, with a free access code
Left out on purpose in search 2: weight management and your population	as before, so protocols that don't name them are not missed

FACT, from [4]: of the 9 reviews (of 206 records examined) that may bear on your focus	Reviews
name body fat	7
name muscle or lean mass	0
name both Asian people and people without diabetes	0

Sources: [1] L000097 PubMed, run 2026-10-25; [2] L000098 PubMed, run 2026-10-25; [3] L000099 PubMed, run 2026-10-25; [4] L000085 computation from earlier searches, run 2026-10-25

Next: I'll ask why this matters for patients or practice.

Your turn, Uday: the case for a review rests on this. Do these findings change how you see your focus?

a) no, keep it
b) yes, say how

For example: "a"
Or: "b, <what changes>"

1 **Step 2 is complete: Step 3 of 12, Certified 2 of 8.** The certification also triggers a search for published review protocols. Step 3 now asks one question per reply.

UDAY (SIMULATED USER) session 1, message 29

pause

STIR-L4 ASSISTANT session 1, message 30

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Paused at your request. Saved a summary of this session and a copy of the whole plan for you to keep. Nothing you have approved is affected.

Session summary (saved; also open next to the chat):

```
# Session summary (S1, 2026-10-25)

Step 3 of 12: Need for a Review | Certified 2 of 8 | Waiting on you: 1 (Uday) | Focus: which kind of
weight is lost: fat or muscle, within Asian people…

Topic: the role of SGLT2 inhibitors in weight management in Asian populations without diabetes
Focus: which kind of weight is lost: fat or muscle, within Asian people without diabetes

Settled this session:
- CERTIFY Individual Motivation | Uday (summarized): interest — which kind of weight is lost: fat or
muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional curiosity;
academic activity | reason: testing the workflow | Uday, solo
- CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle, within
Asian people without diabetes; kinds of purpose — research gaps; finding the right question; the review —
a systematic review on this topic; with the results — decide whether a full review is worth it; maybe
publish | reason: focus is right | Uday, solo

Reopened:
- nothing

Still open: Need for a Review, Review Objective, Feasibility, Research Question, Search Strategy,
Methodology

Next questions:
- Uday (open now): do these findings change how you see your focus? a) no, keep it; b) yes, say how
- Uday: why does your focus matter for patients or practice?
- Uday: what impact do you expect from the review?
- Uday: search PROSPERO and OSF Registries with strings I will prepare, and Cochrane and Epistemonikos if
you have access
- Uday: whether I may use a free OpenAlex access code to read past the first 100 results of each search
- Then: the case for a new review, an update or no review, for you to certify

Notes: Step 3 was started early at your request and is now the current step. Every search count reported
this session was recomputed from the saved results and matches. Changed this session (all new): your plan
and team record, the record of every step, the saved search results and the set-up and Step 3
calculations.

Searches this session: 19
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	now	Uday
4. Review Objective	what the review should achieve	next	—

Step	Settles	Status	Waiting on
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

When convenient: If you like, type a contact email for PubMed searches (PubMed uses it only to reach us if our searches cause a problem); I keep it private.

To save: please keep this copy of the whole plan and every record: `stir-l4-workspace_2026-10-25.zip`, next to the planning folder. If you resume in a new chat, give it to me with the prompt.

Next: when you return, I will read the saved plan and records, show where we are, and continue Step 3 from your answer.

Your turn, Uday: nothing more is needed today. When you return, the case for a review rests on one open answer: do these findings change how you see your focus?

a) no, keep it
b) yes, say how

For example: "a"
Or: "b, <what changes>"

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_S1_summary.md

9 **Pause.** The session summary and the plan map are shown in the chat; the workspace is packaged only now, for the user to keep and bring to the next session.

Part 2. Session 2

Session 2. A new chat resumes from the saved files. The soundness check runs; body water joins the focus; the need for a review and the objective are certified; feasibility says broaden, which reopens three earlier decisions, each certified again; the session ends as the research question is drafted.

Messages 1 to 4 of each session (the tool test and the calibration quiz, scored 16 of 16) are not shown.

UDAY (SIMULATED USER) session 2, message 5

passed 16/16. we paused last time, continue from the saved plan. my answer to your last question: a

STIR-L4 ASSISTANT session 2, message 6

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your answer: your focus stays as certified. Nothing you have approved is affected.

JUDGMENT (Uday, summarized), from your saved plan	
Topic	the role of SGLT2 inhibitors in weight management in Asian populations without diabetes
Focus	which kind of weight is lost: fat or muscle, within Asian people without diabetes
Certified	your motivations; the team's purpose

Step	**Settles**	**Status**	**Waiting on**
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	now	Uday
4. Review Objective	what the review should achieve	next	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.
The line at the top of each reply shows the step, the decisions you have approved (out of 8), the answers or approvals the current step is waiting for, and your research focus.
Quick commands, in any wording: "where are we" · "next" (never accepts or certifies) · "why" (how this answer affects your other decisions) · "show <item>" · "skip" (the question stays open) · "back" (reopens only if the team agrees) · "pause" (summary and stop).

If anything here looks wrong, tell me. Files here may not be kept, so I will give you a copy of the plan when we pause.

Next: I will ask what impact you expect, then check by searches whether your question holds up clinically, scientifically and logically, before drafting the case.

Your turn, Uday: the case for a new review rests on this: why does your focus matter for patients or practice?
For example: "it matters because ..."
Or: "not sure"

R **A new chat resumes from the saved files.** The assistant has no memory of Session 1. It reads the saved plan, team file, log and session summary, restores the status line and plan map, and continues from the open question.

UDAY (SIMULATED USER) session 2, message 7

sglt2i are used off label for weight in asia. if the loss is mostly muscle or water, that matters for older and lean asian patients

STIR-L4 ASSISTANT session 2, message 8

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded why your focus matters, as a summary in my words. Nothing you have approved is affected.

JUDGMENT (Uday, summarized; tell me if I got it wrong)	
Why your focus matters	SGLT2 inhibitors are used off label for weight loss in Asia; if most of the weight lost is muscle or water, that matters for older and lean Asian patients

Your answer leaves your focus as certified, but it names two things the focus does not: water, and older and lean patients. No search so far has checked whether SGLT2 inhibitors are used for weight in Asian people without diabetes, or whether studies can tell lost muscle from lost water; both are unknown for now. I will bring them back with the evidence, and you decide what goes into the focus.

Next: I will check by searches whether your question holds up clinically, scientifically and logically, then ask you to search the registers of planned reviews.

Your turn, Uday: the case for a new review also needs the impact you expect. If the review shows which kind of weight is lost, what would that change?
For example: "it would ..."
Or: "not sure"

UDAY (SIMULATED USER) session 2, message 9

tell clinicians whether the weight lost is fat, and show where trials are missing

STIR-L4 ASSISTANT session 2, message 10

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your expected impact; searched PubMed to check your question. Nothing you have approved is affected.

JUDGMENT (Uday, summarized; tell me if I got it wrong)	
Expected impact	the review would show clinicians whether the weight lost is fat, and point out where trials are lacking

FACT: PubMed (the main index of medical journals), no filter	Records
1. Your whole topic	10 [5]
2. Your focus, people without diabetes, any region	135 [7]
3. Your focus within your population	4 [8]: by PubMed's publication types, 3 reviews and 1 case report [18]
4. How the drug causes weight loss, without diabetes	136 [10]
5. Fat or muscle, with water, any population	81 [12]
6. Off-label use for weight, any population	34 [14]
Still open for your focus	full texts, other sources, and studies these words miss

How to read the strings	
`[Mesh]`	a MeSH heading (Medical Subject Headings, PubMed's subject index), with its narrower headings
`[tiab]`	words in the title or abstract
`*`	any word ending
`"..."`	an exact phrase
`AND` · `OR` · `( )`	every block must match · any term in a block may match · brackets group a block

Search 1: Clinical: your topic, all records (no review filter) (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Weight"[Mesh] OR weight[tiab] OR obes*[tiab] OR overweight[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR asian*[tiab] OR asia[tiab] OR chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*[tiab] OR korea[tiab] OR indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab] OR malaysia*[tiab] OR vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR bangladesh*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab] OR normoglycemic[tiab] OR normoglycaemic[tiab])
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	subject headings, title/abstract words, with truncation	23,338 [1]
2	weight management	topic	subject headings, title/abstract words, with truncation	1,650,180 [2]
3	Asian populations	topic	subject headings, title/abstract words, with truncation	2,263,145 [3]
4	without diabetes	topic	title/abstract words	66,022 [4]
All	combined	—	blocks joined as shown	**10** [5]

Search 2: Scientific: your focus in people without diabetes, any region (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
```

```
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab]
OR "non diabetic"[tiab] OR "without diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"
[tiab] OR "absence of diabetes"[tiab] OR normoglycemic[tiab] OR normoglycaemic[tiab])
```

Block	Covers	From	How it searches	Records
1	fat or muscle (which kind of weight is lost)	focus	subject headings, title/abstract words, with truncation	1,524,751 [6]
All	combined	—	blocks joined as shown	**135** [7]

Search 3: Your focus within Asian people without diabetes (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR
asian*[tiab] OR asia[tiab] OR chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*
[tiab] OR korea[tiab] OR indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab]
OR malaysia*[tiab] OR vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR
bangladesh*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab])
```

Records: **4** [8] (records, not studies).

Search 4: Scientific: how the drug causes weight loss, in people without diabetes (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Glycosuria"[Mesh] OR
glycosuria[tiab] OR glucosuria[tiab] OR "urinary glucose"[tiab] OR "Energy Intake"[Mesh] OR "energy
intake"[tiab] OR "energy balance"[tiab] OR "energy expenditure"[tiab] OR calori*[tiab]) AND ("non-
diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without diabetes"[tiab] OR "without type
2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab] OR normoglycemic[tiab] OR
normoglycaemic[tiab])
```

Block	Covers	From	How it searches	Records
1	glucose lost in urine; energy intake and balance	my suggestion (how the drug causes weight loss)	subject headings, title/abstract words, with truncation	217,544 [9]
All	combined	—	blocks joined as shown	**136** [10]

Search 5: Scientific: can studies tell muscle from water? (any population) (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab]) AND ("Body Water"[Mesh] OR "body water"[tiab]
OR "extracellular water"[tiab] OR "extracellular fluid"[tiab] OR "plasma volume"[tiab] OR "fluid volume"
[tiab] OR hydration[tiab] OR "Diuresis"[Mesh] OR diuresis[tiab] OR "Natriuresis"[Mesh] OR
natriuresis[tiab])
```

Block	Covers	From	How it searches	Records
1	water and fluid loss	Uday's reason (water)	subject headings, title/abstract words	124,598 [11]
All	combined	—	blocks joined as shown	**81** [12]

Search 6: Clinical: off-label use of SGLT2 inhibitors for weight (any population) (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Off-Label Use"[Mesh] OR "off-label"[tiab] OR "off label"[tiab]) AND ("Body Weight"[Mesh] OR weight[tiab] OR obes*[tiab] OR overweight[tiab])
```

Block	Covers	From	How it searches	Records
1	off-label use	Uday's reason (off label)	subject headings, title/abstract words	16,336 [13]
All	combined	—	blocks joined as shown	**34** [14]

Search	Left out on purpose	Why
1	nothing	all four topic concepts
2	Asian populations; weight management	to see evidence from any region (search 3 adds your population); fat or muscle already implies weight change
3	weight management	as in search 2
4	Asian populations; weight management; fat or muscle	to see how the drug works, from any region
5	Asian populations; without diabetes; weight management	to see how studies measure water, in any population
6	Asian populations; without diabetes	to find any report of off-label use; I then read the abstracts
1–4	the phrase "absence of diabetes"	PubMed could not find it and searched without it [4]

Not searched yet	Who can do it
PROSPERO and OSF Registries (registers of planned reviews)	you, Uday, next
OpenAlex (a free index of research papers) and Embase (a subscription database) for this check	me (OpenAlex); someone with Embase access
Ongoing trials, in ClinicalTrials.gov (the US trial register)	me, at Feasibility
Full texts of the 10 records below	me, for open-access copies, if you want
How often clinicians in Asia prescribe these drugs for weight	no source I can search; a clinician colleague or prescribing records

# (FACT, from [15], [16])	Title	Year	Says (quoted from the abstract)	PMID
1	Effect of sodium-glucose cotransporter-2 inhibitors on skeletal muscle.	2025	“are now also recommended for nondiabetic patients with heart failure (HF)”	40707349
2	Multi-Center Health Informatics to Examine the Implementation of Guideline-Directed Medical Therapy in Chronic Kidney Disease.	2025	“SGLT2i uptake doubled but remained below 40%, with lower adoption in non-diabetic and non-obese patients”	40916713
3	Before you start: the safety of SGLT2 inhibitors for anti-obesity treatment.	2026	“have become a viable off-label option for managing obesity”	41064961

# (FACT, from [15], [16])	Title	Year	Says (quoted from the abstract)	PMID
4	Administration of Once-daily Canagliflozin to a Non-diabetic Patient in Addition to Standard Aerobic Exercise: A Case Report.	2019	“the early use of canagliflozin (although on an off-label basis) in patients who are at high risk”	31192057
5	Clinical Physiology of SGLT2 Inhibitors in Non-diabetic and Diabetic Individuals.	2026	“Glycosuria also creates a chronic negative energy balance, although compensatory hyperphagia limits the degree of weight loss”	42730799
6	Shift to Fatty Substrate Utilization in Response to Sodium-Glucose Cotransporter 2 Inhibition in Subjects Without Diabetes and Pa…	2016	“similar in subjects without diabetes and in those with T2D but quantitatively smaller in the former”	26861783
7	Comparison of dapagliflozin and teneligliptin in nonalcoholic fatty liver disease patients without type 2 diabetes mellitus: a pr…	2021	“With dapagliflozin, body weight decreased, which was related to changes in total body water and body fat mass”	33879970
8	Exploring glycosuria as a mechanism for weight and fat mass reduction. A pilot study with remogliflozin etabonate and sergliflozi…	2014	“There was a reduction in TBW measured by D2O dilution in the RE group that was significantly greater than placebo”	29235586
9	Ipragliflozin Improves Glycemic Control and Decreases Body Fat in Patients With Type 2 Diabetes Mellitus.	2017	“Reduction of body fat accounted for more than 70% of the total weight loss and reduction of extracellular water accounted for about 20%”	28611859
10	Glucose-Lowering Therapies in Type 2 Diabetes-Associated Sarcopenia: Mechanisms, Clinical Evidence and a Muscle-Centred Framework.	2026	“lean-mass estimates do not directly quantify contractile skeletal muscle”	42892668

DRAFT: is your question sound? (yours to challenge)	What the records show	Still uncertain
Clinically: do Asian people without diabetes take these drugs, and does the outcome matter?	Without diabetes, the records name heart failure (#1) and kidney disease, in Singapore (#2), as uses. Off-label use is stated for obesity by a review (#3), and for a person without diabetes in India by a case report (#4). Muscle loss is a stated worry: an earlier review, of people with diabetes, urges careful monitoring in older adults [17].	How often clinicians in Asia prescribe these drugs for weight; #3 and #4 state such use but do not measure it.
Scientifically: is there a mechanism and earlier evidence, and can the outcome be measured?	Glucose lost in urine creates an energy deficit, but eating more limits the weight lost; there is also a short-lived loss of salt and water in urine (#5). The body's responses are similar without diabetes but smaller (#6). Two earlier reviews of people without diabetes found modest weight loss (1.62 kg more than placebo in one) and no significant change in fat mass [17]. Heavy-water dilution and magnetic resonance (#8; there TBW is total body water, D2O heavy water and RE one of the drugs), impedance, a weak electric current (#9), and DXA, a low-dose X-ray scan (#10), are used; lean-mass estimates can shift with body water and do not measure muscle directly, so strength or performance is also needed (#10).	Whether the small loss without diabetes splits as it does with diabetes, where fat was over 70% and extracellular water about 20% (#9). Studies of these drugs and muscle loss were mostly in people with diabetes (68%) and Japanese people (53%) (#1). I read abstracts only.
Logically: do population, drug, outcome and purpose fit?	1) "Fat or muscle" leaves out water, which is part of the weight lost (#8, #9). 2) Within your population, search 3 found 4 records [8]: 3 reviews and 1 case report [18]; without the Asian limit, search 2 found 135 [7]. A trial in people without diabetes with a Japanese affiliation (#7) names no country in its abstract and was found only by search 5 [18], so these words miss some studies. 3) No comparison is named yet; it comes with the research question. A thin evidence base fits your purpose (research gaps) and the impact you expect.	How many trials exist; Feasibility will count them.

DRAFT: is your question sound? (yours to challenge)	What the records show	Still uncertain
Plainly	Your question is clinically and scientifically reasonable and measurable, but it assumes two things the records question: that weight is lost only as fat or muscle (part is water, and lean mass is not muscle), and that these drugs are used for weight in your population (the uses these records name without diabetes are heart failure and kidney disease; off-label use is stated, not measured).	You decide whether and how to reframe.

Sources: [1] L000117 PubMed, run 2026-10-25; [2] L000118 PubMed, run 2026-10-25; [3] L000119 PubMed, run 2026-10-25; [4] L000120 PubMed, run 2026-10-25; [5] L000124 PubMed, run 2026-10-25; [6] L000121 PubMed, run 2026-10-25; [7] L000125 PubMed, run 2026-10-25; [8] L000126 PubMed, run 2026-10-25; [9] L000122 PubMed, run 2026-10-25; [10] L000127 PubMed, run 2026-10-25; [11] L000123 PubMed, run 2026-10-25; [12] L000128 PubMed, run 2026-10-25; [13] L000130 PubMed, run 2026-10-25; [14] L000131 PubMed, run 2026-10-25; [15] L000129 PubMed records, run 2026-10-25; [16] L000132 PubMed records, run 2026-10-25; [17] L000084 PubMed records, session 1, run 2026-10-25; [18] L000134 computation from [15], [16], [17] and the saved searches, run 2026-10-25

Next: I will prepare your PROSPERO and OSF Registries searches.

Your turn, Uday: your case rests on the focus. Does this check change it? One or more:

Option	Your focus would be
a	unchanged: which kind of weight is lost: fat or muscle, within Asian people without diabetes
b	with water added (your word): fat, muscle or water; you would certify the changed focus again
c	narrowed to older and lean patients (your words); this goes into the objective, at Step 4
d	other, for example a different population: say how; a changed focus is certified again

For example: "a, because ..."
Or: "b and c"

S **The soundness check of the question.** Before drafting the case for a review, the assistant checks with real, logged PubMed searches whether the question holds up clinically, scientifically and logically: the whole topic, the focus in any region, how the drug causes weight loss, fat or muscle with water, and off-label use.

UDAY (SIMULATED USER) session 2, message 11

good point. b, add body water as an outcome

STIR-L4 ASSISTANT session 2, message 12

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within… | **Reopened: Team's Purpose**

Done: Recorded your choice: body water joins your focus, which widens it. Your Team's Purpose names the focus, so I reopened it for you to certify again, as option b said. This may change what you approved for Individual Motivation (the agreed purpose in turn shapes what each member seeks from the review). Nothing changes unless you agree.

JUDGMENT (Uday, summarized)	
Focus, before	which kind of weight is lost: fat or muscle, within Asian people without diabetes
Focus, now	which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes

JUDGMENT (Uday, summarized)	
Not added	older and lean patients (you chose b only)
Given for the earlier focus	your interest (fat or muscle); your kinds of purpose; why the focus matters (it already names water); the impact you expect (whether the weight lost is fat)

Still open from my soundness check, for the case: lean mass, as studies measure it, is not muscle [1].

Sources: [1] L000129 PubMed records, run 2026-10-25

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	reopened	Uday
3. Need for a Review	whether a new review or an update is needed	started early	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

Next: once you certify, I will search again for existing reviews of your focus with body water added, then prepare your PROSPERO and OSF Registries searches.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes; kinds of purpose — research gaps; finding the right question; the review — a systematic review on this topic; with the results — decide whether a full review is worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose names the focus every later search follows. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change, including any answer in the table given for the earlier focus.
For example: "certify, because ..."
Or: "change <what> to <what>"

F **A feedback loop: the focus changes, so the purpose is reopened.** Uday adds body water to the focus. Because the certified Team's Purpose names the focus, it is reopened and certified again (message 14).

UDAY (SIMULATED USER) session 2, message 13

certify, same purpose, i just added body water to the focus

STIR-L4 ASSISTANT session 2, message 14

Step 3 of 12: Need for a Review	Certified 2 of 8	Waiting on you: 2 (Uday)

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Certified your Team's Purpose; searched again for reviews of your focus.
This may change what you approved for Individual Motivation (the agreed purpose in turn shapes what each member seeks from the review). Nothing changes unless you agree.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes; kinds of purpose — research gaps; finding the right question; the review — a systematic review on this topic; with the results — decide whether a full review is worth it; maybe publish | reason: same purpose, i just added body water to the focus | Uday, solo
```

FACT: existing reviews of your focus, now "which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes"	
PubMed, systematic-review filter	73 reviews [2]; 1 of them a published protocol [3]
OpenAlex, reviews	2,343 [5]; I read the first 100
May bear on your focus (last session's rule, with body water added)	10 [9]: the 9 from last session and 1 new [10]; 5 state a search end date, the latest August 2025 [9]
Name body water	1 of the 10; 3 more, of other groups (table below) [9]
Still open for your focus	no review found names muscle in your population, or body water other than as a possible mechanism; registers of planned reviews not yet searched

Search 1: Your focus, with body water added (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR "Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab] OR "Body Water"[Mesh] OR "body water"[tiab] OR "total body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"[tiab] OR "extracellular fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR "water loss"[tiab] OR "fluid loss"[tiab])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	body water (added to your focus)	focus	subject headings, title/abstract words	106,629 [1]
All	combined	—	blocks joined as shown	**73** [2]

Block counts are before the filter; the combined count is after it.

Search 2: Published protocols for your focus, with body water added (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR "Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab] OR "Body Water"[Mesh] OR "body water"[tiab] OR "total body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"[tiab] OR "extracellular fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR "water loss"[tiab] OR "fluid loss"[tiab]) AND (protocol[ti])) AND systematic[sb]
```

Records: **1** [3] (records, not studies).

Search 3: Your focus, with body water added (OpenAlex; published any to 2026-10-25; filter: type: review)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin) AND (fat OR muscle OR muscles OR adipose OR adiposity OR "body composition" OR "lean mass" OR "lean body mass" OR "fat-free mass" OR sarcopenia OR "body water" OR "total body water" OR "extracellular water" OR "intracellular water" OR "extracellular fluid" OR "fluid volume" OR "plasma volume" OR hydration OR "water loss" OR "fluid loss")
```

Block	Covers	From	How it searches	Records
1	body water (added to your focus)	focus	words in title, abstract or text	1,682,380 [4]
All	combined	—	blocks joined as shown	**2,343** [5]

Block counts are before the filter; the combined count is after it.

Search	Left out on purpose	Why
1–3	weight management; Asian populations; without diabetes	to find reviews of your focus in any group; the rule then looks for your population
—	a new search of your topic	your topic did not change

Not searched yet	Who can do it
PROSPERO and OSF Registries	you, Uday, next
Cochrane (a library of systematic reviews) and Epistemonikos (a database of reviews)	you, if you have access
OpenAlex past the first 100 (2,243 more for search 3) [9]	me, with a free access code

FACT, from [5], [6], [7], [8]: Reviews that name body water

#	Title	Year	Says (quoted from the abstract)	PMID
1	SGLT2 Inhibitors Beyond Glycemic Control: Emerging Pharmacological and Cardiovascular App…	2026	…“feedback, lower intraglomerular pressure, modest changes in plasma and interstitial fluid volume, improved ventricular loading conditions, metabolic adaptation, and possible effects on inflammation, oxidative stress,”…	none; DOI 10.5281/zenodo.22978843
2	Effect of sodium-glucose transporter 2 inhibitors on sarcopenia in patients with type 2 d…	2023	“In addition, body water likewise decreased significantly (WMD=-0.96, 95% CI: -1.68 to -0.23, P=0.009).”	37465122
3	Mechanisms of SGLT inhibitor action and physiological mediators: systematic review and pr…	2026	…“tubular sodium handling, echocardiography, cardiac output monitoring, arterial stiffness and fluid volume will be extracted.”	41748173
4	Sodium-Glucose Cotransporter 2 Inhibitors in Heart Failure	2021	“Consequently, there are decreases in plasma volume, body weight, and blood pressure.”	34516288

DRAFT: what these reviews show for your focus (abstracts only)	
Your population	#1 names people who do not have diabetes, so it may bear on your focus; #2 to #4 name neither part [9]
Nearest to your question	#2, in type 2 diabetes: fat mass, lean mass, skeletal muscle mass and body water all fell [7]
The protocol found (#3)	names no population limit; main outcome kidney filtration; fluid volume to be extracted [6]
Search end dates	#2 February 2023; #1, #3, #4: none stated, so unknown [9]

DRAFT: what these reviews show for your focus (abstracts only)	
Parts of your focus in the 10	body fat named in 7; muscle or lean mass in 0; body water in 1 (#1), only as one possible mechanism; Asian people without diabetes, named together, in 0 [9]
Rule (my terms, yours to edit)	last session's rule, plus body water words (not water lost as a treatment or in urine) and the wording "do not have diabetes"

Sources: [1] L000148 PubMed, run 2026-10-25; [2] L000149 PubMed, run 2026-10-25; [3] L000150 PubMed, run 2026-10-25; [4] L000151 OpenAlex, run 2026-10-25; [5] L000152 OpenAlex, run 2026-10-25; [6] L000153 PubMed records, run 2026-10-25; [7] L000084 PubMed records, session 1, run 2026-10-25; [8] L000080 OpenAlex, session 1, run 2026-10-25; [9] L000155 computation from [2], [3], [5] to [8], run 2026-10-25; [10] L000160 computation from [9] and session 1's grouping, run 2026-10-25

Steps: 1 done · 2 done · 3 now · 4–12 open

Next: I will show your two register searches.

Your turn, Uday: registers list planned reviews, which no search here reaches. Say "next" for your PROSPERO and OSF Registries strings.
For example: "next"
Or: "skip"

UDAY (SIMULATED USER) session 2, message 15

cant search prospero or the others today. record them as not searched

STIR-L4 ASSISTANT session 2, message 16

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: I've recorded PROSPERO, OSF Registries, Cochrane and Epistemonikos as not searched today. So we still don't know whether anyone has a review planned. I drafted the case for your review from the searches so far. Nothing you have approved is affected.

FACT, from [1]: the two reviews nearest your focus (people without diabetes, weight in the title, no liver fat) [2]

#	Title	Year	Says (quoted from the abstract)	PMID
1	Effect of Sodium-Glucose Cotransporter 2 Inhibitors on Weight Reduction in Overweight and…	2021	“There were no significant changes in fat mass, blood pressure, low-density lipoprotein cholesterol or high-density lipoprotein cholesterol with SGLT2 inhibitor treatment.”	34897070
2	Sodium-glucose co-transporter 2 inhibitors on weight change and cardiometabolic profiles …	2021	…“were observed, but not for the changes in waist circumference, fat mass, blood pressure, and lipid profile compared with placebo.”	34542222

FACT, from [2]	#1	#2
Search end	May 2021, stated in its abstract	not stated in its abstract; unknown, as I read abstracts only
Names muscle or lean mass; body water; an Asian population	none of these	none of these

DRAFT: grounds for your case (question 2; yours to edit)	What bears on it
a. Previous reviews (what existing reviews cover)	Of the 10 reviews that may bear on your focus, 6 also name liver or other organ fat [2]. None names muscle or lean mass, and 1 names body water, only as a possible mechanism [3]. The two above report weight and fat mass, and name no Asian population [2].
b. Knowledge gaps (what is not yet known)	None of the 10 names Asian people and people without diabetes together [3]. Within your population, the earlier check found 4 records: 3 reviews and 1 case report [4].
c. Clinical justification (why it matters for patients or practice)	Yours (summarized): these drugs are used off label for weight in Asia, and losing muscle or water matters for older and lean patients. Records state off-label use [5] but do not measure how often it happens.
d. Impact analysis (what the review would change)	Yours (summarized): tell clinicians whether the weight lost is fat, and show where trials are missing. Lean mass, as studies measure it, is not muscle [6].
e. Other	say what
Not known yet	planned reviews (registers not searched); Cochrane and Epistemonikos; OpenAlex reviews past the first 100; full texts; how many trials exist (the Feasibility step will count them)

Question 1: the case	It would say	What bears on it
a. New review	no existing review answers your focus	none of the 10 that may bear on it names muscle or lean mass, or Asian people and people without diabetes together [3]
b. Update	an existing review answers your focus but needs newer studies	the two nearest (2021; one searched to May 2021, the other unknown) [2] and the one review your topic search found (2022) [7] are of people without diabetes. That 2022 review names none of your focus's fat, muscle or water terms [3], and names Indonesia only in “written in Bahasa Indonesia” [1]
c. No review	a review already answers it, or there is too little to review	in your population, 4 records: 3 reviews and 1 case report [4]; planned reviews unknown; trials not yet counted
d. Other	say what	—

Sources: [1] L000084 PubMed records, session 1, run 2026-10-25; [2] L000166 computation from [1] and [3], run 2026-10-25; [3] L000155 computation, run 2026-10-25; [4] L000134 computation, run 2026-10-25; [5] L000132 PubMed records, run 2026-10-25; [6] L000129 PubMed records, run 2026-10-25; [7] L000072 PubMed, session 1, run 2026-10-25

Next: once you choose, I will show your case as one line for you to certify. When you can search the registers, say "show register searches".

Your turn, Uday: your case says why this review is worth doing, and it shapes the objective in the next step.

1. Which case do you make: a new review, b update, c no review, or d other?
2. Which grounds apply, one or more: a to e in the draft above?

For example: "1 a; 2 a, b, c, d"
Or: "1 b, because ..."

H **Honest about what was not searched.** PROSPERO, OSF Registries, Cochrane and Epistemonikos are recorded as not searched, so whether a planned review exists stays unknown.

UDAY (SIMULATED USER) session 2, message 17

1 a; 2 a, b, c, d

STIR-L4 ASSISTANT session 2, message 18

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your choices: a new review, on four grounds: previous reviews, knowledge gaps, clinical justification and impact analysis. Nothing you have approved is affected.

DRAFT: your case as one line, built from your choices and the grounds drafted in my last reply. "Examined so far" means the 223 records from the review searches whose titles and abstracts I read; of the 10 that may bear on your focus, none names muscle or lean mass, or Asian people and people without diabetes together [1]. If the registers later show a planned review of your focus, I will bring this case back for you to decide whether to reopen it.

Sources: [1] L000155 computation, run 2026-10-25

Next: once you certify, I will show the chain so far and ask which Review Objective applies.

```
CERTIFY Need for a Review | Uday (summarized): case — new review: no review examined so far answers
the focus, which kind of weight is lost: fat, muscle or body water, within Asian people without
diabetes; grounds — previous reviews; knowledge gaps; clinical justification (used off label for
weight in Asia; muscle or water loss matters for older and lean patients); impact analysis (tell
clinicians whether the weight lost is fat; show where trials are missing); not searched — PROSPERO,
OSF Registries, Cochrane, Epistemonikos | reason: ___ | Uday, solo
```

Your turn, Uday: your case is why this review is worth doing, and the objective builds on it. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

UDAY (SIMULATED USER) session 2, message 19

certify, no review answers fat vs muscle in this population

STIR-L4 ASSISTANT session 2, message 20

Step 4 of 12: Review Objective | **Certified 3 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Certified your Need for a Review: a new review. Nothing you have approved is affected.

```
CERTIFY Need for a Review | Uday (summarized): case — new review: no review examined so far answers
the focus, which kind of weight is lost: fat, muscle or body water, within Asian people without
diabetes; grounds — previous reviews; knowledge gaps; clinical justification (used off label for
weight in Asia; muscle or water loss matters for older and lean patients); impact analysis (tell
clinicians whether the weight lost is fat; show where trials are missing); not searched — PROSPERO,
OSF Registries, Cochrane, Epistemonikos | reason: no review answers fat vs muscle in this
population | Uday, solo
```

JUDGMENT (Uday, summarized): the chain so far	
Topic	the role of SGLT2 inhibitors in weight management in Asian populations without diabetes
Focus and purpose (certified)	which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes; for research gaps and finding the right question; to decide whether a full review is worth it, maybe publish
Need for a Review (certified)	a new review; grounds: previous reviews, knowledge gaps, clinical justification, impact analysis; registers of planned reviews not searched

Step	**Settles**	**Status**	**Waiting on**
1. Topic	the provisional topic	done	—

Step	Settles	Status	Waiting on
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	now	Uday
5. Feasibility	go, narrow, broaden or defer	next	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

Your objective says what the review should achieve. More than one may apply, and none is pre-selected.

Option	Review Objective	What bears on it (FACT, abstracts only)
a	understand a topic	a 2026 paper on how these drugs act in people with and without diabetes [1]
b	compare interventions	the two nearest reviews compared these drugs with placebo [2]; one trial in people with fatty liver and without type 2 diabetes compared dapagliflozin with another drug, teneligliptin [1]
c	summarize available evidence	within your population, the earlier check found 4 records: 3 reviews and 1 case report [3]
d	measure evidentiary trust (how far the evidence can be trusted)	not checked yet
e	evaluate quality of evidence	lean-mass estimates do not directly measure muscle [1]
f	question a clinical decision	a 2026 review says they “have become a viable off-label option for managing obesity” [4]; how often they are used this way is unknown
g	identify new areas for research	a 2025 review: “There is an urgent need for research, especially in nondiabetic, non-Japanese and HF-patients” (HF: heart failure) [1]
h	update a previous SRMA (systematic review and meta-analysis)	your case, just certified, is a new review; of the two nearest reviews, one searched to May 2021 and the other's end date is unknown [5]
i	other	say what

Sources: [1] L000129 PubMed records, run 2026-10-25; [2] L000084 PubMed records, session 1, run 2026-10-25; [3] L000134 computation, run 2026-10-25; [4] L000132 PubMed records, run 2026-10-25; [5] L000166 computation, run 2026-10-25

Next: I will draft the wording of your objective from your choice and the chain above, showing where each phrase came from, for you to edit and certify.

Your turn, Uday: the objective sets what the review must achieve, and your research question will be built from it.
Which objectives apply: one or more of a to i?
For example: "<letters>, because ..."
Or: "i: <your own words>"

C **Need for a Review certified: 3 of 8.** A new review, on four grounds the user chose, with the user's reason.

UDAY (SIMULATED USER) session 2, message 21

a, g

STIR-L4 ASSISTANT session 2, message 22

Step 4 of 12: Review Objective | **Certified 3 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your choices: understand a topic, and identify new areas for research. Nothing you have approved is affected.

DRAFT: your objective, built from your answers (yours to edit)	Came from
to understand	your choice a
which kind of weight is lost … (fat, muscle or body water) within Asian people without diabetes	your focus, as certified
with SGLT2 inhibitors	your topic
and to identify new areas for research	your choice g
where trials are missing	the impact you expect, in your certified case
Left out	clinicians, named in your expected impact; say if you want them in

Added by me: nothing.

Next: once you certify, I will start Feasibility: count the records for your objective, with no review filter, and list ongoing trials.

```
CERTIFY Review Objective | Uday (summarized): objectives — understand a topic; identify new areas
for research; wording — to understand which kind of weight is lost with SGLT2 inhibitors (fat,
muscle or body water) within Asian people without diabetes, and to identify new areas for research:
where trials are missing | reason: ___ | Uday, solo
```

Your turn, Uday: your research question will be built from this wording. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

UDAY (SIMULATED USER) session 2, message 23

Certify

STIR-L4 ASSISTANT session 2, message 24

Step 4 of 12: Review Objective | **Certified 3 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certify; it needs your reason.

Your turn, Uday: What is your reason, in a few words?

C **A bare "Certify" gets one question.** Only the reason is asked for.

UDAY (SIMULATED USER) session 2, message 25

matches my focus

STIR-L4 ASSISTANT session 2, message 26

Step 5 of 12: Feasibility | **Certified 4 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Certified your Review Objective; searched for Feasibility. Nothing you have approved is affected.

```
CERTIFY Review Objective | Uday (summarized): objectives — understand a topic; identify new areas
for research; wording — to understand which kind of weight is lost with SGLT2 inhibitors (fat,
muscle or body water) within Asian people without diabetes, and to identify new areas for research:
where trials are missing | reason: matches my focus | Uday, solo
```

FACT: what the Feasibility searches found (no review filter)	
PubMed: your objective as certified	5 records [1]; PubMed tags 4 as reviews and 1 as a case report, none as a trial [10]
PubMed: the same, any region	154 records [2]; 51 with trial words or tags [4]; 13 of the 51 tagged as randomized controlled trials, none of the 13 naming an Asian population in title or abstract [10]
ClinicalTrials.gov: ongoing studies	473 match the drug words [7]; 361 of them list one of the drugs as an intervention, 38 of the 361 name fat, muscle or body water in their outcomes, and 8 of the 38 have a site in Asia [8]
Still open for your focus	whether any trial reports fat, muscle and water separately for Asian people without diabetes; that needs full texts and better search words

FACT: checks on the search words	
Without diabetes	searches 1–3 miss the trial with a Japanese affiliation from my soundness check: its abstract says “patients without type 2 diabetes mellitus were enrolled” [12], yet that phrase finds it 0 times [6]
Asian people	10 of the 51 trial records list a first affiliation in Asia but name no Asian population [10]
The drugs	registrations in South Korea and China name enavogliflozin and ganagliflozin, which your drug words lack [8]; in PubMed this misses 1 record, and no record names ganagliflozin [5]
So	the counts above may be too low; by how much is unknown

How to read the strings	
`[pt]`	publication type
`[pmid]`	a PubMed identifier
`NOT`	excludes records that match
`query.intr`	the register's intervention field

Search 1: Your objective as certified: fat, muscle or body water, Asian people without di… (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab] OR "Body Water"[Mesh] OR "body water"[tiab] OR
"total body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"[tiab] OR "extracellular
fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR "water loss"[tiab] OR
"fluid loss"[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR asian*[tiab] OR asia[tiab] OR
chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*[tiab] OR korea[tiab] OR
indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab] OR malaysia*[tiab] OR
```

```
vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR bangladesh*[tiab]) AND
("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without diabetes"[tiab] OR
"without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab] OR
normoglycemic[tiab] OR normoglycaemic[tiab])
```

Records: **5** [1] (records, not studies).

Search 2: Broader: the same, any region (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab] OR "Body Water"[Mesh] OR "body water"[tiab] OR
"total body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"[tiab] OR "extracellular
fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR "water loss"[tiab] OR
"fluid loss"[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab])
```

Records: **154** [2] (records, not studies).

Search 3: Trials: the broader search, trial records only (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab] OR "Body Water"[Mesh] OR "body water"[tiab] OR
"total body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"[tiab] OR "extracellular
fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR "water loss"[tiab] OR
"fluid loss"[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab])) AND ("Randomized Controlled Trial"[pt] OR "Controlled
Clinical Trial"[pt] OR "Clinical Trial"[pt] OR randomized[tiab] OR randomised[tiab] OR randomly[tiab] OR
placebo[tiab] OR trial[ti])
```

Block	Covers	From	How it searches	Records
1	trials (my suggestion)	my suggestion (where trials are missing)	title/abstract words, publication filter	2,016,460 [3]
All	combined	—	blocks joined as shown	**51** [4]

Search 4: Drug names in Asian registrations that your drug block misses (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
(enavogliflozin[tiab] OR ganagliflozin[tiab]) NOT ("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR
SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter
2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR
dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR
luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR
bexagliflozin[tiab])
```

Records: **1** [5] (records, not studies).

Search 5: Known-study check: the no-diabetes phrase on the Japanese trial (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
"without type 2 diabetes"[tiab] AND 33879970[pmid]
```

Records: **0** [6] (records, not studies).

Search 6: Ongoing studies of these drugs	ClinicalTrials.gov; data as of 2026-10-23; filter: recruiting, not yet recruiting, enrolling by invitation, or active but not recruiting

```
query.intr: SGLT2 OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin
```

Block	Covers	From	How it searches	Studies
1	SGLT2 inhibitors	topic	words in the intervention field	**473** [7]

Search	Left out on purpose	Why
1	nothing	all parts of your objective
2–3	Asian populations	to see trials from any region; place words miss some trials done in Asia (checks above)
4–5	all but the words tested	to test one set of words each
6	condition, population, outcome	registrations word these freely, so I read each record by code (rule below)
1–3	the phrase "absence of diabetes"	PubMed again could not find it

Not searched yet	Who can do it
Embase	someone with access
Other trial registers, through the WHO portal (ICTRP, the World Health Organization's search across trial registers)	you, Uday, with a search I write
Full texts of the 13 trials, to see whether they report fat, muscle and water separately	me, for open-access copies
OpenAlex for these counts	me, if you want

# (FACT, from [7], [8]: ongoing, site in Asia, focus in outcomes)	Study (registered title, cut)	Country	Status	Focus words in outcomes	First eligibility line naming diabetes (quoted)	NCT
1	Empagliflozin vs linagliptin… in renal transplant recipients with diabetes mellitus	India	recruiting	lean mass; total fat	inclusion: “…with the diagnosis of diabetes mellitus (pre-transplantation type 2 diabetes or post-transplantation…”	NCT06095492
2	Enavogliflozin for the management of patients with amyloid cardiomyopathy	South Korea	recruiting	body fat; extracellular water; total body water	exclusion: “Type 1 diabetes mellitus or insulin-dependent diabetes”	NCT07240844
3	Henagliflozin's impact on prediabetes remission	China	not yet recruiting	body fat	exclusion: “Previously diagnosed with diabetes ;”	NCT06448130
4	Dapagliflozin delays the loss of renal function in peritoneal dialysis patients	China	recruiting	body water	exclusion: “Patients with type 1 diabetes mellitus.”	NCT06398977
5	SGLT-2i combined with probiotic preparations in elderly patients with HFpEF (heart failure with preserved ejection fraction)	China	not yet recruiting	adiposity; body fat; sarcopenia	not named in the eligibility text	NCT07687212

# (FACT, from [7], [8]: ongoing, site in Asia, focus in outcomes)	Study (registered title, cut)	Country	Status	Focus words in outcomes	First eligibility line naming diabetes (quoted)	NCT
6	Korean study… of OAD (oral antidiabetic drug) triple therapy in type 2 diabetes	South Korea	recruiting	body composition; body fat	inclusion: “Patients with type 2 diabetes who are 19 years of age or…”	NCT06838286
7	Effects of enavogliflozin on coronary microvascular and cardiac function in obesity	South Korea	recruiting	body composition; sarcopenia; skeletal muscle; visceral fat	inclusion: “Diabetes : Hemoglobin A1c ≥ 6.5%, or Fasting…”	NCT06782139
8	Efficacy and safety of canagliflozin in the treatment of type 2 diabetes with mild cognitive impairment	China	recruiting	body composition; body fat; skeletal muscle; visceral fat	exclusion: “duration of diabetes is less than 6 years;”	NCT07711171

DRAFT: rule for this table (my terms, yours to edit)	
Tests one of these drugs	a name with "gliflozin" or "SGLT" among the interventions
Names your focus	your focus words in the outcomes, in their own sense (fat around the heart or the coronary arteries does not count); words only, outcomes not checked
Site in Asia	a site in a country your population words name
Diabetes	quoted only; whether a study lets in people without diabetes is yours to judge

Steps	1 done · 2 done · 3 done · 4 done · 5 now · 6–12 open

Sources: [1] L000192 PubMed, run 2026-10-25; [2] L000193 PubMed, run 2026-10-25; [3] L000194 PubMed, run 2026-10-25; [4] L000195 PubMed, run 2026-10-25; [5] L000199 PubMed, run 2026-10-25; [6] L000200 PubMed, run 2026-10-25; [7] L000198 ClinicalTrials.gov, run 2026-10-25 (data as of 2026-10-23); [8] L000205 computation from [7], run 2026-10-25; [9] L000196 PubMed records, run 2026-10-25; [10] L000202 computation from [1], [2], [4], [9], run 2026-10-25; [11] L000129 PubMed records, run 2026-10-25; [12] L000203 computation from [2], [4], [6], [11], run 2026-10-25

Next: I will draft your feasibility summary, with risks.

Your turn, Uday: the summary needs your resources.

1. When is your protocol due?
2. Which other databases can you search?
3. Will you screen alone, or with someone?

For example: "1 March 2027; 2 PubMed only; 3 alone"

UDAY (SIMULATED USER) session 2, message 27

solo. about 5 hours a week. 3 months. pubmed and openalex only, no embase. risk: very few trials

STIR-L4 ASSISTANT session 2, message 28

Step 5 of 12: Feasibility | **Certified 4 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your resources and the risk you named, and drafted your feasibility summary. This may change what you approved if feasibility falls short: limits can reshape your Review Objective, and a severe shortfall can send you back to your Team's Purpose and, through it, your motivations and Need for a Review. Nothing changes unless you agree.

JUDGMENT (Uday, summarized)	
People	you alone; you will screen alone
Time	about 5 hours a week; protocol due in about 3 months
Databases	PubMed and OpenAlex; no Embase
Risk you named	very few trials

DRAFT: your feasibility summary (yours to edit)	**What bears on it**
Requirements clarity (is what the review needs clear?)	your objective is certified; not yet settled: the comparison (Step 6), and what counts as muscle, since lean mass, as studies measure it, is not muscle [6]
Literature availability (do enough studies exist?)	your objective: 5 records [1], none tagged as a trial [2]. Without the Asian limit: 13 records tagged as randomized trials; in title or abstract, 6 name body fat, 2 muscle or lean mass and 2 body water; none names an Asian population; 1 lists a first affiliation in Asia [2]. Ongoing: 8 studies with a site in Asia name your focus in their outcomes [3]
Resource availability (time, people, access)	about 66 hours before the protocol is due (5 hours a week for about 13 weeks; my arithmetic from your answers); one screener; PubMed and OpenAlex
FMEA (failure mode and effects analysis: what could go wrong, and its effect)	the risks below

DRAFT: risks (yours to edit)	**Raised by**	**What bears on it**	**Effect if it happens**	**What would catch or reduce it (my suggestion)**
1. Very few trials	you	as above [2]	the review may not show which weight is lost in your population; it would still show where trials are missing, as your objective asks	count the trials after the full search (Step 8)
2. The search words miss studies	me, from my checks	a trial in people without type 2 diabetes was missed by your searches [4]; 10 of 51 trial records list a first affiliation in Asia but name no Asian population [2]; drug names used in Korea and China are missing from the drug words [3]	studies missed; counts too low	known-study checks and added words (Step 8)
3. Muscle cannot be told from water	me, from my soundness check	lean-mass estimates do not directly measure muscle [6]	the review may report lean mass, not muscle	record how each study measured fat, muscle and water (extraction form, Step 7)
4. One screener	you	—	errors in including or excluding go unchecked	a second person checks a sample
5. Sources not searched	you	no Embase; other trial registers not searched	studies found only there are missed; how many is unknown	state it as a limit in the protocol

Option	**Your decision would be**	**What bears on it**	**It would change**
a	go: plan the review as certified	5 records for your objective [1], none tagged as a trial [2]; about 66 hours	nothing you approved

Option	Your decision would be	What bears on it	It would change
b	narrow: a smaller question within your focus; you say how	5 records already [1]	your objective, if you agree to reopen it
c	broaden: a wider question; you say how	without the Asian limit, 154 records [5], 13 tagged as randomized trials [2]	your objective; your purpose too, if the focus changes
d	defer: wait for ongoing studies	8 ongoing studies with a site in Asia name your focus in their outcomes [3]; whether any admits people without diabetes is yours to judge	your objective, if you agree to reopen it
e	other	say what	—

Sources: [1] L000192 PubMed, run 2026-10-25; [2] L000202 computation from PubMed records, run 2026-10-25; [3] L000205 computation from ClinicalTrials.gov, run 2026-10-25; [4] L000203 computation from PubMed records, run 2026-10-25; [5] L000193 PubMed, run 2026-10-25; [6] L000129 PubMed records, run 2026-10-25

Next: once you choose, I will show your feasibility decision as one line for you to certify.

Your turn, Uday: this decides whether planning goes on as certified or your objective changes first.

1. Which decision: a go, b narrow, c broaden, d defer, or e other?
2. Which kinds of feasibility does it rest on, one or more: a requirements clarity, b literature availability, c resource availability, d risks; you may also edit any risk.

For example: "1 a; 2 b, c, d"
Or: "1 c: <how>, because ..."

UDAY (SIMULATED USER) session 2, message 29

broaden. all adults without diabetes, any country; asian people as a subgroup

STIR-L4 ASSISTANT session 2, message 30

Step 5 of 12: Feasibility | **Certified 4 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your decision: broaden. It widens your focus beyond Asian people, and beyond your topic, so it may change your Team's Purpose, Need for a Review and Review Objective, which name Asian people. Your motivations name no population, so they still stand. Nothing changes unless you agree.

JUDGMENT (Uday, summarized)	
Your decision	broaden: all adults without diabetes, any country; Asian people as a subgroup
Focus, certified	which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes
Focus, it would be	which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, any country; Asian people as a subgroup
Added by me	nothing

Earlier answers that name Asian people (summarized)	What bears on it with the wider focus (FACT)
a. Topic: the role of SGLT2 inhibitors in weight management in Asian populations without diabetes	—
b. Team's Purpose (certified): your focus as certified; research gaps; finding the right question	—

Earlier answers that name Asian people (summarized)	What bears on it with the wider focus (FACT)
c. Need for a Review (certified): a new review, as none examined answers your focus in Asian people without diabetes; used off label for weight in Asia; your reason: no review answers fat vs muscle in this population	the two nearest reviews shown earlier (2021), of people with overweight or obesity and without diabetes, now fall inside your population: both report fat mass; neither names muscle, lean mass or body water [1], [2]. Without diabetes, the records also name heart failure and kidney disease as uses [3]
d. Review Objective (certified): which kind of weight is lost … within Asian people without diabetes; where trials are missing	any region: 154 records [4]; of the 13 tagged as randomized trials, none names an Asian population in title or abstract [5]

Sources: [1] L000084 PubMed records, session 1, run 2026-10-25; [2] L000166 computation from [1], run 2026-10-25; [3] L000129 PubMed records, run 2026-10-25; [4] L000193 PubMed, run 2026-10-25; [5] L000202 computation from PubMed records, run 2026-10-25

Next: I will show your feasibility decision as one line to certify, then redraft the answers that no longer hold, asking whether "all adults" includes people taking these drugs for heart failure or kidney disease.

Your turn, Uday: I need both answers before your feasibility decision can be certified.

1. Which kinds of feasibility does broadening rest on, one or more: a requirements clarity, b literature availability, c resource availability, d risks?
2. Which of the answers a to d above still hold as written, if any? I will redraft the others with your wider focus, for you to certify.

For example: "1 b, c; 2 none"
Or: "1 b, d; 2 c, because ..."

F **Feasibility says broaden, and the flows say what it reopens.** Widening to all adults without diabetes changes three certified decisions that name Asian people. The assistant lists each with the evidence and asks before reopening anything.

UDAY (SIMULATED USER) session 2, message 31

1 b, d; 2 none. yes reopen

STIR-L4 ASSISTANT session 2, message 32

Step 5 of 12: Feasibility | **Certified 4 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: I've recorded your answers, including that you agree to reopen your earlier decisions. Broadening is what reopens your Team's Purpose, Need for a Review and Review Objective. A feasibility shortfall can make you revisit your purpose and reshape your objective. So I will reopen them once you certify broadening. Your motivations still stand.

JUDGMENT (Uday, summarized)	
Broadening rests on	literature availability; FMEA (what could go wrong)
Still hold as written	none of your topic, Team's Purpose, Need for a Review and Review Objective
Reopen them	yes

The line below includes only the risk you named, very few trials. The four risks I drafted stay drafts unless you add them.

Next: once you certify, I will reopen those three decisions. Then I will redraft them and your topic with the wider focus. First I will ask whether "all adults" includes people taking these drugs for heart failure or kidney disease.

```
CERTIFY Feasibility | Uday (summarized): decision — broaden: all adults without diabetes, any country; Asian people as a subgroup; kinds of feasibility — literature availability; FMEA (very few trials) | reason: ___ | Uday, solo
```

Your turn, Uday: your earlier decisions are reopened because of this one. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

UDAY (SIMULATED USER) session 2, message 33

certify, broaden is right

STIR-L4 ASSISTANT session 2, message 34

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Reopened: Team's Purpose, Need for a Review, Review Objective

Done: Certified your Feasibility decision: broaden. As you agreed, I reopened your Team's Purpose, Need for a Review and Review Objective. Your motivations name no population, so they still stand; they and this decision stay provisional until those three are certified again.

```
CERTIFY Feasibility | Uday (summarized): decision — broaden: all adults without diabetes, any country; Asian people as a subgroup; kinds of feasibility — literature availability; FMEA (very few trials) | reason: broaden is right | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	reopened	Uday
3. Need for a Review	whether a new review or an update is needed	reopened	—
4. Review Objective	what the review should achieve	reopened	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

FACT, from [1], [2]: condition named in the title, of the 13 records tagged as randomized trials (people without diabetes, any region)	Records
overweight or obesity	4 [2]
fatty liver	3 [2]

FACT, from [1], [2]: condition named in the title, of the 13 records tagged as randomized trials (people without diabetes, any region)	Records
heart failure	2 [2]
prediabetes or insulin resistance	2 [2]
kidney disease	1 [2]
polycystic ovary syndrome	1 [2]
none of these	1 [2]

A title can name more than one condition; these are records, not studies.

# (FACT, from [1], [2])	Heart failure and kidney disease trials (title, cut)	Year	Focus words, title or abstract	PMID
1	“Mechanistic Insights of Empagliflozin in Nondiabetic Patients With HFrEF: From the EMPA-TROPISM”…	2021	adiposity; subcutaneous adipose	34325888
2	…“estimated plasma volume, and measured glomerular filtration rate in patients with heart failure (Empire HF”…	2021	fluid volume; plasma volume	33357505
3	…“dapagliflozin on volume status and systemic haemodynamics in patients with chronic kidney disease without diabetes:”…	2022	water loss	35478433
Rule (DRAFT, my terms, yours to edit)	condition words in the title only; “risk of heart failure” not counted; abstracts not read for this; HFrEF is a form of heart failure			

DRAFT (my reading, yours to challenge): "all adults without diabetes" takes in people treated for quite different conditions. In heart failure and kidney disease, titles #2 and #3 name fluid volume, and a heart-failure review shown earlier names falls in “plasma volume, body weight” [3]. So how much of the weight lost is water may differ by condition.

Option	Who “all adults without diabetes” takes in	What bears on it (FACT)
a	any adult without diabetes, whatever the drug is given for (weight, heart failure, kidney disease, fatty liver or other)	all 13 trial records above [2]
b	only adults treated for weight management (your topic's words)	4 of the 13 name overweight or obesity in their titles [2]
c	other: say who	—

Sources: [1] L000196 PubMed records, run 2026-10-25; [2] L000237 computation from [1] and the earlier trial tags, run 2026-10-25; [3] L000152 OpenAlex, run 2026-10-25

Next: I will redraft your topic and Team's Purpose with the wider focus for you to certify, then your case and objective.

Your turn, Uday: your purpose will name who the review covers, and every later search follows it. Which applies: a, b or c?
For example: "a, because ..."
Or: "c: <who>"

F **Reopened with the user's agreement: Certified drops from 4 to 2.** Feasibility is certified; the Team's Purpose, Need for a Review and Review Objective are reopened and marked in the status line. They are certified again at messages 38, 48 and 52, and the count climbs back to 5.

UDAY (SIMULATED USER) session 2, message 35

a

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 2 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Reopened: Team's Purpose, Need for a Review, Review Objective

Done: I recorded your answer in your focus and redrafted your topic and Team's Purpose to match. Your motivations name no population, so they still stand.

JUDGMENT (Uday, summarized)	
Who is in	any adult without diabetes, whatever the drug is given for (your choice a)
Focus, before	which kind of weight is lost: fat, muscle or body water, within Asian people without diabetes
Focus, now	which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

DRAFT (my reading, yours to challenge): your topic says "weight management", but your focus also takes in people given these drugs for heart failure, kidney disease or fatty liver. Your case's reason, off-label use for weight in Asia, has the same gap; it returns with the case.

Option	Your topic would be	Words from
a	the role of SGLT2 inhibitors in weight management in all adults without diabetes, any country; Asian people as a subgroup	your topic ("weight management") and your broadening
b	the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup	your focus ("weight … lost"), your broadening and your choice a
c	other: your wording	—

DRAFT: your Team's Purpose, redrafted (yours to edit)	**Came from**
focus — which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup	your focus, now
kinds of purpose — research gaps; finding the right question	your earlier choice
the review — a systematic review on this topic; with the results — decide whether a full review is worth it; maybe publish	your earlier answers

Added by me: nothing.

Steps: 1 done · 2 now · 3 started early · 5 done · 4–12 open

Next: once you certify, I will search again for reviews of your wider focus and recheck whether your question holds up, then redraft your case.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup; kinds of purpose — research gaps; finding the right question; the review — a systematic review on this topic; with the results — decide whether a full review is worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: every later search follows who your topic and purpose cover.

1. Your topic: a, b or c?
2. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.

For example: "1 b; certify, because ..."
Or: "1 a; change <what> to <what>"

UDAY (SIMULATED USER) session 2, message 37

1 b; certify, broadened for feasibility

STIR-L4 ASSISTANT session 2, message 38

Step 3 of 12: Need for a Review | **Certified 3 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within… | **Reopened: Need for a Review, Review Objective**

Done: Recorded your topic (b); certified your Team's Purpose; searched for reviews of your new topic; rechecked your question. Your interest and motivations name no population, so they still stand.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup; kinds of purpose — research gaps; finding the right question; the review — a systematic review on this topic; with the results — decide whether a full review is worth it; maybe publish | reason: broadened for feasibility | Uday, solo
```

FACT: existing reviews of your focus, now "which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup"	
PubMed, your new topic, systematic-review filter	19 reviews [2]; all read
OpenAlex, your new topic, reviews	1,279 [4]; I read the first 100
May bear on your focus (rule below)	9 of the 241 review records read, from these and the earlier review searches [5]; 3 state a search end, the latest August 2025 [5]
Parts of your focus they name	body fat 7; muscle or lean mass 0; body water 1, as a possible mechanism; an Asian population 0 [5]
Reviews of any kind, no review filter	of the 153 records read from your Feasibility search (any region) [9], PubMed tags 51 as reviews; 5 name muscle or lean mass, none of them tagged a systematic review [8]
Still open for your focus	no systematic review found names muscle or lean mass in people without diabetes; planned reviews unknown

Search 1: Your topic, now: SGLT2 inhibitors, weight lost, without diabetes (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Weight"[Mesh] OR weight[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab] OR normoglycemic[tiab] OR normoglycaemic[tiab])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	weight lost (your topic, now)	topic	subject headings, title/abstract words	1,468,706 [1]
All	combined	—	blocks joined as shown	**19** [2]

Block counts are before the filter; the combined count is after it.

Search 2: Your topic, now: SGLT2 inhibitors, weight lost, without diabetes (OpenAlex; published any to 2026-10-25; filter: type: review)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin) AND (weight) AND ("non-diabetic" OR nondiabetic OR "without diabetes" OR "without type 2 diabetes" OR "absence of diabetes" OR normoglycemic)
```

Block	Covers	From	How it searches	Records
1	weight lost (your topic, now)	topic	words in title, abstract or text	14,226,412 [3]
All	combined	—	blocks joined as shown	**1,279** [4]

Block counts are before the filter; the combined count is after it.

The drug and without-diabetes blocks are as shown earlier this session.

Search	Left out on purpose	Why
1–2	adults	age is set with the eligibility criteria (Step 6); age words would drop reviews that do not state it
1–2	Asian people	a subgroup, not a limit; the rule marks it
—	new focus searches	your focus words did not change: 73 PubMed reviews [6], 2,343 in OpenAlex [7]

Not searched yet	Who can do it
PROSPERO, OSF Registries, Cochrane, Epistemonikos	you, Uday (recorded as not searched today)
OpenAlex past the first 100 (1,179 more for search 2) [5]	me, with a free access code

FACT, from [5]: Reviews that may bear on your focus (abstracts only)

#	Title	Year	Search end, as the abstract states it	Focus words	Population words	PMID
1	Selective SGLT2 inhibitors in MASLD/MASH: an outcome-specific systematic review…	2026	not stated; unknown	visceral adiposity	non-diabetic	42517896
2	Effectiveness of sodium-glucose cotransporter 2 inhibitors in patients with met…	2026	not stated; unknown	visceral adipose	without diabetes	42215421
3	Emerging Effects of Glucagon-Like Peptide-1 Receptor Agonists and Sodium-Glucos…	2026	August 2025	body composition	without diabetes	42011503
4	SGLT2 Inhibitors Beyond Glycemic Control: Emerging Pharmacological and Cardiova…	2026	not stated; unknown	fluid volume	do not have diabetes	none; DOI 10.5281/zenodo.22978843
5	Effects of sodium-glucose cotransporter 2 inhibitors on body weight and metabol…	2026	not stated; unknown	adipose tissue; visceral adipose	without t2dm; without type 2 diabetes	42339604
6	Comparison of sodium-glucose cotransporter-2 inhibitors and thiazolidinediones …	2023	not stated; unknown	visceral fat	without type 2 diabetes	36931466
7	GLP-1 RAs and SGLT-2 Inhibitors for Insulin Resistance in Nonalcoholic Fatty Li…	2022	March 2021	visceral fat	without t2dm	35909522
8	Effect of Sodium-Glucose Cotransporter 2 Inhibitors on Weight Reduction in Over…	2021	May 2021	fat mass	non-diabetic; without diabetes	34897070

#	Title	Year	Search end, as the abstract states it	Focus words	Population words	PMID
9	Sodium-glucose co-transporter 2 inhibitors on weight change and cardiometabolic…	2021	not stated; unknown	fat mass	without diabetes	34542222

DRAFT: what these reviews show (yours to challenge)	
Reasons for treatment, by title	fatty liver 4; overweight or obesity 3; prediabetes 1; none named 2; heart failure or kidney disease 0 [5]
Muscle	none of the 9 names it; of the 5 reviews of any kind that do, PubMed tags none as systematic; one is the 2025 skeletal-muscle review shown earlier [8]
Rule (my terms, yours to edit)	a record may bear if it names fat, muscle or body water in their own sense and people without diabetes; Asian words mark the subgroup

DRAFT: does your wider question hold up? (yours to challenge)	What the records show	Still uncertain
Clinically	Without diabetes, these drugs are recommended in heart failure (shown earlier). In a review of people without diabetes, those with heart failure had a 20% lower risk of cardiovascular death or heart-failure admission than those not treated (PMID 33625242) [10]. In heart failure, losing water may be part of the benefit: “Fluid volume changes might be an important mechanism underlying the beneficial clinical effects” (PMID 33357505) [9]. Muscle loss is a stated worry, but the evidence is mixed: muscle mass fell in 15 studies and did not change in 6, and physical performance did not change (PMID 40707349) [9].	How many people without diabetes take these drugs for weight; most muscle studies were in people with diabetes (shown earlier).
Scientifically	Water: in a heart-failure trial (88% without diabetes), estimated plasma volume fell 7.3% more than with placebo (PMID 33357505) [9]; in kidney disease without diabetes, dapagliflozin lowered blood pressure without raising salt and water output in urine, in a study of six patients (PMID 35478433) [9]. Fat: fat under the skin fell in heart failure without diabetes (PMID 34325888) [9]. The total to split is small: body weight 1.21 kg lower than without the drug, in the review above (PMID 33625242) [10].	Whether body-composition scans can split a loss this small; I have not searched for how precise they are. I read abstracts only.
Logically	1) Of the 13 records tagged as randomized trials (searched as people without diabetes, any region), the 4 in overweight or obesity all name fat and none names water; of the 3 in heart failure or kidney disease, 2 name water [8]. One pooled answer could mix water lost as part of treatment with water lost as a side effect. 2) 3 of the 4 obesity trials give the SGLT2 inhibitor together with another drug, exenatide or topiramate [8], [9]; in one, the loss was mostly fat with no lean change (PMID 27550386) [9], but that cannot be credited to the SGLT2 inhibitor alone. The comparison is settled with the research question. 3) None of the 13 names fat, muscle and water together [8], or an Asian population [11].	Full texts may report parts that abstracts leave out.
Plainly	Your question is reasonable and can be measured. But it assumes that weight lost means the same thing whatever the drug is given for. The records suggest it does not: in heart failure, water loss may be part of the benefit; when weight is the aim, it is a side effect. And 3 of the 4 weight trials test the drug with another drug.	You decide whether and how to reframe.

Sources: [1] L000254 PubMed, run 2026-10-25; [2] L000255 PubMed, run 2026-10-25; [3] L000256 OpenAlex, run 2026-10-25; [4] L000257 OpenAlex, run 2026-10-25; [5] L000259 computation from [2], [4] and the earlier review searches, run 2026-10-25; [6] L000149 PubMed, run 2026-10-25; [7] L000152 OpenAlex, run 2026-10-25; [8] L000262 computation from [9], [10], run 2026-10-25; [9] L000196 PubMed records, run 2026-10-25; [10] L000258 PubMed records, run 2026-10-25; [11] L000202 computation from PubMed records, run 2026-10-25

Steps: 1 done · 2 done · 3 now · 5 done · 4–12 open

Next: I will ask why your wider focus matters for patients (your earlier reason was about Asia), then draft the case.

Your turn, Uday: your case rests on your focus. Does this check change it? One or more:

Option	Your focus would be
a	unchanged: which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for
b	unchanged, but answered separately for each reason for treatment (for example heart failure, kidney disease, weight); this goes into the objective, at Step 4 (my suggestion)
c	narrowed to people given the drug for weight (your earlier words, "weight management"); you would certify your purpose again
d	other: say how

For example: "b, because ..."
Or: "a"

S **The soundness check is repeated for the wider focus.** With new searches, before the case is certified again.

UDAY (SIMULATED USER) session 2, message 39

a

STIR-L4 ASSISTANT session 2, message 40

Step 3 of 12: Need for a Review | Certified 3 of 8 | Waiting on you: 1 (Uday)

Focus: which kind of weight is lost: fat, muscle or body water, within… | Reopened: Need for a Review, Review Objective

Done: Recorded your answer: your focus stays as certified. It is one question for all adults without diabetes, whatever the drug is given for. Nothing you have approved is affected.

DRAFT, still open from my check (yours to challenge): one pooled answer would join water lost as a possible part of the benefit, in heart failure [1], [2], with water lost as a side effect when weight is the aim. I will list this in your case as an uncertainty; you decide what it means.

Sources: [1] L000196 PubMed records, run 2026-10-25; [2] L000262 computation from [1], run 2026-10-25

Next: I will ask what impact you expect with the wider focus, then draft your case: a new review, an update or no review.

Your turn, Uday: your case needs a reason that fits the wider focus; your earlier one was about off-label use for weight in Asia. Why does it matter for patients or practice which kind of weight people without diabetes lose on these drugs?
For example: "it matters because ..."
Or: "same as before, plus ..."

UDAY (SIMULATED USER) session 2, message 41

same as before, plus it matters for any adult without diabetes on these drugs, not just asians

STIR-L4 ASSISTANT session 2, message 42

Step 3 of 12: Need for a Review | Certified 3 of 8 | Waiting on you: 1 (Uday)

Focus: which kind of weight is lost: fat, muscle or body water, within… | Reopened: Need for a Review, Review Objective

Done: I've recorded why your wider focus matters, summarized in my own words. Nothing you have approved is affected.

JUDGMENT (Uday, summarized; tell me if I got it wrong)	
Why your focus matters	as before: these drugs are used off label for weight in Asia, and losing muscle or water matters for older and lean patients; now also: it matters for any adult without diabetes who takes these drugs, not only Asian people

DRAFT (yours to challenge): your added reason takes in adults who take these drugs for heart failure or kidney disease. The records name both as uses in people without diabetes [1]. We still don't know how many adults without diabetes take them for weight.

Sources: [1] L000129 PubMed records, run 2026-10-25

Next: I will draft your case from your answers and the searches so far: a new review, an update or no review, with the open points listed.

Your turn, Uday: your case also needs the impact you expect. You gave it for the earlier focus (summarized): the review would show clinicians whether the weight lost is fat, and point out where trials are lacking. Does it still hold for the wider focus?
a. yes, as given
b. change it: say how
For example: "a"
Or: "b: it would also ..."

UDAY (SIMULATED USER) session 2, message 43

a

STIR-L4 ASSISTANT session 2, message 44

Step 3 of 12: Need for a Review | **Certified 3 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within… | **Reopened: Need for a Review, Review Objective**

Done: I recorded that the impact you expect still holds for the wider focus, and drafted your case from your answers and the searches so far. Your earlier case covered Asian people only, so I am asking the questions below again. Nothing you have approved is affected.

DRAFT: grounds for your case, wider focus (question 2; yours to edit)	**What bears on it**
a. Previous reviews (what existing reviews cover)	Of the 9 reviews that may bear on your focus, 7 name body fat; none names muscle or lean mass; 1 names body water, only as a possible mechanism [1]. None names heart failure or kidney disease in its title [2]. Of the 51 reviews of any kind in your Feasibility search, the 5 that name muscle or lean mass are not tagged as systematic reviews [3].
b. Knowledge gaps (what is not yet known)	Of the 13 records tagged as randomized trials (people without diabetes, any region), none names fat, muscle and water together [3], and none names an Asian population [4].
c. Clinical justification (why it matters for patients or practice)	Yours (summarized): these drugs are used off label for weight in Asia; losing muscle or water matters for older and lean patients; it matters for any adult without diabetes on these drugs. Records: recommended without diabetes in heart failure [5]; off-label use for obesity is stated, not measured [6].
d. Impact analysis (what the review would change)	Yours (summarized): show clinicians whether the weight lost is fat, and point out where trials are lacking. Lean mass, as studies measure it, is not muscle [5].
e. Other	say what

DRAFT: open points your case would list (yours to edit)	What bears on it
One answer for every reason for treatment	in heart failure, water loss may be part of the benefit [7]; when weight is the aim, it is a side effect (my reading)
The drug's own effect	3 of the 4 obesity trials give it with another drug [3]
Can body scans measure each part of a small loss?	not searched yet
Not searched	PROSPERO, OSF Registries, Cochrane, Epistemonikos; OpenAlex reviews past the first 100; full texts

Question 1: the case	It would say	What bears on it
a. New review	no existing review answers your focus	none of the 9 names muscle or lean mass, or all three parts [1]
b. Update	an existing review answers your focus but needs newer studies	the 3 weight reviews among the 9 (2026, 2021, 2021) name fat, but not muscle, lean mass or body water; 1 states a search end, May 2021; the 2026 one also covers people with type 2 diabetes [2]. The latest stated search end, August 2025, belongs to a review of male sex hormones that names only body composition [2]
c. No review	a review already answers it, or there is too little to review	154 records for your focus, any region [8]; 13 tagged as randomized trials [4]; planned reviews unknown
d. Other	say what	—

Sources: [1] L000259 computation, run 2026-10-25; [2] L000279 computation from [1], run 2026-10-25; [3] L000262 computation from PubMed records, run 2026-10-25; [4] L000202 computation from PubMed records, run 2026-10-25; [5] L000129 PubMed records, run 2026-10-25; [6] L000132 PubMed records, run 2026-10-25; [7] L000196 PubMed records, run 2026-10-25; [8] L000193 PubMed, run 2026-10-25

Next: once you choose, I will show your case as one line for you to certify. Then I will redraft your objective.

Your turn, Uday: your case says why a review of the wider focus is worth doing, and your objective builds on it.

1. Which case do you make: a new review, b update, c no review, or d other?
2. Which grounds apply, one or more: a to e above? You may also edit the open points.

For example: "1 a; 2 a, b, c, d"
Or: "1 b, because ..."

UDAY (SIMULATED USER) session 2, message 45

1 a; 2 a, b, c, d

STIR-L4 ASSISTANT session 2, message 46

Step 3 of 12: Need for a Review | **Certified 3 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within… | **Reopened: Need for a Review, Review Objective**

Done: Recorded your choices for the wider focus: a new review, on four grounds: previous reviews, knowledge gaps, clinical justification and impact analysis. Nothing you have approved is affected.

DRAFT: your case as one line, from your choices and my last reply; the uncertainties and sources not searched are my open points, which you did not edit. "Examined so far" means the 241 review records whose titles and abstracts I read; of the 9 that may bear on your focus, none names muscle or lean mass, or all three parts [1]. If the registers later show a planned review, I will bring this case back.

Sources: [1] L000259 computation, run 2026-10-25

Next: once you certify, I will show the chain so far and ask which objectives apply to your wider focus.

```
CERTIFY Need for a Review | Uday (summarized): case — new review: no review examined so far answers
the focus, which kind of weight is lost: fat, muscle or body water, within all adults without
diabetes, whatever the drug is given for, any country; Asian people as a subgroup; grounds —
previous reviews; knowledge gaps; clinical justification (used off label for weight in Asia; muscle
or water loss matters for older and lean patients; it matters for any adult without diabetes on
these drugs); impact analysis (tell clinicians whether the weight lost is fat; show where trials
are missing); uncertainties — whether one answer fits every reason for treatment; the drug's own
effect, when trials give it with another drug; whether body scans can measure each part of a small
loss; not searched — PROSPERO, OSF Registries, Cochrane, Epistemonikos; OpenAlex reviews past the
first 100; full texts | reason: ___ | Uday, solo
```

Your turn, Uday: your objective builds on this case. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

UDAY (SIMULATED USER) session 2, message 47

certify, broadened for feasibility

STIR-L4 ASSISTANT session 2, message 48

Step 4 of 12: Review Objective | **Certified 4 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within… | **Reopened: Review Objective**

Done: Certified your Need for a Review for the wider focus: a new review. Nothing you have approved is affected. It stays provisional, like your other approved decisions, until your Review Objective is certified again.

```
CERTIFY Need for a Review | Uday (summarized): case — new review: no review examined so far answers
the focus, which kind of weight is lost: fat, muscle or body water, within all adults without
diabetes, whatever the drug is given for, any country; Asian people as a subgroup; grounds —
previous reviews; knowledge gaps; clinical justification (used off label for weight in Asia; muscle
or water loss matters for older and lean patients; it matters for any adult without diabetes on
these drugs); impact analysis (tell clinicians whether the weight lost is fat; show where trials
are missing); uncertainties — whether one answer fits every reason for treatment; the drug's own
effect, when trials give it with another drug; whether body scans can measure each part of a small
loss; not searched — PROSPERO, OSF Registries, Cochrane, Epistemonikos; OpenAlex reviews past the
first 100; full texts | reason: broadened for feasibility | Uday, solo
```

JUDGMENT (Uday, summarized): the chain so far	
Topic	the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup
Focus and purpose (certified)	which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup; for research gaps and finding the right question; to decide whether a full review is worth it, maybe publish
Need for a Review (certified)	a new review; grounds: previous reviews, knowledge gaps, clinical justification, impact analysis; three uncertainties listed; registers of planned reviews not searched

Step	**Settles**	**Status**	**Waiting on**
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—

Step	Settles	Status	Waiting on
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	reopened	Uday
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

Your objective says what the review should achieve for the wider focus. More than one may apply, and none is pre-selected. For the Asian focus you chose a and g; they carry over only if you say so.

Option	Review Objective	What bears on it, for your wider focus (FACT, abstracts only)
a	understand a topic	a 2026 chapter, “Clinical Physiology of SGLT2 Inhibitors in Non-diabetic and Diabetic Individuals” [1]
b	compare interventions	the two nearest reviews compared these drugs with placebo [1]; of the 13 records tagged as randomized trials (row c), 3 of the 4 in obesity give the drug with another drug [2]
c	summarize available evidence	your focus, any region: 154 records [3], 13 of them tagged as randomized trials [4]
d	measure evidentiary trust (how far the evidence can be trusted)	not checked yet
e	evaluate quality of evidence	lean-mass estimates “do not directly quantify contractile skeletal muscle” [5]
f	question a clinical decision	a 2025 review: these drugs “are now also recommended for nondiabetic patients with heart failure (HF)” [1]; off-label use for obesity is stated [6], how often is unknown
g	identify new areas for research	none of the 13 trial records names fat, muscle and water together [7], or an Asian population [4]; the 2025 review: “There is an urgent need for research, especially in nondiabetic, non-Japanese and HF-patients” [1]
h	update a previous SRMA (systematic review and meta-analysis)	your case, just certified, is a new review; of the 3 weight reviews that may bear on your focus, 1 states a search end, May 2021 [7]
i	other	say what

Sources: [1] L000196 PubMed records, run 2026-10-25; [2] L000262 computation from PubMed records, run 2026-10-25; [3] L000193 PubMed, run 2026-10-25; [4] L000202 computation from PubMed records, run 2026-10-25; [5] L000129 PubMed records, run 2026-10-25; [6] L000132 PubMed records, run 2026-10-25; [7] L000279 computation from PubMed records, run 2026-10-25

Next: I will draft the wording of your objective from your choice and the chain above, showing where each phrase came from, for you to edit and certify.

Your turn, Uday: your research question will be built from this objective, so it must fit the wider focus. Which objectives apply: one or more of a to i?

For example: "<letters>, because ..."
Or: "i: <your own words>"

UDAY (SIMULATED USER) session 2, message 49

a, g

STIR-L4 ASSISTANT session 2, message 50

Step 4 of 12: Review Objective | **Certified 4 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within… | **Reopened: Review Objective**

Done: Recorded your choices for the wider focus: understand a topic, and identify new areas for research, the same two as before. Your motivations, Team's Purpose, Need for a Review and Feasibility decision already fit this wider focus, so they still stand.

DRAFT: your objective, redrafted for the wider focus (yours to edit)	Came from
within all adults without diabetes, whatever the drug is given for, any country (Asian people as a subgroup)	your focus, as certified; it replaces "within Asian people without diabetes"
the rest, unchanged	your choices a and g, your topic, and your expected impact ("where trials are missing")
Left out, as in the wording you certified before	clinicians, named in your expected impact

Added by me: nothing.

Next: once you certify, I will show the plan map and start your Research Question: which of your words become each part of it.

```
CERTIFY Review Objective | Uday (summarized): objectives — understand a topic; identify new areas
for research; wording — to understand which kind of weight is lost with SGLT2 inhibitors (fat,
muscle or body water) within all adults without diabetes, whatever the drug is given for, any
country (Asian people as a subgroup), and to identify new areas for research: where trials are
missing | reason: ___ | Uday, solo
```

Your turn, Uday: your research question will be built from this wording. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

UDAY (SIMULATED USER) session 2, message 51

certify, broadened for feasibility

STIR-L4 ASSISTANT session 2, message 52

Step 6 of 12: Research Question | **Certified 5 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: I certified your Review Objective. All three reopened decisions are certified again, so none of your decisions is provisional now. Your other decisions already cover this wider focus, so they still stand. I also looked up PubMed's subject headings for each part of your question.

```
CERTIFY Review Objective | Uday (summarized): objectives — understand a topic; identify new areas
for research; wording — to understand which kind of weight is lost with SGLT2 inhibitors (fat,
muscle or body water) within all adults without diabetes, whatever the drug is given for, any
```

```
country (Asian people as a subgroup), and to identify new areas for research: where trials are
missing | reason: broadened for feasibility | Uday, solo
```

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 now · 7–12 open

DRAFT: your words, as the parts of your question (yours to edit)	Your words	Came from
Population	all adults without diabetes, whatever the drug is given for, any country	your focus
Subgroup	Asian people	your focus
Intervention	SGLT2 inhibitors	your topic
Comparison	not given	—
Outcomes	which kind of weight is lost: fat, muscle or body water	your focus
Study types	not given; your objective names "where trials are missing"	your objective

Added by me: nothing.

FACT, from [1], [2]: PubMed's subject headings for each part	Looked up as	Heading	What PubMed's record says (cut)
Intervention	SGLT2 inhibitors	Sodium-Glucose Transporter 2 Inhibitors	“…are used in the treatment of TYPE 2 DIABETES MELLITUS”
Population	adults	Adult	“Adults are of 19 through 44 years of age”; narrower headings returned: Young Adult, Frail Elderly
Without diabetes (to exclude)	diabetes mellitus	Diabetes Mellitus	narrower headings returned include Type 1, Type 2 and Gestational
Subgroup	Asian people	Asian People	narrower headings returned: East, Southeast, South, West, Central and North Asian, and Middle Eastern People
Weight lost (your topic)	weight loss	Weight Loss	“Decrease in existing BODY WEIGHT.”
Fat	body fat (my wording of "fat")	Adipose Tissue	lists “Body Fat” as another name
Muscle	skeletal muscle (my wording of "muscle")	Muscle, Skeletal	narrower heading returned: Muscle Fibers, Skeletal
Body water	body water	Body Water	“Fluids composed mainly of water found within the body.”
All three (my suggestion)	body composition	Body Composition	“The relative amounts of various components in the body”

Say "show all" for the other headings each lookup returned.

DRAFT (my reading, yours to challenge): headings that differ from your words	Why it matters
Adult is defined as 19 to 44 [1], though Frail Elderly sits under it [2]	the search, at Step 8
Asian People takes in Middle Eastern People [2]	the search, and who your subgroup is

Question	Option	What bears on it (FACT)
1. Comparison	a placebo	11 of the 13 records tagged as randomized trials (without diabetes, any region) name placebo [3]
	b another drug or usual care	the other 2 compare with other drugs: pioglitazone; exenatide or phentermine/topiramate [3]

Question	Option	What bears on it (FACT)
	c either a or b	—
	d also studies with no comparison group (before and after)	not counted
	e other	say what
2. Study types	a randomized trials only	of the 153 records read from your Feasibility search, PubMed tags 13 as randomized trials [3]
	b any trial, randomized or not	51 records with trial words or tags (shown earlier) [4]
	c trials and observational studies	PubMed tags 1 as observational and 4 as case reports [3]
	d other	say what
3. What counts as muscle	a lean or fat-free mass, as studies report it	lean-mass estimates do not directly measure muscle (shown earlier) [5]
	b only measures of muscle itself	in a 2025 review, muscle mass fell in 15 studies and did not change in 6 (shown earlier) [6]
	c a and b, kept apart	—
	d also strength or physical performance	the same review: performance tests did not change [6]
	e other	say what

Sources: [1] L000308, L000310, L000312, L000314, L000316, L000318, L000320, L000322, L000324 MeSH (PubMed's subject index), run 2026-10-25; [2] L000325 computation from [1], run 2026-10-25; [3] L000326 computation from PubMed records, run 2026-10-25; [4] L000202 computation from PubMed records, run 2026-10-25; [5] L000129 PubMed records, run 2026-10-25; [6] L000196 PubMed records, run 2026-10-25

Next: I will ask about criteria your words still leave open (my suggestions: prediabetes, the drug given with another drug, treatment length), then draft your question.

Your turn, Uday: your criteria depend on these parts, and your words do not give them yet.

1. Comparison: a, b, c, d or e?
2. Study types: a, b, c or d?
3. What counts as muscle: a, b, c, d or e?

For example: "1 c; 2 a; 3 c"
Or: "1 a; 2 b; 3 e: <how>"

Part 2. Session 3

Session 3. The research question is certified; the extraction form, property registry and atlas are designed and certified before the search; the search strategy is refined against known studies and certified; Uday screens a pilot of 26 records to test the criteria.

Messages 1 to 4 of each session (the tool test and the calibration quiz, scored 16 of 16) are not shown.

UDAY (SIMULATED USER) session 3, message 5

passed 16/16. continue from the saved plan

STIR-L4 ASSISTANT session 3, message 6

Step 6 of 12: Research Question | **Certified 5 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Read your saved plan and record; every saved search still matches its logged count. Session 2 left no summary, so this comes from the record. Nothing you have approved is affected. Tell me if anything looks wrong.

JUDGMENT (Uday, summarized)

Topic	the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup
Focus	which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	now	Uday
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.
Quick commands, in any wording: "where are we" · "next" (never accepts or certifies) · "why" (how this answer affects your other decisions) · "show <item>" · "skip" (the question stays open) · "back" (reopens only if the team agrees) · "pause" (summary and stop).

Open since last session: three parts of your question; the others come from your focus and topic.

Question	Option	What bears on it
1. Comparison	a placebo	FACT: 11 of the 13 records PubMed tags as randomized controlled trials (people without diabetes, any region) name placebo [1]
	b another drug or usual care	FACT: the other 2 compare it with pioglitazone, or with exenatide or phentermine/topiramate, by their titles [1]
	c either a or b	—
	d also studies with no comparison group (before and after)	not counted
	e other	say what
	bears on every option	FACT: 3 of the 13 give it with a second drug, by their titles [1], for example “dapagliflozin 10 mg once daily plus subcutaneous long-acting exenatide 2 mg once weekly or placebo” [1]. DRAFT: against placebo, such trials cannot show the SGLT2 inhibitor's own share of the loss
2. Study types	a randomized trials only	FACT: of the 153 records fetched from your Feasibility search (154 found), PubMed tags 13 as randomized controlled trials [2]
	b any trial, randomized or not	FACT: 51 records match trial words or tags [1]
	c trials and observational studies	FACT: of the same 153, PubMed tags 1 as observational and 4 as case reports [2]
	d other	say what
3. What counts as muscle	a lean or fat-free mass, as studies report it	—
	b only measures of muscle itself	—
	c a and b, kept apart	FACT: a 2026 meta-analysis in type 2 diabetes did this: “Primary outcomes were skeletal muscle mass (SMM) and lean mass (LM)”; lean mass fell with some drugs, but “No SGLT2 inhibitor significantly reduced SMM compared with placebo” [4]
	d also strength or physical performance	FACT: a 2026 review: falls in lean mass alone are “insufficient to infer clinically meaningful muscle loss without concurrent assessment of muscle strength” [3]; a 2025 review: “Physical-performance-tests remained unchanged under SGLT2i” [3]
	e other	say what
	bears on every option	FACT: the 2026 review: “lean-mass estimates do not directly quantify contractile skeletal muscle” and “may be influenced by alterations in body water” [3]. DRAFT: your focus keeps muscle and body water apart, so lean mass alone may mix the two

Sources: [1] L000195 PubMed, run 2026-10-25; [2] L000193 PubMed, run 2026-10-25; [3] L000129 PubMed, run 2026-10-25; [4] L000084 PubMed, run 2026-10-25

Next: I will draft your question, criteria and subject headings for you to edit and certify.

Your turn, Uday: these three answers complete your question. 1. Comparison: a, b, c, d or e? 2. Study types: a, b, c or d? 3. What counts as muscle: a, b, c, d or e?
For example: "1 a, 2 a, 3 c" or "1 e: say what, 2 c, 3 d".

R **Session 3 resumes at the Research Question.** From the saved files, with the draft question for the user to edit.

UDAY (SIMULATED USER) session 3, message 7

1 e: placebo, no treatment or usual care. 2 d: rcts and controlled

cohorts. 3 d

STIR-L4 ASSISTANT session 3, message 8

Step 6 of 12: Research Question | **Certified 5 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: I recorded your answers, looked up subject headings for the new parts and drafted your question. Nothing you have approved is affected.

DRAFT: your question, with its criteria (yours to edit)

Part	Criterion	Came from	PubMed subject heading
Population	adults without diabetes, whatever the drug is given for, any country	your focus	Adult; Diabetes Mellitus, to leave out [1]
	people with prediabetes count as without diabetes	my suggestion; 3 of the 13 trial records enrol them [2] (two seem to report one trial), e.g. “73.5% of participants had prediabetes” [4]	—
	studies that also enrol people with diabetes	question 2 below	—
Subgroup	Asian people	your focus	Asian People [1]
Intervention	any SGLT2 inhibitor	your topic	Sodium-Glucose Transporter 2 Inhibitors [1]
	given with a second drug	question 1 below	—
	shortest treatment time	question 3 below	—
Comparison	placebo, no treatment or usual care	your answer 1	Placebos; no heading matches usual care or no treatment [3]
Outcomes	which kind of weight is lost: fat, muscle or body water	your focus	Adipose Tissue; Muscle, Skeletal; Body Water; Body Composition (my suggestion) [1]
	muscle: lean or fat-free mass, muscle itself, and also strength or physical performance	your answer 3	none for lean or fat-free mass [5]; Muscle Strength; Physical Functional Performance [3]
	fat, lean mass, muscle itself, body water and strength, each reported apart	my suggestion; lean mass may include body water (shown earlier) [7]	—
	any measuring method, recorded for each study	my suggestion; only 5 of the 153 abstracts name a method beside fat, muscle or body water [2]	—
Study types	randomized controlled trials and controlled cohort studies	your answer 2	Randomized Controlled Trial (a type PubMed gives records); Cohort Studies [3]

Added by me: the rows marked “my suggestion”.

DRAFT: do your three answers fit your focus? (my check, yours to challenge)

Your answer	Why it fits	What to watch
1. Placebo, no treatment or usual care	the difference between groups is then the drug's own effect	it leaves out the 2 trials against other drugs [6]; a second drug given only with the SGLT2 inhibitor blurs this (question 1)
2. Randomized trials and controlled cohorts	your Feasibility decision notes very few trials	in a cohort, who gets the drug is not random, so groups may differ in why they were treated; the methods step will set how this is judged. Few may come: PubMed tags 1 of the 153 records as observational [6]

Your answer	Why it fits	What to watch
3. Muscle, with strength or physical performance	a 2026 review says lean mass alone is not enough without strength (shown earlier) [7]	strength is not a kind of weight; I list it beside the muscle measures, not in place of them

Still open: three criteria.

Question	Option	What bears on it
1. A second drug	a include only when both groups get the same other treatment	FACT: background treatment is common, e.g. “in addition to optimal medical treatment” [4]
	b also the drug given with a second drug against placebo, kept apart	FACT: 3 of the 13 trial records do this [2]; one finds the loss “mostly attributable to adipose tissue reduction without lean tissue change” [4]
	c other	say what
2. People with diabetes in the same study	a leave the study out	—
	b include it if it reports those without diabetes apart	FACT: 3 of the 13 trial records enrol both [2], e.g. “Thirty individuals with T2D and 27 without were randomly assigned” [4]
	c include it if most are without diabetes (say what share)	FACT: “including 105 (88%) without diabetes” [4]
	d other	say what
3. Shortest treatment time	a no minimum	FACT: treatment in the 13 trial records lasted 2 to 52 weeks, read from their abstracts [2]
	b at least a number of weeks (say how many)	FACT: 2 of the 13 lasted under 12 weeks [2]
	c other	say what
	bears on every option	not yet searched: whether the shares of fat, muscle and water lost change with treatment time

The 13 trial records, with the passage read for each, are open beside the chat; say "show all" to see them here.

Sources: [1] L000325 MeSH (PubMed's subject index), run 2026-10-25; [2] L000358 PubMed, run 2026-10-25; [3] L000355 MeSH, run 2026-10-25; [4] L000196 PubMed, run 2026-10-25; [5] L000347, L000354 MeSH, run 2026-10-25; [6] L000326 PubMed, run 2026-10-25; [7] L000129 PubMed, run 2026-10-25

Next: I will put your choices and edits into the question and show it, with its criteria, for you to certify.

Your turn, Uday: these choices decide which studies your question admits. 1. A second drug: a, b or c? 2. People with diabetes in the same study: a, b, c or d? 3. Shortest treatment time: a, b or c? Edit any draft row in the same reply.
For example: "1 a, 2 b, 3 a" or "1 b, 2 c: at least 80%, 3 b: 12 weeks; drop the prediabetes row".

Opened beside the chat: stir-l4-workspace/data/derived/step6_criteria_S3_r3.csv

> C **Subject headings by retrieval.** Each concept of the question is mapped to MeSH headings looked up in PubMed, not recalled.

UDAY (SIMULATED USER) session 3, message 9

1 a, 2 b, 3 a

STIR-L4 ASSISTANT session 3, message 10

Step 6 of 12: Research Question | **Certified 5 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your three choices and put them into your question; rows you did not edit stay as drafted. Nothing you have approved is affected.

DRAFT: what your three choices do (my reading, yours to challenge)

Your choice	Now in your question	What it keeps in or leaves out
1 a	another treatment only when both groups get the same one	leaves out the 3 trial records that give it with a second drug [1], including the one on fat and lean tissue (shown earlier): by their abstracts, the other group gets placebo alone, e.g. “an empagliflozin placebo (daily) plus a topiramate placebo” [2]
	my suggestion: a pair of groups within a larger trial counts if it fits	one trial gives “exenatide (EQW) and dapagliflozin (DAPA) alone and coadministered” [2], so exenatide with and without the drug would fit
2 b	studies that also enrol people with diabetes only if they report those without diabetes apart	keeps the 3 trial records that enrol both [1] only if they do; whether they do is read in the full text at screening
3 a	no minimum treatment time	keeps all 13 trial records, including the 2 that lasted under 12 weeks [1]; not yet searched: whether the shares of fat, muscle and water lost change with treatment time

Added by me, and in the line: prediabetes counts as without diabetes; the five measures kept apart; the measuring method recorded; the pair of groups. Say which to drop.

Sources: [1] L000358 PubMed, run 2026-10-25; [2] L000196 PubMed, run 2026-10-25

Next: once you certify, I will draft what to record from each study, the study features to count and a map of the evidence you expect, for you to certify before the search.

```
CERTIFY Research Question | Uday (summarized): question — which kind of weight is lost with SGLT2
inhibitors, compared with placebo, no treatment or usual care: fat, muscle or body water, in adults
without diabetes (Asian people as a subgroup); framework — PICO; context — any country, whatever
the drug is given for; Asian people as a subgroup; population — adults without diabetes; people
with prediabetes count as without diabetes; intervention — any SGLT2 inhibitor, no minimum
treatment time; comparison — placebo, no treatment or usual care; outcomes — fat, muscle and body
water; muscle as lean or fat-free mass, muscle itself, and also strength or physical performance;
fat, lean mass, muscle itself, body water and strength each kept apart; the measuring method
recorded for each study; inclusion and exclusion criteria — randomized controlled trials and
controlled cohort studies; another treatment only when both groups get the same one; studies that
also enrol people with diabetes only if they report those without diabetes apart; a pair of groups
within a larger trial counts if it meets these criteria | reason: ___ | Uday, solo
```

Your turn, Uday: every step from the search on is built on this question. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "drop the prediabetes part"

UDAY (SIMULATED USER) session 3, message 11

certify, this is the question

STIR-L4 ASSISTANT session 3, message 12

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certify and reason; your question is certified. Nothing you have approved is affected.

```
CERTIFY Research Question | Uday (summarized): question — which kind of weight is lost with SGLT2
inhibitors, compared with placebo, no treatment or usual care: fat, muscle or body water, in adults
without diabetes (Asian people as a subgroup); framework — PICO; context — any country, whatever
the drug is given for; Asian people as a subgroup; population — adults without diabetes; people
with prediabetes count as without diabetes; intervention — any SGLT2 inhibitor, no minimum
```

```
treatment time; comparison — placebo, no treatment or usual care; outcomes — fat, muscle and body
water; muscle as lean or fat-free mass, muscle itself, and also strength or physical performance;
fat, lean mass, muscle itself, body water and strength each kept apart; the measuring method
recorded for each study; inclusion and exclusion criteria — randomized controlled trials and
controlled cohort studies; another treatment only when both groups get the same one; studies that
also enrol people with diabetes only if they report those without diabetes apart; a pair of groups
within a larger trial counts if it meets these criteria | reason: this is the question | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	now	Uday
8. Search Strategy	search strings and stopping rule	next	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

Before the search, you certify what to record from each study, the features to count and a map of expected evidence.

FACT: none of the 13 trial records names muscle itself, strength or physical performance in title or abstract; 6 name fat inside the liver or heart [1].

DRAFT: what to record from each study (yours to edit)

Part	What to record	Came from	What bears on it
Study	identifiers; one key linking every report of the same study	the standard study features you uploaded	—
	design: randomized trial (parallel or crossover) or controlled cohort; how groups were formed; who was blinded	your question; standard features	—
	trial registration number; funding source	standard features	—
	country where the study was done	your question (any country)	—
People	number enrolled and analysed in each group; age; percent women	standard features	—
	share with prediabetes; if people with diabetes were also enrolled, whether results for those without are given apart	your question	—
	the Asian subgroup	question 2 below	—

Part	What to record	Came from	What bears on it
	what the drug was given for	your question	FACT: the titles of the 13 name overweight or obesity (4), fatty liver (3), heart failure (2), prediabetes or insulin resistance (2), kidney disease (1), polycystic ovary syndrome (1) [2]
	starting weight and body mass index	my suggestion	—
Drug	which SGLT2 inhibitor, dose, treatment time	your question	—
	other treatment both groups get	your question	—
Comparison	placebo, no treatment or usual care, as the study describes it	your question	—
Outcomes, each kept apart	total weight change	my suggestion: needed to tell what share of the loss is fat, muscle or water	—
	whole-body fat: fat mass, percent body fat	your question	FACT: 4 of the 13 [1]
	fat in particular places	question 1 below	—
	lean or fat-free mass	your question	FACT: 2 of the 13 [1]
	muscle itself: muscle mass, area or volume	your question	FACT: none of the 13 [1]
	body water: total; outside or inside cells and plasma volume when given	your question; the parts are my suggestion	FACT: 2 of the 13 [1]
	strength; physical performance, as each study tests them	your question	FACT: none of the 13 [1]
	share of the weight lost that is fat, lean or water, when given	my suggestion	FACT: one record says most of its loss was fat tissue (shown earlier) [3]
Every outcome	measuring method	your question	FACT: the 5 abstracts that name one (shown earlier) name magnetic resonance (3), DXA, a low-dose X-ray body scan (1), and bioimpedance, a weak current through the body (1) [4]
	every time point; per group: number analysed, starting value, change or end value with its spread; difference between groups with its confidence interval; how missing data were handled	my suggestion, from standard features	—
Every value	its exact passage, where it is in the paper, and whether reported, not reported or unclear	how I keep every record	—

Added by me: rows marked “my suggestion”.

Still open: two rows.

Question	Option	What bears on it
1. Fat in particular places	a whole-body fat only	—
	b also fat under the skin or around the belly organs, kept apart	FACT: 1 of the 13 names it, “reductions in subcutaneous adipose tissue area” [1]

Question	Option	What bears on it
	c b, and also fat inside organs such as the liver or heart, kept apart	FACT: 6 of the 13 name it, e.g. “Empagliflozin Reduces Liver Fat in Individuals With and Without Diabetes” [1]
	d other	say what
	bears on every option	DRAFT: fat inside organs is a different sense from the body fat in your focus; if kept, it stays apart and never counts as body fat
2. What marks the Asian subgroup	a race or ethnicity as the study reports it	—
	b the country where the study was done	—
	c a and b, kept apart	—
	d other	say what
	bears on every option	FACT: none of the 13 names Asian people or an Asian country (my word list: East, South and Southeast Asia) in title or abstract [1]. DRAFT: so this will be read from full texts

The 13 records and the passage read for each are beside the chat; say "show all" to see them here.

Sources: [1] L000372 PubMed, run 2026-10-25; [2] L000237 PubMed, run 2026-10-25; [3] L000196 PubMed, run 2026-10-25; [4] L000358 PubMed, run 2026-10-25

Next: I will update the form for you to certify, then the features to count.

Your turn, Uday: this sets what is read from each study. 1. Fat in particular places: a, b, c or d? 2. Asian subgroup: a, b, c or d? 3. Edit any row, or say it stands.
For example: "1 b, 2 c, 3 stands" or "1 a, 2 a, 3 drop starting weight".

Opened beside the chat: stir-l4-workspace/data/derived/step7_measures.csv

C **Research Question certified: 6 of 8.**

UDAY (SIMULATED USER) session 3, message 13

1 a, 2 c. body water and how it was measured are already in there, so 3 stands

STIR-L4 ASSISTANT session 3, message 14

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: I recorded your answers. The other rows stay as they are, and body water and how it was measured are already among them. Nothing you have approved is affected: your choices only set what is read for fat and for the Asian subgroup.

FACT: none of the 6 trial records that name fat inside the liver or heart (shown earlier) also names whole-body fat in its title or abstract [1].

DRAFT: the two rows you settled (my reading, yours to challenge)

Row	Now in the form	What it leaves out or keeps apart
Fat	whole-body fat only: fat mass, percent body fat	fat under the skin, around the belly organs or inside organs is not recorded; the 6 records above give a fat value only if their full texts report whole-body fat [1]

Row	Now in the form	What it leaves out or keeps apart
Asian subgroup	race or ethnicity as the study reports it, with any results for Asian participants given apart	the country where the study was done stays in its own row; a study done in an Asian country does not count as reporting Asian participants unless it says so

The full form is open beside the chat; say "show all" to see it here. In the line below, "Paper as an Object" means the record kept for each study.

Sources: [1] L000376 PubMed, run 2026-10-25

Next: once you certify, I will draft the study features to count, then the map of the evidence you expect, each for you to certify.

```
CERTIFY Paper as an Object | data-extraction form, version 1, before the search: study —
identifiers and one key linking every report of the same study; design (randomized trial, parallel
or crossover, or controlled cohort), how groups were formed and who was blinded; trial registration
number and funding source; the country where the study was done; people — number enrolled and
analysed in each group, age and percent women; share with prediabetes and, if people with diabetes
were also enrolled, whether results for those without diabetes are given apart; race or ethnicity
as the study reports it, kept apart from country, with any results for Asian participants given
apart; what the drug was given for; starting weight and body mass index; drug — which SGLT2
inhibitor, dose and treatment time; other treatment both groups get; comparison — placebo, no
treatment or usual care, as the study describes it; outcomes, each kept apart — total weight
change; whole-body fat only (fat mass, percent body fat); lean or fat-free mass; muscle itself
(mass, area or volume); body water (total; outside or inside cells and plasma volume when given);
strength and physical performance, as each study tests them; the share of the weight lost that is
fat, lean or water, when given; for every outcome — the measuring method; every time point; for
each group the number analysed, the starting value and the change or end value with its spread; the
difference between groups with its confidence interval; how missing data were handled; for every
value — its exact passage, where it is in the paper, and whether it is reported, not reported or
unclear | reason: ___ | Uday, solo
```

Your turn, Uday: this form sets what is read from every study you include. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "drop starting weight and body mass index"

Opened beside the chat: stir-l4-workspace/protocol/extraction_form_v1.yaml

UDAY (SIMULATED USER) session 3, message 15

certify, covers my outcomes

STIR-L4 ASSISTANT session 3, message 16

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certify and reason; your form is certified. Nothing you have approved is affected.


```
CERTIFY Paper as an Object | data-extraction form, version 1, before the search: study —
identifiers and one key linking every report of the same study; design (randomized trial, parallel
or crossover, or controlled cohort), how groups were formed and who was blinded; trial registration
number and funding source; the country where the study was done; people — number enrolled and
analysed in each group, age and percent women; share with prediabetes and, if people with diabetes
were also enrolled, whether results for those without diabetes are given apart; race or ethnicity
as the study reports it, kept apart from country, with any results for Asian participants given
apart; what the drug was given for; starting weight and body mass index; drug — which SGLT2
inhibitor, dose and treatment time; other treatment both groups get; comparison — placebo, no
treatment or usual care, as the study describes it; outcomes, each kept apart — total weight
change; whole-body fat only (fat mass, percent body fat); lean or fat-free mass; muscle itself
(mass, area or volume); body water (total; outside or inside cells and plasma volume when given);
strength and physical performance, as each study tests them; the share of the weight lost that is
fat, lean or water, when given; for every outcome — the measuring method; every time point; for
```

```
each group the number analysed, the starting value and the change or end value with its spread; the
difference between groups with its confidence interval; how missing data were handled; for every
value — its exact passage, where it is in the paper, and whether it is reported, not reported or
unclear | reason: covers my outcomes | Uday, solo
```

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 now · 8–12 open

FACT: titles or abstracts of the 13 trial records name empagliflozin (7) or dapagliflozin (6); 2 describe a crossover design [1].

DRAFT: features to count, with fixed levels, to show combinations no study covers (yours to edit)

Feature	Levels	Counted per	Came from
Identifiers	PubMed and DOI numbers; one key per study, linking its reports	report; study	standard features
Publication	year; journal; language; first author's country	report	standard features
Publication type	as PubMed tags it; conference abstract; thesis	report	standard features
Funding	public / charity / industry / mixed / none stated / not reported	study	standard features
Competing interests	declared / none declared / not reported	report	standard features
Registration	registry and number; registered before the start: yes / no / not reported	study	standard features
Design	randomized parallel / randomized crossover / randomized cluster / prospective cohort / retrospective cohort / secondary analysis of trial data	study	standard features, limited to your question; crossover: FACT above [1]
How groups were formed	random, method described / random, method not described / not random / not reported	study	standard features
Who was blinded	participants / care providers / outcome assessors / analysts; open label; not reported	study	standard features
Sites; study period	one / several / not reported; start and end dates	dataset	standard features
Data source (cohorts)	trial data / cohort / registry / health records / insurance claims / other / not reported	dataset	standard features
Number of people	enrolled, analysed, lost; analysis set: all randomized / modified / per protocol / complete case	group	standard features
Age; sex	mean or median age; adult / older adult / mixed; percent women	group	standard features
Diabetes status	without diabetes only / some or all with prediabetes / with and without diabetes, those without given apart	study	your question
What the drug was given for	overweight or obesity / fatty liver / heart failure / prediabetes or insulin resistance / kidney disease / polycystic ovary syndrome / other / not reported	study	your form; levels from the titles (shown earlier) [3]
Asian participants	results for Asian participants given apart / share given, no separate results / race or ethnicity not reported	group	your form
Country where people were studied	the country; in an Asian country: yes / no / partly / not reported	dataset	your form; which countries: question 2
Setting	outpatient / inpatient / community / other / not reported	dataset	standard features
SGLT2 inhibitor	each drug as named; dose; how often	group	your question; FACT above [1]
Treatment time	weeks, counted while a comparison group lasts	group	your question; “while…lasts”: my suggestion; grouping: question 1

Feature	Levels	Counted per	Came from
Other treatment both groups get	none / diet or exercise advice / medicines for the condition / other / not reported	group	your form; e.g. “Participants were instructed to follow a balanced diet and exercise moderately” [2]; levels my suggestion
Comparison	placebo / usual care / no treatment	group	standard features, limited to your question
Part of the weight	total weight / whole-body fat / lean or fat-free mass / muscle itself / body water: total, outside cells, inside cells, plasma volume / strength / physical performance / share of the loss that is fat, lean or water	outcome	your form
Measuring method	DXA / magnetic resonance / bioimpedance / computed tomography / tracer dilution / air displacement / tape or skinfolds / a named strength or performance test / other / not reported	outcome	your form; the first three named in the 5 abstracts (shown earlier) [4]; the rest my suggestion
Primary outcome	declared primary / not	outcome	standard features
Results given	per group: change or end value with its spread; difference between groups with its confidence interval	group × outcome × time point	standard features
Missing data	complete case / single imputation / multiple imputation / model-based / not reported	dataset	standard features
Checklist; data sharing	CONSORT / STROBE / other / none; data public / on request / not available / not stated	report	standard features
Left out by me	outcome-event counts and outcome type (every outcome here is a measurement); eligibility flags (covered by diabetes status and what the drug was given for); the eight features for prediction models	—	they do not fit your question; say "keep <name>" to keep one

Added by me: the levels marked “my suggestion”.

Still open: three choices.

Question	Option	What bears on it
1. Treatment time	a weeks as reported, not grouped (then never crossed)	—
	b grouped: under 12, 12 to 25, 26 weeks or more	FACT: 2 to 52 weeks, 2 under 12 (shown earlier) [4]
	c other	say what
	bears on every option	FACT: one trial's comparison ends early: “entered an open-label extension from 24 to 52 weeks during which all participants received active treatment” [2]. DRAFT: so weeks count only while a comparison group lasts
2. Which countries count as Asian	a East, Southeast and South Asia, Iran not included (my word list so far)	—
	b a, and also Iran	—
	c all of Asia, Western and Central Asia included	—
	d other	say what

Question	Option	What bears on it
	bears on every option	FACT: by first author's affiliation, 2 of the 13 come from Iran and 1 from Hong Kong; the others from the United States (3), Denmark, the Netherlands and Sweden (2 each) and Egypt (1) [1]. DRAFT: an author's country is not always where people were studied; that is read from full texts
3. Which features to cross, to find combinations no study covers	a part of the weight × measuring method × what the drug was given for	—
	b a, and also the two Asian marks	FACT: none of the 13 names Asian people or an Asian country in title or abstract (shown earlier) [5]
	c a, and also treatment time (with 1 b)	—
	d other	say what
	bears on every option	DRAFT: each added feature multiplies the combinations, so with few studies most will be empty; an empty one counts as a gap only after a targeted search finds nothing

The full list is beside the chat; say "show all" for its details here.

Sources: [1] L000382 PubMed, run 2026-10-25; [2] L000383 PubMed, run 2026-10-25; [3] L000237 PubMed, run 2026-10-25; [4] L000358 PubMed, run 2026-10-25; [5] L000372 PubMed, run 2026-10-25

Next: I will put your choices in for you to certify, then draft the map of expected evidence.

Your turn, Uday: these set how studies are counted. 1. Treatment time: a, b or c? 2. Asian countries: a, b, c or d? 3. Features to cross: a, b, c or d? Edit any row in the same reply.
For example: "1 b, 2 b, 3 b" or "1 a, 2 c, 3 d: say what".

Opened beside the chat: stir-l4-workspace/lat/properties.yaml

> L **Step 7, before the search: the data-extraction form.** Paper as an Object: the fields to extract from each study are fixed and certified before anything is screened. Designs only: building paper objects for each study is not available in the prompt version (it is being developed for a dedicated planning tool).

UDAY (SIMULATED USER) session 3, message 17

1 a, 2 b, 3 b. measurement method dxa, bia, mri is already in, so rows stand

STIR-L4 ASSISTANT session 3, message 18

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your three answers; the other rows stand, with DXA, bioimpedance and magnetic resonance (MRI) already among the measuring methods. Nothing you have approved is affected.

FACT: with your list (Iran included), none of the 13 trial records names Asian people or a listed country in its title or abstract; by first author's affiliation, 3 of the 13 come from a listed country, Iran (2) and Hong Kong (1) [1].

DRAFT: what your three choices do (my reading, yours to challenge)

Your choice	Now in the list	What to watch
1 a	treatment time: weeks as reported, counted while a comparison group lasts; recorded, never crossed	—

Your choice	Now in the list	What to watch
2 b	country where people were studied, with an Asian-country mark: yes / no / partly (some of its countries) / not reported	race or ethnicity stays a separate mark: a study in Iran counts as done in an Asian country, whatever race or ethnicity it reports
3 b	crossed per study and outcome: part of the weight × measuring method × what the drug was given for × Asian participants × Asian-country mark	10,560 combinations, of which 2,208 are impossible by construction (below), leaving 8,352 [2]

Added by me, say which to drop:

Item	My suggestion
Asian countries, by region	East Asia: China, Hong Kong, Macao, Taiwan, Japan, North and South Korea, Mongolia. Southeast Asia: Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand, Timor-Leste, Vietnam. South Asia: Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka. Your addition: Iran
Impossible combinations	a strength or performance test goes only with strength or physical performance; these two go only with such a test, “other” or “not reported”
Studies whose groups differ	for Asian participants, a study takes the most detailed level any of its groups gives

The full list is beside the chat; say "show all" to see it here. Below, "Literature as a Tensor" means the features counted and the table crossing them.

Sources: [1] L000387 PubMed, run 2026-10-25; [2] L000388 computed from your list, run 2026-10-25

Next: once you certify, I will draft the map of expected evidence for you to certify.

```
CERTIFY Literature as a Tensor | property registry, version 1, before the search: features counted,
each with the fixed levels shown and the grain it is counted per — identifiers and one key per
study; publication (year, journal, language, first author's country); publication type; funding;
competing interests; registration; design; how groups were formed; who was blinded; sites and study
period; data source (cohorts); number of people; age and sex; diabetes status; what the drug was
given for; Asian participants (race or ethnicity as reported); country where people were studied,
with an Asian-country mark (East, Southeast and South Asia, and also Iran, countries as listed);
setting; SGLT2 inhibitor, dose and how often; treatment time in weeks as reported, counted while a
comparison group lasts, never crossed; other treatment both groups get; comparison; part of the
weight; measuring method; primary outcome; results given; missing data; checklist and data sharing;
crossed per study and outcome — part of the weight × measuring method × what the drug was given for
× Asian participants × Asian-country mark, every combination kept, impossible ones marked, and an
empty one called a gap only after a targeted search; left out — outcome-event counts, outcome type,
eligibility flags and the features for prediction models | reason: ___ | Uday, solo
```

Your turn, Uday: this fixes how studies are counted. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "drop Afghanistan from the Asian countries".

Opened beside the chat: stir-l4-workspace/lat/properties.yaml

UDAY (SIMULATED USER) session 3, message 19

certify, properties are fine

STIR-L4 ASSISTANT session 3, message 20

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certify and reason; your features to count are certified. Nothing you have approved is affected.

```
CERTIFY Literature as a Tensor | property registry, version 1, before the search: features counted, each with the fixed levels shown and the grain it is counted per — identifiers and one key per study; publication (year, journal, language, first author's country); publication type; funding; competing interests; registration; design; how groups were formed; who was blinded; sites and study period; data source (cohorts); number of people; age and sex; diabetes status; what the drug was given for; Asian participants (race or ethnicity as reported); country where people were studied, with an Asian-country mark (East, Southeast and South Asia, and also Iran, countries as listed); setting; SGLT2 inhibitor, dose and how often; treatment time in weeks as reported, counted while a comparison group lasts, never crossed; other treatment both groups get; comparison; part of the weight; measuring method; primary outcome; results given; missing data; checklist and data sharing; crossed per study and outcome — part of the weight × measuring method × what the drug was given for × Asian participants × Asian-country mark, every combination kept, impossible ones marked, and an empty one called a gap only after a targeted search; left out — outcome-event counts, outcome type, eligibility flags and the features for prediction models | reason: properties are fine | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	2 of 3 certified	Uday
8. Search Strategy	search strings and stopping rule	next	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

Last before the search: a map of the evidence you expect; empty regions will point to missing evidence.

FACT: a 2014 study against placebo in obese adults without diabetes is not among the 13 trial records; PubMed does not tag it as randomized [1]. I will suggest it as a known study to test the search.

DRAFT: sides of the map (yours to edit)

Side	Levels, in order	Came from	What bears on it
1. Part of the weight	body fat · lean or fat-free mass · muscle itself · body water · strength or physical performance	your focus and form; a study giving total weight alone goes in no region (my suggestion)	the next table
2. What the drug was given for	overweight or obesity · fatty liver · heart failure · prediabetes or insulin resistance · kidney disease · polycystic ovary syndrome · other · not reported	your certified features	FACT: in heart failure, “Fluid volume changes might be an important mechanism”; in kidney disease without diabetes, dapagliflozin “did not increase 24-hour sodium and volume excretion” [2]. DRAFT: so the water share may differ by reason

Side	Levels, in order	Came from	What bears on it
3. Asian results (question 1)	results for Asian participants given apart, or done only in your listed countries · other	your form and features; this rule is my suggestion	question 1
How studies are placed	by rule from the counted features; failing that, by likeness of text to each region's description (matching tool recorded first); then my judgment, labelled; leftovers go to you	the standard rules	an empty region is a gap only after a targeted search for it

DRAFT: what to expect, by part

Part	How the drug may change it (theory, earlier studies)	Earlier reviews naming it
Any part	—	of 73 earlier review records on these drugs and fat, muscle or body water (any population), 12 name people without diabetes, 5 Asian people or an Asian country [3]
Body fat	sugar lost in urine “provides an energy and osmotic drain that could alter body composition”, but “compensatory hyperphagia limits the degree of weight loss” (hyperphagia: eating more) [4]	14; 2 also name people without diabetes, one: “There were no significant changes in fat mass” [3]
Lean or fat-free mass	it may include body water (shown earlier) [4]	6; none name people without diabetes [3]
Muscle itself	a 2025 review: muscle mass “decreased in 15 and remained unchanged in 6 studies” [4]	6; none [3]
Body water	the drug “transiently promotes osmotic diuresis and natriuresis” (more urine and salt) [4]; the 2014 study: “a reduction in TBW measured by D2O dilution in the RE group that was significantly greater than placebo” (TBW, total body water; D2O, a tracer; RE, one of its two drugs) [4]	2; none [3]
Strength or physical performance	the 2025 review: performance tests unchanged (shown earlier) [4]	strength 2, performance 2; none; one: “muscle strength and physical performance outcomes were rarely reported” [3]

Still open: two choices.

Question	Option	What bears on it
1. Sides of the map	a part of the weight × what the drug was given for: 40 regions [5]	—
	b a, and also Asian results: 80 regions [5]	FACT: the 2025 review's muscle studies were mostly “with diabetes-mellitus (68 %) and Japanese (53 %)”; it calls for research “especially in nondiabetic, non-Japanese and HF-patients” (HF, heart failure) [4]
	c other	say what
	bears on every option	FACT: none of the 13 trial records names Asian people or a listed country (shown earlier) [6]. DRAFT: with so few trials, most regions will start empty
2. Studies per region for empty, thin and crowded	a empty 0 · thin 1–2 · crowded 5 or more	—
	b empty 0 · thin 1 · crowded 3 or more	—
	c other	say the numbers

Question	Option	What bears on it
	bears on every option	FACT: the titles of the 13 trial records name six reasons for treatment (shown earlier) [7]. DRAFT: crowded marks where results might later be pooled

The draft map is beside the chat; say "show all" for its 40 regions here.

Sources: [1] L000395 PubMed, run 2026-10-25; [2] L000196 PubMed, run 2026-10-25; [3] L000394 PubMed, run 2026-10-25; [4] L000129 PubMed, run 2026-10-25; [5] L000397 computed from the draft map, run 2026-10-25; [6] L000387 PubMed, run 2026-10-25; [7] L000237 PubMed, run 2026-10-25

Next: I will put your choices in for you to certify, then draft the search.

Your turn, Uday: this sets how the evidence found is mapped. 1. Sides: a, b or c? 2. Empty, thin, crowded: a, b or c? 3. Edit any row, or say it stands.
For example: "1 b, 2 a, 3 stands" or "1 a, 2 c: thin 1 to 3".

Opened beside the chat: stir-l4-workspace/red/atlas_v1.yaml

> L **The property registry.** Literature as a Tensor: the features to count across studies, with fixed levels, certified before the search. A design only, not applied to studies in the prompt version.

UDAY (SIMULATED USER) session 3, message 21

1 b, 2 a. facets: population (asian / not asian), compartment (fat / lean / water / function)

STIR-L4 ASSISTANT session 3, message 22

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your choices: three sides, with Asian results; empty 0, thin 1–2, crowded 5 or more; and your names for the sides. Nothing you have approved is affected.

FACT: none of the 13 trial records names anyone's race or ethnicity in its title or abstract [1].

DRAFT: your edits in the map (my reading, yours to challenge)

Side	Levels	What each holds, from your form	What to watch
Compartment	fat · lean · water · function	fat: whole-body fat only; lean: lean or fat-free mass; water: total, outside or inside cells, plasma volume; function: strength or physical performance	muscle itself has no level yet (question 1)
What the drug was given for	the eight reasons, unchanged	your certified features	your 1 b keeps it; your named sides leave it out (question 3)
Population	Asian · not Asian	Asian: results for Asian participants given apart, or done only in your listed countries (my suggestion, as before)	who counts as not Asian (question 2)
Studies per region	empty 0 · thin 1–2 · crowded 5 or more	your answer 2	3 or 4 studies is neither thin nor crowded

Still open: three choices.

Question	Option	What bears on it
1. Muscle itself	a under lean	—
	b its own level: fat · lean · muscle · water · function	your certified question keeps lean mass and muscle itself apart

Question	Option	What bears on it
	bears on every option	FACT: none of the 13 trial records names muscle itself; 2 name lean or fat-free mass (shown earlier) [2]. In type 2 diabetes, lean mass fell with some of these drugs; skeletal muscle mass fell significantly with none (shown earlier) [3]. DRAFT: the form records both either way; with a, the map cannot show a missing muscle study on its own
2. Not Asian	a every other study, including those that do not report race or ethnicity	—
	b three levels: Asian · not Asian (race or ethnicity reported, no separate results for Asian participants) · not reported	your certified features already keep these three apart
	bears on every option	FACT: no trial abstract names race or ethnicity (above) [1]. DRAFT: so it will be read from full texts; with a, a study that never says is called not Asian
3. What the drug was given for	a keep it as a side	—
	b drop it	DRAFT: the map then cannot show, for example, water measured only in heart failure
	bears on every option	FACT: the 13 trial titles name six reasons (shown earlier) [4]

FACT: regions, by your answers [5]

Your answers	3 a (keep)	3 b (drop)
1 a, 2 a	64	8
1 a, 2 b	96	12
1 b, 2 a	80	10
1 b, 2 b	120	15

The updated draft map is beside the chat; say "show all" to see it here.

Sources: [1] L000405 PubMed, run 2026-10-25; [2] L000372 PubMed, run 2026-10-25; [3] L000084 PubMed, run 2026-10-25; [4] L000237 PubMed, run 2026-10-25; [5] L000406 computed from the draft map, run 2026-10-25

Next: I will put your three answers into the map and show it for you to certify; then I draft the search.

Your turn, Uday: these settle the map's sides before you certify it. 1. Muscle itself: a or b? 2. Not Asian: a or b? 3. What the drug was given for: a or b?
For example: "1 b, 2 b, 3 a" or "1 a, 2 a, 3 b".

Opened beside the chat: stir-l4-workspace/red/atlas_v1.yaml

UDAY (SIMULATED USER) session 3, message 23

1 a, 2 a, 3 b

STIR-L4 ASSISTANT session 3, message 24

Step 7 of 12: Before the search | **Certified 6 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your three answers in the map. Nothing you have approved is affected.

FACT: your map has 8 regions, none filled yet; each part of the weight you certified goes to one compartment, or as below [1].

DRAFT: your map, ready to certify

Compartment	Holds	Asian	not Asian
fat	whole-body fat only	region 1	region 2
lean	lean or fat-free mass, and muscle itself (your 1 a)	region 3	region 4
water	total, outside or inside cells, plasma volume	region 5	region 6
function	strength or physical performance	region 7	region 8

Rule	What it says	Came from
Asian	results for Asian participants given apart, or done only in your listed countries	my suggestion
not Asian	every other study, including those that never report race or ethnicity	your 2 a
What the drug was given for	not a side of the map	your 3 b
Where a study goes	each compartment it reports; total weight alone, no region; the share of the loss, each compartment it gives	my suggestion

DRAFT, my reading: the map cannot show a missing muscle study on its own, studies silent on race, or water measured for one reason for treatment only (shown earlier); your certified features still record and cross all three, so such gaps can still be found.

The map is beside the chat; say "show all" to see its evidence here. In the line below, "Registering Evidence Dynamically" means this map.

Sources: [1] L000412 computed from your map and your certified features, run 2026-10-25

Next: once you certify, I will draft the search for each source and test it on the known study shown earlier.

```
CERTIFY Registering Evidence Dynamically | atlas, version 1, before the search: two sides, counted in studies — compartment (fat: whole-body fat only; lean: lean or fat-free mass, with muscle itself counted here; water: total, outside or inside cells, plasma volume; function: strength or physical performance) × population (Asian: results for Asian participants given apart, or done only in the listed Asian countries; not Asian: every other study, including those that do not report race or ethnicity), giving 8 regions, with the theory and earlier reviews shown for each compartment; what the drug was given for is not a side; a study is placed in each compartment it reports, one giving total weight alone in none, and one giving the share of the loss in each compartment it gives; studies per region: empty 0, thin 1–2, crowded 5 or more; placed by rule from the counted features, then by likeness of text to each region's description (matching tool recorded first), then by the assistant's labelled judgment, with any study left over going to Uday; an empty region called a gap only after a targeted search for it | reason: ___ | Uday, solo
```

Your turn, Uday: this is the last approval before the search. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "make muscle its own compartment".

Opened beside the chat: stir-l4-workspace/red/atlas_v1.yaml

UDAY (SIMULATED USER) session 3, message 25

certify, atlas is fine

STIR-L4 ASSISTANT session 3, message 26

Step 8 of 12: Search Strategy | **Certified 6 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certify; ran a draft search. Nothing you have approved is affected.

```
CERTIFY Registering Evidence Dynamically | atlas, version 1, before the search: two sides, counted in studies — compartment (fat: whole-body fat only; lean: lean or fat-free mass, with muscle itself counted here; water: total, outside or inside cells, plasma volume; function: strength or physical performance) × population (Asian: results for Asian participants given apart, or done only in the listed Asian countries; not Asian: every other study, including those that do not report race or ethnicity), giving 8 regions, with the theory and earlier reviews shown for each compartment; what the drug was given for is not a side; a study is placed in each compartment it reports, one giving total weight alone in none, and one giving the share of the loss in each compartment it gives; studies per region: empty 0, thin 1–2, crowded 5 or more; placed by rule from the counted features, then by likeness of text to each region's description (matching tool recorded first), then by the assistant's labelled judgment, with any study left over going to Uday; an empty region called a gap only after a targeted search for it | reason: atlas is fine | Uday, solo
```

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 now · 9–12 open

FACT: PubMed, no filter: search 1, 739 records [5]; search 2, 72 [7]. OpenAlex, no filter: 45,239 works [11].

Open: which records report fat, muscle or water in adults without diabetes.

Search 1: Search 1: no words for without diabetes (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab] OR enavogliflozin[tiab] OR ganagliflozin[tiab] OR remogliflozin[tiab] OR sergliflozin[tiab] OR janagliflozin[tiab] OR licogliflozin[tiab]) AND ("Body Composition"[Mesh] OR "Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR "Body Water"[Mesh] OR "Muscle Strength"[Mesh] OR "Physical Functional Performance"[Mesh] OR fat[tiab] OR adipos*[tiab] OR muscle*[tiab] OR muscular[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "lean tissue"[tiab] OR "fat-free mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab] OR "body water"[tiab] OR "extracellular water"[tiab] OR "intracellular water"[tiab] OR "extracellular fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"[tiab] OR hydration[tiab] OR "water loss"[tiab] OR "fluid loss"[tiab] OR "grip strength"[tiab] OR handgrip[tiab] OR "hand grip"[tiab] OR dynamomet*[tiab] OR "physical performance"[tiab] OR "physical function"[tiab] OR "walk test"[tiab] OR "minute walk"[tiab] OR "gait speed"[tiab] OR "walking speed"[tiab] OR "chair stand"[tiab] OR DXA[tiab] OR DEXA[tiab] OR "dual-energy x-ray"[tiab] OR bioimpedance[tiab] OR "bioelectrical impedance"[tiab]) AND ("Randomized Controlled Trial"[pt] OR "Controlled Clinical Trial"[pt] OR "Clinical Trial"[pt] OR randomized[tiab] OR randomised[tiab] OR randomly[tiab] OR placebo[tiab] OR trial[tiab] OR crossover[tiab] OR "cross-over"[tiab] OR "Cohort Studies"[Mesh] OR cohort*[tiab] OR "Observational Study"[pt] OR propensity[tiab] OR retrospective*[tiab] OR prospective*[tiab])) NOT ("Animals"[Mesh] NOT "Humans"[Mesh])
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors, by heading and by each drug's name	topic	subject headings, title/abstract words, with truncation	23,388 [1]
2	fat, lean or muscle, water, strength or performance, methods	focus; your question	subject headings, title/abstract words, with truncation	1,792,382 [2]
3	randomized trials and cohort studies	your question	subject headings, title/abstract words, with truncation, publication filter	6,066,499 [3]
4	animal-only records, left out (NOT)	my suggestion	subject headings	5,511,444 [4]
All	combined	—	blocks joined as shown	**739** [5]

Search 2: Search 2: search 1 with words for without diabetes (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
((("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab] OR enavogliflozin[tiab] OR ganagliflozin[tiab] OR remogliflozin[tiab] OR sergliflozin[tiab] OR janagliflozin[tiab] OR
```

```
licogliflozin[tiab]) AND ("Body Composition"[Mesh] OR "Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh]
OR "Sarcopenia"[Mesh] OR "Body Water"[Mesh] OR "Muscle Strength"[Mesh] OR "Physical Functional
Performance"[Mesh] OR fat[tiab] OR adipos*[tiab] OR muscle*[tiab] OR muscular[tiab] OR "body composition"
[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "lean tissue"[tiab] OR "fat-free mass"[tiab] OR
"fat free mass"[tiab] OR sarcopeni*[tiab] OR "body water"[tiab] OR "extracellular water"[tiab] OR
"intracellular water"[tiab] OR "extracellular fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"
[tiab] OR hydration[tiab] OR "water loss"[tiab] OR "fluid loss"[tiab] OR "grip strength"[tiab] OR
handgrip[tiab] OR "hand grip"[tiab] OR dynamomet*[tiab] OR "physical performance"[tiab] OR "physical
function"[tiab] OR "walk test"[tiab] OR "minute walk"[tiab] OR "gait speed"[tiab] OR "walking speed"
[tiab] OR "chair stand"[tiab] OR DXA[tiab] OR DEXA[tiab] OR "dual-energy x-ray"[tiab] OR
bioimpedance[tiab] OR "bioelectrical impedance"[tiab]) AND ("Randomized Controlled Trial"[pt] OR
"Controlled Clinical Trial"[pt] OR "Clinical Trial"[pt] OR randomized[tiab] OR randomised[tiab] OR
randomly[tiab] OR placebo[tiab] OR trial[tiab] OR crossover[tiab] OR "cross-over"[tiab] OR "Cohort
Studies"[Mesh] OR cohort*[tiab] OR "Observational Study"[pt] OR propensity[tiab] OR retrospective*[tiab]
OR prospective*[tiab])) NOT ("Animals"[Mesh] NOT "Humans"[Mesh])) AND ("non-diabetic"[tiab] OR
nondiabetic[tiab] OR "non diabetic"[tiab] OR "without diabetes"[tiab] OR "without type 2 diabetes"[tiab]
OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab] OR normoglycemic[tiab] OR normoglycaemic[tiab] OR
prediabet*[tiab] OR "pre-diabetes"[tiab] OR "impaired glucose tolerance"[tiab] OR "impaired fasting
glucose"[tiab])
```

Block	Covers	From	How it searches	Records
1	without diabetes, or prediabetes	topic; your question	title/abstract words, with truncation	95,296 [6]
All	combined	—	blocks joined as shown	**72** [7]

Search 3: OpenAlex: drug, parts of the weight, study types (OpenAlex; published any to 2026-10-25; filter: none)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR "sodium-
glucose co-transporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin
OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR
bexagliflozin OR enavogliflozin OR ganagliflozin OR remogliflozin OR sergliflozin OR janagliflozin OR
licogliflozin) AND (fat OR adipose OR adiposity OR muscle OR muscles OR muscular OR "body composition" OR
"lean mass" OR "lean body mass" OR "lean tissue" OR "fat-free mass" OR "fat free mass" OR sarcopenia OR
"body water" OR "extracellular water" OR "intracellular water" OR "extracellular fluid" OR "fluid volume"
OR "plasma volume" OR hydration OR "water loss" OR "fluid loss" OR "grip strength" OR handgrip OR "hand
grip" OR "physical performance" OR "physical function" OR "walk test" OR "minute walk" OR "gait speed" OR
"walking speed" OR "chair stand" OR DXA OR DEXA OR "dual-energy x-ray" OR bioimpedance OR "bioelectrical
impedance") AND (randomized OR randomised OR randomly OR placebo OR trial OR crossover OR "cross-over" OR
cohort OR "observational study" OR propensity OR retrospective OR prospective)
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors, by each drug's name	topic	words in title, abstract or text	124,265 [8]
2	fat, lean or muscle, water, strength or performance, methods	focus; your question	words in title, abstract or text	10,751,276 [9]
3	randomized trials and cohort studies	your question	words in title, abstract or text	29,702,854 [10]
All	combined	—	blocks joined as shown	**45,239** [11]

Check	Result
Trials found earlier	FACT: both PubMed searches keep all 13 trial records found earlier [12]
Known studies	FACT: each PubMed search was run again with `AND (29235586[pmid] OR 33879970[pmid])` added; search 1 finds 2 of 2 [13], search 2 finds 1 of 2 [14] (below). OpenAlex: not checked (question 3)
Left out on purpose	DRAFT: adults, the comparison and Asian people; they are judged at screening, so abstracts silent on them are not lost
Not searched yet	ClinicalTrials.gov for finished trials (I can run it); other trial registries through the WHO portal (you); paid databases such as Embase (a large drug-focused index), which you cannot reach

Suggested known study (my suggestion)	What its abstract says	Search 1	Search 2
2014 pilot study: remogliflozin or sergliflozin against placebo	“with placebo (n = 12) in obese non-diabetic subjects” [15]	found [13]	found [14]
2021 trial: dapagliflozin against teneligliptin, a drug of another class, in fatty liver	“NAFLD patients without type 2 diabetes mellitus were enrolled” [16]	found [13]	missed [14]

Passages beside the chat ("show all").

Added by me (say which to drop)	Came from
drug names enavogliflozin, ganagliflozin, remogliflozin, sergliflozin	names from Asian trial registrations, checked in PubMed [17], and the 2014 study's abstract [15]; PubMed reports no record with ganagliflozin, nor with “absence of diabetes” [18]
drug names janagliflozin, licogliflozin	my suggestion
lean tissue; strength, performance and measuring-method words	a trial abstract (shown earlier) [15]; your question and form
study-type words	your question; the words are my suggestion
animal-only records left out	my suggestion
no language or date limit	my suggestion

DRAFT: finishing the search (yours to edit)	
Stopping rule	stop changing the strings when every known study you confirm is found, and the last change adds no record that meets your question among the records it adds (you screen them)
Peer review	a second searcher checks the strings with PRESS, a standard checklist for search strings; if no one can, the protocol says so

Still open	Option	What bears on it
1. Words for without diabetes	a leave them out (search 1); diabetes status is judged at screening	FACT: 739 records [5]; finds both suggested studies [13]
	b keep them (search 2)	FACT: 72 records [7]; misses the 2021 trial, whose title says “without type 2 diabetes mellitus” [14][16]
	c other	say what
	bears on every option	DRAFT: abstracts may not say whether people had diabetes; with a, you screen more records
2. Known studies to test the search	a the two above	DRAFT: only the 2014 study seems to fit your question by its abstract; the 2021 trial's comparison is another drug: “allocated to receive either dapagliflozin (SGLT2i) or teneligliptin (DPP4i) for 12 weeks” [16]
	b a, and also studies you know (give their titles)	—
	c other	say what
3. OpenAlex	a search its titles and abstracts only; reading past its first 100 results needs a free access code (a key you create on OpenAlex's website)	FACT: 45,239 works [11]
	b use it only to check the references and citations of included studies	—
	c other	say what
	bears on every option	DRAFT: its search also reads full texts, so its count is not comparable with PubMed's; today it refused more requests because its free daily allowance for this network is used up

Sources: [1] L000420 PubMed, run 2026-10-25; [2] L000421 PubMed, run 2026-10-25; [3] L000422 PubMed, run 2026-10-25; [4] L000423 PubMed, run 2026-10-25; [5] L000424 PubMed, run 2026-10-25; [6] L000425 PubMed, run 2026-10-25; [7] L000426 PubMed, run 2026-10-25; [8] L000429 OpenAlex, run 2026-10-25; [9] L000430 OpenAlex, run 2026-10-25; [10] L000431 OpenAlex, run 2026-10-25; [11] L000432 OpenAlex, run 2026-10-25; [12] L000435 computed from the saved searches, run 2026-10-25; [13] L000427 PubMed, run 2026-10-25; [14] L000428 PubMed, run 2026-10-25; [15] L000196 PubMed, run 2026-10-25; [16] L000129 PubMed, run 2026-10-25; [17] L000199 PubMed, run 2026-10-25; [18] L000434 computed from the saved searches, run 2026-10-25

Next: I apply your choices, then show strategy and stopping rule to certify.

Your turn, Uday: these shape your strings. 1. Without-diabetes words: a, b or c? 2. Known studies: a, b or c? 3. OpenAlex: a, b or c?
For example: "1 a, 2 a, 3 b".

Opened beside the chat: stir-l4-workspace/data/derived/step8_known_items_suggested.csv

> L **The atlas.** Registering Evidence Dynamically: the map of the field (sides and thresholds for empty, thin and crowded regions) certified before the search; then the first draft search is run. A design only, not applied to studies in the prompt version.

UDAY (SIMULATED USER) session 3, message 27

2 b: known studies: pmid 42104798 and 39361103

STIR-L4 ASSISTANT session 3, message 28

Step 8 of 12: Search Strategy | **Certified 6 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your four known studies; checked both searches and read your 2026 trial's free full text. Nothing you have approved is affected.

FACT: search 1 finds 3 of your 4 known studies [1], search 2 finds 2 [2]; both miss your 2026 trial [1][2], whose full text, free in PubMed Central (PubMed's full-text archive), names no fat, muscle, lean mass, body water or strength (my word list) [3].

Open: whether the searches miss studies meeting your question.

Known-study check (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none): searches 1 and 2 (shown earlier), each in parentheses, then:

```
AND (29235586[pmid] OR 33879970[pmid] OR 42104798[pmid] OR 39361103[pmid])
```

Variant	Added block	From	Records
search 1 + known studies	your four known studies, by PubMed identifier	your answer	**3** [1]
search 2 + known studies	the same	your answer	**2** [2]
OpenAlex	not checked (question 2)	—	—

Your known studies (FACT, from [6])

Study	What its text says	Meets your question? (DRAFT)	Search 1	Search 2
2014 pilot study (shown earlier)	against placebo, adults without diabetes (shown earlier)	yes, by its abstract	found [1]	found [2]
2021 trial (shown earlier)	against another drug (shown earlier)	no: the comparison	found [1]	missed [2]

Study	What its text says	Meets your question? (DRAFT)	Search 1	Search 2
2026 trial, Hong Kong: empagliflozin or placebo, lifestyle advice in both groups, 16 weeks	“overweight or obese adults with schizophrenia spectrum disorders (non-diabetic or pre-diabetic)” [5]; outcomes: body weight and BMI [5]; “Secondary outcomes included changes in waist circumference” [4]; “in Hong Kong between November 2023 and April 2024” [4]	no: none of your parts of the weight [3]	missed [1]	missed [2]
2024 trial: empagliflozin with topiramate, against placebo alone (shown earlier)	“had a greater percent change of fat mass and fat percent at week 12” [5]	no: the second drug goes to one group only	found [1]	found [2]

Still open	Option	What bears on it
3. Your two studies as known studies	a keep both	DRAFT: the search could then find the 2026 trial only with words for body weight, which your question does not name
	b keep the 2024 trial only	DRAFT: it names fat mass [5], so it still tests your words for parts of the weight
	c drop both	—
	d other	say what
	bears on every option	DRAFT: my stopping rule asks that every known study be found. Only the 2014 study meets your question by its text, so the test rests on it; a study your question leaves out tests only the words. If you want trials like the 2024 one in the review, that reopens your question; say so

Known studies are beside the chat; say "show all" for titles.

Sources: [1] L000443 PubMed, run 2026-10-25; [2] L000444 PubMed, run 2026-10-25; [3] L000447 computed from the saved full text, run 2026-10-25; [4] L000445 PubMed Central, run 2026-10-25; [5] L000441 PubMed, run 2026-10-25; [6] L000442 computed from your answer and the saved records, run 2026-10-25

Next: I put your answers into the strings and stopping rule for you to certify.

Your turn, Uday: these fix what the search must find. 1. Without-diabetes words: a leave out, b keep, c other? 2. OpenAlex: a titles and abstracts only, b checking citations only, c other? 3. Your two studies: a, b, c or d?
For example: "1 a, 2 b, 3 b".

Opened beside the chat: stir-l4-workspace/data/derived/known_items.csv

V **Known studies test the search.** The draft searches find 3 of 4 known studies; the assistant reads the free full text of the missed trial. The strategy is refined until a search finds all 4 (certified at message 36).

UDAY (SIMULATED USER) session 3, message 29

1 a, 2 a, 3 a

STIR-L4 ASSISTANT session 3, message 30

Step 8 of 12: Search Strategy | **Certified 6 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your three answers. Nothing you have approved is affected.

FACT: search 1 plus four phrases for body weight finds all 4 known studies [1]; 2,026 records [2], 1,287 not in search 1 [3].

FACT: ClinicalTrials.gov, a public register of studies: 540 name a drug word among treatments and a weight-part word among outcomes, any status [4]; 225 completed, 90 with results posted [5].

Open: which records report your parts of the weight in adults without diabetes.

Search 4: search 1 plus body-weight words (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab] OR enavogliflozin[tiab] OR
ganagliflozin[tiab] OR remogliflozin[tiab] OR sergliflozin[tiab] OR janagliflozin[tiab] OR
licogliflozin[tiab]) AND ("Body Composition"[Mesh] OR "Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh]
OR "Sarcopenia"[Mesh] OR "Body Water"[Mesh] OR "Muscle Strength"[Mesh] OR "Physical Functional
Performance"[Mesh] OR fat[tiab] OR adipos*[tiab] OR muscle*[tiab] OR muscular[tiab] OR "body composition"
[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "lean tissue"[tiab] OR "fat-free mass"[tiab] OR
"fat free mass"[tiab] OR sarcopeni*[tiab] OR "body water"[tiab] OR "extracellular water"[tiab] OR
"intracellular water"[tiab] OR "extracellular fluid"[tiab] OR "fluid volume"[tiab] OR "plasma volume"
[tiab] OR hydration[tiab] OR "water loss"[tiab] OR "fluid loss"[tiab] OR "grip strength"[tiab] OR
handgrip[tiab] OR "hand grip"[tiab] OR dynamomet*[tiab] OR "physical performance"[tiab] OR "physical
function"[tiab] OR "walk test"[tiab] OR "minute walk"[tiab] OR "gait speed"[tiab] OR "walking speed"
[tiab] OR "chair stand"[tiab] OR DXA[tiab] OR DEXA[tiab] OR "dual-energy x-ray"[tiab] OR
bioimpedance[tiab] OR "bioelectrical impedance"[tiab] OR "body weight"[tiab] OR "weight loss"[tiab] OR
"weight reduction"[tiab] OR "weight change"[tiab]) AND ("Randomized Controlled Trial"[pt] OR "Controlled
Clinical Trial"[pt] OR "Clinical Trial"[pt] OR randomized[tiab] OR randomised[tiab] OR randomly[tiab] OR
placebo[tiab] OR trial[tiab] OR crossover[tiab] OR "cross-over"[tiab] OR "Cohort Studies"[Mesh] OR
cohort*[tiab] OR "Observational Study"[pt] OR propensity[tiab] OR retrospective*[tiab] OR prospective*
[tiab])) NOT ("Animals"[Mesh] NOT "Humans"[Mesh])
```

Block	Covers	From	How it searches	Records
2, added	body weight, weight loss, reduction or change	my suggestion	title/abstract words	405,014 [6]
All	combined	—	blocks joined as shown	**2,026** [2]
Known studies	search 4 AND your four, by PubMed identifier (as before)	your answer	—	**4** [1]

Search 5: ClinicalTrials.gov (data as of 2026-10-23; any status; filter: none)

```
Interventions: "SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter
2" OR "sodium-glucose co-transporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR
canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR
henagliflozin OR bexagliflozin OR enavogliflozin OR ganagliflozin OR remogliflozin OR sergliflozin OR
janagliflozin OR licogliflozin
Outcome measures: fat OR adipose OR adiposity OR muscle OR muscles OR muscular OR "body composition" OR
"lean mass" OR "lean body mass" OR "lean tissue" OR "fat-free mass" OR "fat free mass" OR sarcopenia OR
"body water" OR "extracellular water" OR "intracellular water" OR "extracellular fluid" OR "fluid volume"
OR "plasma volume" OR hydration OR "water loss" OR "fluid loss" OR "grip strength" OR handgrip OR "hand
grip" OR "physical performance" OR "physical function" OR "walk test" OR "minute walk" OR "gait speed" OR
"walking speed" OR "chair stand" OR DXA OR DEXA OR "dual-energy x-ray" OR bioimpedance OR "bioelectrical
impedance"
```

Block	Covers	From	How it searches	Studies
1	SGLT2 inhibitors, by each drug's name	topic	words in the treatments	1,892 [7]
2	fat, lean or muscle, water, strength or performance, methods	focus; your question	words in the outcomes	133,420 [8]
All	combined	—	both must match	**540** [4]
Known studies	not checked: register entries carry no PubMed identifier	—	—	—

OpenAlex	What changes	Records
Search 3, titles and abstracts only	search 3 (shown earlier), published any to 2026-10-25, searched in titles and abstracts	pending: refused today, its free daily allowance for this network used up until midnight UTC (question 2)

Not searched yet: the WHO trial portal (you); Embase and other paid databases (out of reach).

Still open	Option	What bears on it
1. Your 2026 trial, which search 1 misses	a add the four phrases (search 4)	FACT: finds 4 of 4 [1]; 1,287 more records to screen [3]
	b keep search 1; the stopping rule leaves this trial out	FACT: search 1 finds the other 3 [9]; the trial's full text names no part of the weight you chose (shown earlier) [10]
	c other	say what
	bears on every option	DRAFT: with b, the protocol records that search 1 misses it, and why
2. OpenAlex	a you create a free access code on OpenAlex's website and type it here; I keep it private	DRAFT: it also lets me read past its first 100 results, which screening needs
	b I run it in our next session, after the reset	—
	c other	say what

Added by me (say which to drop)	Came from
the four body-weight phrases (only with 1 a)	my suggestion; the trial's title names body weight [11]
search 5: any status, no population or study-type words (judged at screening)	my suggestion; a register may hold trials not yet published as papers; 90 have results posted there [5]

The 540 studies are beside the chat; say "show all" here.

Sources: [1] L000460 PubMed, run 2026-10-25; [2] L000459 PubMed, run 2026-10-25; [3] L000461 computed from the saved searches, run 2026-10-25; [4] L000468 ClinicalTrials.gov, run 2026-10-25; [5] L000469 computed from the saved register entries, run 2026-10-25; [6] L000458 PubMed, run 2026-10-25; [7] L000464 ClinicalTrials.gov, run 2026-10-25; [8] L000466 ClinicalTrials.gov, run 2026-10-25; [9] L000443 PubMed, run 2026-10-25; [10] L000447 computed from the saved full text, run 2026-10-25; [11] L000441 PubMed, run 2026-10-25

Next: I put your answers into the strings and stopping rule for you to certify.

Your turn, Uday: these settle the search. 1. Your 2026 trial: a, b or c? 2. OpenAlex: a, b or c?
For example: "1 b, 2 b" or "1 a, 2 a".

Opened beside the chat: stir-l4-workspace/data/derived/step8_ctgov_registered.csv

UDAY (SIMULATED USER) session 3, message 31

1 a, 2 b. cant make an openalex key today

STIR-L4 ASSISTANT session 3, message 32

Step 8 of 12: Search Strategy | **Certified 6 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your two answers: PubMed uses search 4, and OpenAlex waits for our next session. Nothing you have approved is affected: the search only widens, and screening applies your certified question.

DRAFT: your search strategy, version 1 (yours to edit)

Source	What it searches	Where it stands
PubMed	search 4 (shown earlier): drug × parts of the weight, their measuring methods or body weight × trials or cohorts; animal-only records left out	FACT: run; 2,026 records [1]; finds all 4 known studies [2]
ClinicalTrials.gov	search 5 (shown earlier): drug among treatments × parts of the weight among outcomes, any status	FACT: run; 540 studies [3]

Source	What it searches	Where it stands
OpenAlex	search 3 (shown earlier), titles and abstracts only, plus the four body-weight phrases	next session; reading past its first 100 records needs a free access code
WHO trial portal (one search across other trial registers)	the drug names of search 5	you, by hand, next session; I give the steps then
Embase and other paid databases	—	out of reach; the protocol says so
Every source	no language, start-date or population limit	adults, the comparison, diabetes status and Asian people judged at screening

DRAFT: finishing the search

Part	Draft	What bears on it
Stopping rule	the strings stop changing when every known study is found and none of the records the last change added meets your question	FACT: all 4 found [2]; the body-weight phrases added 1,287 records [4]. DRAFT: you screen these with the rest, so screening tests the rule; if one meets your question, the strings are revisited, if you agree
OpenAlex and the WHO portal	run next session, OpenAlex with the same known-study check; if either changes the strings, they are revisited, if you agree	—
Peer review	a second searcher checks the strings with the PRESS checklist; if no one can, the protocol says so	—

DRAFT, what to watch: alone, you would screen 2,026 PubMed records and 540 register entries [1][3], plus OpenAlex's; the methods step weighs this against your 5 hours a week.

Added by me (say which to drop)	Came from
the four body-weight phrases in OpenAlex too	my suggestion: without them it may miss your 2026 trial, whose full text names no part of the weight you chose (shown earlier) [5]
the WHO trial portal, searched by you	my suggestion
the known-study check in OpenAlex	my suggestion

The full strategy, with every string, is beside the chat; say "show all" to see it here.

Sources: [1] L000459 PubMed, run 2026-10-25; [2] L000460 PubMed, run 2026-10-25; [3] L000468 ClinicalTrials.gov, run 2026-10-25; [4] L000461 computed from the saved searches, run 2026-10-25; [5] L000447 computed from the saved full text, run 2026-10-25

Next: once you certify, I remove duplicate records and show you the first batch to screen; OpenAlex and the WHO portal follow next session.

```
CERTIFY Search Strategy | version 1: PubMed — search 4 as shown, records created up to 2026-10-25:
SGLT2 inhibitors, by subject heading and each drug's name, AND fat, lean or muscle, body water,
strength or performance, their measuring methods or body weight, AND randomized trials or cohort
studies, animal-only records left out; ClinicalTrials.gov — search 5 as shown, any status: the drug
names among treatments AND the parts of the weight among outcomes; OpenAlex — search 3 as shown, in
titles and abstracts only, with the four body-weight phrases added, published up to the run date,
run in the next session; WHO trial portal — the drug names of search 5, searched by Uday by hand in
the next session; not searched — Embase and other paid databases, out of reach; filters — no
language, start-date or population limit, with adults, the comparison, diabetes status and Asian
people judged at screening; known studies — PubMed 29235586, 33879970, 39361103 and 42104798,
checked in PubMed and OpenAlex; stopping rule — the strings stop changing when every known study is
found and none of the records added by the last change meets the question, as Uday screens them;
```

```
peer review — a second searcher checks the strings with PRESS, a standard checklist, or the
protocol says no one did | reason: ___ | Uday, solo
```

Your turn, Uday: this fixes where and how the review searches. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "wait until OpenAlex has run".

Opened beside the chat: stir-l4-workspace/data/derived/step8_strategy_v1.md

> H **No key, no pretending.** Without an OpenAlex access code, the OpenAlex search waits rather than being reported as done.

UDAY (SIMULATED USER) session 3, message 33

no second searcher, skip. cant run embase or cinahl, note it

STIR-L4 ASSISTANT session 3, message 34

Step 8 of 12: Search Strategy | **Certified 6 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your two edits: no peer review of the strings, as there is no second searcher, and Embase and CINAHL (a nursing and allied-health database) noted as not searched, because you cannot reach them. Nothing you have approved is affected.

DRAFT: the two rows you changed (the rest stands as shown)

Part	Before	Now
Peer review	a second searcher checks the strings with PRESS, or the protocol says no one did	none: no second searcher; the protocol says the strings were not peer reviewed
Not searched	Embase and other paid databases, out of reach	Embase and CINAHL, which you cannot search, and other paid databases; the protocol names Embase and CINAHL and says why

DRAFT, what to watch: without peer review, the check on your strings rests on your known studies, all 4 found [1], and on the stopping rule as you screen.

The updated strategy is beside the chat; say "show all" to see it here.

Sources: [1] L000460 PubMed, run 2026-10-25

Next: once you certify, I remove duplicate records and show you the first batch to screen; OpenAlex and the WHO portal follow next session.

```
CERTIFY Search Strategy | version 1: PubMed — search 4 as shown, records created up to 2026-10-25:
SGLT2 inhibitors, by subject heading and each drug's name, AND fat, lean or muscle, body water,
strength or performance, their measuring methods or body weight, AND randomized trials or cohort
studies, animal-only records left out; ClinicalTrials.gov — search 5 as shown, any status: the drug
names among treatments AND the parts of the weight among outcomes; OpenAlex — search 3 as shown, in
titles and abstracts only, with the four body-weight phrases added, published up to the run date,
run in the next session; WHO trial portal — the drug names of search 5, searched by Uday by hand in
the next session; not searched — Embase and CINAHL, which Uday cannot search, named in the protocol
with the reason, and other paid databases, out of reach; filters — no language, start-date or
population limit, with adults, the comparison, diabetes status and Asian people judged at
screening; known studies — PubMed 29235586, 33879970, 39361103 and 42104798, checked in PubMed and
OpenAlex; stopping rule — the strings stop changing when every known study is found and none of the
records added by the last change meets the question, as Uday screens them; peer review — none, as
there is no second searcher; the protocol says the strings were not peer reviewed | reason: ___ |
Uday, solo
```

Your turn, Uday: this fixes where and how the review searches. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.

For example: "certify, because ..." or "add Scopus to the databases not searched".

Opened beside the chat: stir-l4-workspace/data/derived/step8_strategy_v1_r2.md

UDAY (SIMULATED USER) session 3, message 35

certify, finds both known studies

STIR-L4 ASSISTANT session 3, message 36

Step 9 of 12: Scoping and pilot screening | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certify and reason; your search strategy is certified. Nothing you have approved is affected. I gathered the records to screen and checked them for duplicates.

```
CERTIFY Search Strategy | version 1: PubMed — search 4 as shown, records created up to 2026-10-25:
SGLT2 inhibitors, by subject heading and each drug's name, AND fat, lean or muscle, body water,
strength or performance, their measuring methods or body weight, AND randomized trials or cohort
studies, animal-only records left out; ClinicalTrials.gov — search 5 as shown, any status: the drug
names among treatments AND the parts of the weight among outcomes; OpenAlex — search 3 as shown, in
titles and abstracts only, with the four body-weight phrases added, published up to the run date,
run in the next session; WHO trial portal — the drug names of search 5, searched by Uday by hand in
the next session; not searched — Embase and CINAHL, which Uday cannot search, named in the protocol
with the reason, and other paid databases, out of reach; filters — no language, start-date or
population limit, with adults, the comparison, diabetes status and Asian people judged at
screening; known studies — PubMed 29235586, 33879970, 39361103 and 42104798, checked in PubMed and
OpenAlex; stopping rule — the strings stop changing when every known study is found and none of the
records added by the last change meets the question, as Uday screens them; peer review — none, as
there is no second searcher; the protocol says the strings were not peer reviewed | reason: finds
both known studies | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	3 of 3 certified	—
8. Search Strategy	search strings and stopping rule	certified by Uday	—
9. Scoping and pilot screening	first included studies	now	—
10. Methodology	methods for the protocol	next	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

FACT	
Records to screen	2,566: 2,026 PubMed records [1] and 540 ClinicalTrials.gov register entries [2]
Duplicates	none share a PubMed number or DOI (a paper's permanent web identifier); one pair shares title and year (question 3), so 2,565 if it is one paper [3]
Register entries linked	96 list one of these PubMed records as their own report; each is linked to it, not removed [3]
Most likely to fit	26 of the 2,026 PubMed records name a trial or cohort design, people without diabetes and a part of the weight in your form [4]

Open: which records meet your question; only you decide.

DRAFT: batch 1, the first 10 by my order; each decision is yours [6]

#	Record	What the abstract says	My suggestion (include = read in full), and why
1	PubMed 29235586 (2014): pilot study, remogliflozin or sergliflozin, healthy obese subjects	“with placebo (n = 12) in obese non-diabetic subjects”	include: placebo, adults without diabetes, body water and fat-free mass measured; the abstract does not say the groups were randomized, so the full text decides
2	PubMed 27550386 (2017): dapagliflozin and exenatide dual therapy, 24 weeks	“Of 25 dapagliflozin/exenatide- and 25 placebo-treated participants”	exclude: exenatide goes to one group only
3	PubMed 28345814 (2017): the same dual therapy over 1 year	“Of the original 25 dapagliflozin + exenatide-treated and 25 placebo-treated participants”	exclude: the trial of record 2
4	PubMed 30675708 (2019): EMPA-TROPISM trial, rationale and design	“empagliflozin in non-diabetic HF patients … change in the distance in the 6-min walk test”	with 1 a: include, as the plan of record 9 (no results); with 1 b: exclude. The fat it names is heart fat, not body fat
5	PubMed 31218819 (2019): EMPERIAL trials, rationale and design	“with or without type 2 diabetes … change from baseline in 6-min walk test distance”	with 1 a: unsure, whether people without diabetes are reported apart (plan of record 10); with 1 b: exclude
6	PubMed 32568464 (2020): empagliflozin in adults with obesity, randomized trial	“adults with obesity without type 2 diabetes mellitus (T2DM) … magnetic resonance imaging assessment of body fat”	include: against placebo; whether whole-body fat is reported, the full text decides
7	PubMed 33091173 (2021): DAPA-HF population, mathematical modelling	“in a virtual population of HF-rEF patients generated by varying model parameters”	exclude: a computer simulation, not a study of people
8	PubMed 33186500 (2021): SUGAR-DM-HF trial, heart failure	“82 (78.1%) diabetes and 23 (21.9%) prediabetes … 6-minute walk distance”	with 1 a: unsure, whether those with prediabetes are reported apart; with 1 b: exclude
9	PubMed 33197559 (2021): empagliflozin in nondiabetic heart failure, randomized trial	“nondiabetic HFrEF patients (n = 84) were randomized to empagliflozin 10 mg daily or placebo … 6-min walk test”	with 1 a: include; with 1 b: exclude
10	PubMed 33351892 (2021): EMPERIAL trials, results	“with and without T2D, were randomized to empagliflozin 10 mg or placebo … 6-minute walk test distance”	with 1 a: unsure, as record 8; with 1 b: exclude

Full titles are beside the chat; say "show all" to see them here.

Still open	Option	What bears on it
1. A study reporting only strength or physical performance	a keep it, as your question reads	FACT: 80 PubMed records name strength or physical performance but no fat, lean or muscle mass, or body water; 68 of these 80 name heart failure [5]

Still open	Option	What bears on it
	b keep it only when it also reports fat, lean or muscle mass, or body water (this reopens your question)	FACT: the 2026 review that found lean mass alone not enough without strength (shown earlier) [7]
	c other	say what
	bears on every option	FACT: records 4, 5 and 8 to 10 are heart-failure trials whose only outcome in your form is a walking test, studied as a heart-failure outcome: “could potentially improve congestion symptoms and exercise capacity in patients with HF” [6]. DRAFT, my reading: such a study cannot show which kind of weight is lost, and in heart failure a walking test may reflect the heart as much as muscle
2. Batch 1	include, exclude or unsure, for each record	the table above
3. PubMed 25166023 and 25268802	the same paper: yes or no	FACT: same title and year, nearly the same abstract, different DOIs; PubMed tags the second as a published erratum (a correction notice) [6]

Question 1 may change the research question you approved; nothing changes unless you agree.

Added by me (say which to drop)	
Order	records naming more of the three marks first, then oldest first; batches of 10
Register entries	after the articles; one that lists an article as its own report is screened beside it

Sources: [1] L000459 PubMed, run 2026-10-25; [2] L000468 ClinicalTrials.gov, run 2026-10-25; [3] L000492 computed from the saved records, run 2026-10-25; [4] L000494 computed from the saved records, run 2026-10-25; [5] L000495 computed from the saved records, run 2026-10-25; [6] L000491 PubMed, run 2026-10-25; [7] L000129 PubMed, run 2026-10-25

Next: I record your decisions and show batch 2; with 1 b, I first show your question again to certify.

Your turn, Uday: these decisions start your included studies. 1. Strength or physical performance alone: a, b or c? 2. Records 1–10: include, exclude or unsure for each? 3. PubMed 25166023 and 25268802: one paper, yes or no? For example: "1 a, 2: 1 include, 2–3 exclude, 4–10 unsure, 3 yes" or "1 c: say what, 2: 1–10 as suggested, 3 no".

Opened beside the chat: stir-l4-workspace/data/derived/step9_batch1.csv

C **Search Strategy certified: 7 of 8.** Then the records are gathered for pilot screening.

UDAY (SIMULATED USER) session 3, message 37

1 a. include 1, 6, 9. unsure 4, 5, 8, 10. exclude 2, 3 exenatide combo, 7 simulation. 3 yes

STIR-L4 ASSISTANT session 3, message 38

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded batch 1: 3 include, 4 unsure, 3 exclude [1]; the two 2014 records are one paper, leaving 2,565 to screen [2]. Your question, search strategy and feasibility decision still stand: the search found your known 2014 study (shown earlier); the stopping rule waits on records the body-weight phrases added, none screened yet [1].

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 done · 10 now · 11–12 open

With first studies in, the methods step opens; screening continues beside it.

DRAFT: batch 2, by my order; each decision is yours [3]

#	Record	What the abstract says	My suggestion (include = read in full), and why
11	PubMed 33357505 (2021): Empire HF Renal, heart failure, against placebo	“randomly assigned to empagliflozin or placebo, including 105 (88%) without diabetes”	unsure: some have diabetes; whether those without are reported apart, the full text decides. It estimates extracellular and plasma volume
12	PubMed 33879970 (2021): dapagliflozin against teneligliptin, fatty liver	against another drug (shown earlier)	exclude: the comparison; it stays a known study for testing the search
13	PubMed 34097062 (2021): five drug groups, polycystic ovary syndrome	“Nondiabetic women (n = 119 … body composition evaluated by dual-energy x-ray absorptiometry (DXA)”	include: exenatide with and without dapagliflozin is a pair your question admits; whether its results are given apart, the full text decides
14	PubMed 34897070 (2021): systematic review and meta-analysis	“Randomized controlled trials published in English that compared SGLT2 inhibitors with placebo”	exclude: a review, not a trial or cohort
15	PubMed 34716213 (2022): brain insulin sensitivity in prediabetes, against placebo, 8 weeks	“with prediabetes were randomized to receive 25 mg empagliflozin every day or placebo”	unsure: the abstract names liver fat, which your form leaves out; whether whole-body fat was measured, the full text decides
16	PubMed 35391843 (2022): blood pressure in people without diabetes, against placebo	“placebo-controlled study … in untreated normotensive non-diabetic subjects … suggesting plasma volume contraction”	unsure: plasma volume is inferred from hormone levels, not measured; whether any part of the weight was, the full text decides
17	PubMed 35478433 (2022): DAPASALT and DIAMOND, kidney disease without diabetes	“placebo-controlled double-blind crossover trial (DIAMOND) … markers of volume status in 53 patients with CKD”	unsure: DIAMOND fits by design; whether its markers include body water, not only hormones, the full text decides. DAPASALT seems to have no comparison group
18	PubMed 36266774 (2022): DAPA-DIET cohort	“Participants (n = 18) with type 2 diabetes (T2D) and class 3 obesity”	exclude: everyone has diabetes, and there is no comparison group
19	PubMed 37313125 (2023): case report, ketoacidosis in an older man	“We present a case of an elderly male”	exclude: a case report
20	PubMed 37436962 (2023): a liver hormone in prediabetes, from a randomized trial	“115 individuals with prediabetes … compared with control (habitual lifestyle)”	unsure: fat and lean mass appear only as links to the hormone; whether their change is given by group, the full text decides

Full titles are beside the chat; say "show all" here.

Added by me (say which to drop)	
Reports of one trial	records 4 and 9 both name the EMPA-TROPISM trial, and 5 and 10 the EMPERIAL trials [4]: each pair gets one study key, so each trial counts once; your decision on each record stands
The 2014 pair	PubMed 25166023 kept as the paper; 25268802, which PubMed tags as a correction notice, linked to it [2]
Register entries	screened on their own, after the articles, not beside them: NCT05580510 and NCT06668389, entries of other trials in congenital heart disease, list records 8 and 9 among the 5 and 19 papers they give as their results [5]

Sources: [1] L000518 computed from your screening decisions, run 2026-10-25; [2] L000517 computed from the saved records, run 2026-10-25; [3] L000491 PubMed, run 2026-10-25; [4] L000519 computed from the saved records, run 2026-10-25; [5] L000520 ClinicalTrials.gov, run 2026-10-25

Next: I record your decisions and show batch 3, the last of the 26 likeliest records (shown earlier); then the methods.

Your turn, Uday: this decides what you read in full. Records 11–20: include, exclude or unsure for each? Edit my added rows too.
For example: "11–20 as suggested" or "include 13, 15; unsure 11, 16, 17, 20; exclude the rest".

Opened beside the chat: stir-l4-workspace/data/derived/step9_batch2.csv

> P **Pilot screening: the person decides.** Uday screens each record shown; the assistant only suggests, as a DRAFT, and points out duplicates. Pilot screening tests the eligibility criteria and the search; it is not the review's screening and stays out of the final report. (The status line shows Step 10 here, a slip fixed before release.)

UDAY (SIMULATED USER) session 3, message 39

unsure 11, 13, 15, 17, 20. exclude 12 other drug, 14 review, 16 not measured, 18 diabetes, 19 case report

STIR-L4 ASSISTANT session 3, message 40

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 2 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your batch 2 decisions; so far 20 screened: 3 include, 9 unsure, 8 exclude [1]. Your feasibility decision, question and search strategy still stand: 3 includes so far fit “very few trials” [1], and the stopping rule still waits on the records the body-weight phrases added (shown earlier).

DRAFT: batch 3, the last of the likeliest records; each decision is yours [2]

#	Record	What the abstract says	My suggestion (include = read in full), and why
21	PubMed 38219335 (2024): semaglutide, empagliflozin or both; kidney sodium by magnetic resonance	“in 65 participants with type 2 diabetes and high risk of cardiovascular disease”	exclude: everyone treated has diabetes; people without it appear only in a separate, untreated comparison
22	PubMed 38365073 (2024): Empire Prevent trial program, its plan (protocol)	“but without diabetes or established HF … randomized 1:1 to 180 days treatment with empagliflozin 10 mg daily or placebo” (HF, heart failure)	unsure, as you chose for the plan in record 4: no results of its own; it plans record 26's trial and a second one on physical capacity
23	PubMed 39361103 (2024): your known study, empagliflozin with topiramate	topiramate in one group only (shown earlier)	exclude: as records 2 and 3; it stays a known study for testing the search
24	PubMed 40994705 (2025): SiRENA, fluid balance, crossover against placebo	“three identical trials examining patients with T2D with and without CKD and non-diabetic CKD” (T2D, type 2 diabetes; CKD, chronic kidney disease)	unsure: one trial is without diabetes; whether its results are given apart, the full text decides. It measured body composition and water outside cells
25	PubMed 41347301 (2026): dapagliflozin at cardiovascular risk, against placebo	“Patients at Cardiovascular Risk With or Without Type 2 Diabetes … body fluid composition”	unsure: whether those without diabetes are reported apart, the full text decides. It measured fluid outside cells
26	PubMed 41507062 (2026): Empire Prevent Metabolic trial, against placebo	“non-diabetic patients with body mass index (BMI) >28kg/m2 and risk of HF … 180-days empagliflozin 10 mg or placebo”	include: water outside cells is a main outcome, but “estimated”: whether that counts as measured (your reason for record 16), the full text decides. The heart fat it names is not body fat

Full titles are beside the chat; say "show all" to see them here.

Added by me (say which to drop)	
Records 22 and 26	one study key: 26's title names “the Empire Prevent Metabolic Trial”, whose register entry lists 22 as its report [3]; a later record in your order, “a substudy from the Empire Prevent Metabolic trial”, joins them [4]
From batch 2, not yet answered	one study key for records 4 and 9, and for 5 and 10; the 2014 pair; register entries screened after the articles (shown earlier)

Sources: [1] L000540 computed from your screening decisions, run 2026-10-25; [2] L000491 PubMed, run 2026-10-25; [3] L000546 computed from the saved ClinicalTrials.gov entries, run 2026-10-25; [4] L000545 PubMed, run 2026-10-25

Next: I record your decisions, then open the methods: the type of review and whether results can be pooled, with published guidance for your kind of question; screening goes on in batches beside it.

Your turn, Uday: these close the records most likely to fit. 1. Records 21–26: include, exclude or unsure for each? 2. My added rows: keep all, or say which to drop?
For example: "21–26 as suggested; keep all" or "include 26; unsure 22, 24, 25; exclude 21, 23; drop register entries last".

Opened beside the chat: stir-l4-workspace/data/derived/step9_batch3.csv

UDAY (SIMULATED USER) session 3, message 41

include 26. unsure 22, 24, 25. exclude 21 diabetes, 23 topiramate combo. keep all

STIR-L4 ASSISTANT session 3, message 42

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded batch 3 and your kept rows: 26 screened, 4 include, 12 unsure, 10 exclude [1]. Feasibility, question and search strategy still stand: 4 includes fit “very few trials”; the stopping rule awaits the 1,287 records added by body-weight words [1].

FACT: PubMed, methods guidance in titles: 116 records [5], 146 with the guideline's short name [8].
FACT: abstracts of your 4 included studies name fat (2), lean (1), water (2), function (1) [11]; 2,539 records remain to screen [12].

Open: how your review is done.

Search 6: methods guidance (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
(("scoping review"[ti] OR "scoping reviews"[ti]) AND ("systematic review"[ti] OR "systematic reviews"
[ti]) AND (guidance[ti] OR choosing[ti] OR choose[ti] OR difference*[ti] OR "when to"[ti])) OR
("synthesis without meta-analysis"[ti] OR ("few studies"[ti] AND (meta-analy*[ti] OR synthesis[ti]))) OR
(("single screening"[ti] OR "single-reviewer"[ti] OR "single reviewer"[ti] OR "single screener"[ti] OR
"one reviewer"[ti] OR "dual screening"[ti] OR "double screening"[ti]) AND (screening[ti] OR selection[ti]
OR review*[ti]))
```

Block	Covers	From	How it searches	Records
1	choosing between a systematic and a scoping review	methods question 1	words in the title, with truncation	5 [2]
2	synthesis without pooling, or with very few studies	methods question 2	words in the title, with truncation	17 [3]
3	screening by one person rather than two	methods question 3	words in the title, with truncation	94 [4]
All	combined	—	blocks joined by OR	**116** [5]

Search 6b: search 6 in parentheses, then:

```
OR (SWiM[ti] AND (synthesis[ti] OR "meta-analysis"[ti]))
```

Block	Covers	From	How it searches	Records
4, added	the reporting guideline for synthesis without meta-analysis, by its short name (SWiM); it also matches records on swimming	methods question 2	words in the title	30 [7]
All	combined	—	blocks joined by OR	**146** [8]

Check	Result
Phrases PubMed ignored	`"single screener"` and `"synthesis without meta-analysis"`: PubMed found no such phrase [6]; search 6b adds the guideline's short name instead
Not searched yet	OpenAlex (next session); guidance published outside PubMed

FACT, from [9][10], chosen by me:

#	Record	What its abstract says
1	PubMed 30453902 (2018): choosing a systematic or a scoping review	scoping reviews suit reviews “where the purpose of the review is to identify knowledge gaps, scope a body of literature”
2	PubMed 42702311 (2026): scoping reviews as precursors to systematic reviews	one kind: “a complementary scoping review run alongside an SR to contextualize sparse evidence” (SR, systematic review)
3	PubMed 29504289 (2018): evidence synthesis with very few (2–4) studies	“heterogeneity cannot be reliably estimated if only very few studies are available” (heterogeneity: how much results differ between studies)
4	PubMed 34550585 (2022): meta-analyses with very few studies	when estimation methods disagree, “it might be the best solution to refrain from a quantitative meta-analysis”
5	PubMed 31948937 (2020): SWiM, reporting guideline for synthesis without meta-analysis	for reviews “in which alternative synthesis methods to meta-analysis of effect estimates are used”
6	PubMed 31972274 (2020): randomized trial, one screener against two	“single-reviewer abstract screening missed 13% of relevant studies … dual-reviewer abstract screening missed 3%”
7	PubMed 31253092 (2019): methods review, one screener against two	missed studies: “3% for the 6 experienced reviewers … 13% for the 3 reviewers with less experience”
8	PubMed 41942890 (2026): studies one screener wrongly excludes	“nonrandomized studies of interventions (NRSI) constituted the highest proportion of false exclusions”
9	PubMed 41484785 (2026): AI tools as second screeners, a study plan	“yet empirical evidence comparing their performance is limited”

Say "show all" for the other 137, beside the chat.

Still open	Option	What bears on it
1. Type of review	a systematic review: what the studies found, and how far each can be trusted	—
	b scoping review: which studies exist, and where they are missing	FACT: row 1 [9]
	c both: b for everything, then a for the parts with enough studies	FACT: row 2 [9]
	d other	say what
	bears on every option	JUDGMENT (your certified purpose, summarized): research gaps; finding the right question; a systematic review on this topic, its results to decide whether a full review is worth it. DRAFT: with b, check that this purpose still holds

Still open	Option	What bears on it
2. Pooling results	a meta-analysis for each part of the weight with enough studies (say how many); the rest described	FACT: rows 3 and 4 [9]
	b no meta-analysis: results tabled and described part by part, reported as row 5 sets out	FACT: row 5 [10]
	c other	say what
	bears on every option	FACT: your included studies so far, above [11]; 12 unsure records still to read in full [1]. JUDGMENT: your case lists as uncertain whether one answer fits every reason for treatment (shown earlier)
3. Who checks screening, reading and judging	a you alone, with my suggestion beside each record, as now; the protocol says so	FACT: row 6 [9]
	b a, and I check again every record you exclude and flag any that may fit; you decide each	FACT: row 9 [9]
	c a second person checks a share of your decisions (say who and how many)	FACT: row 7 [9]
	d other	say what
	bears on every option	FACT: 2,539 records left [12]; row 8 [9]: your question admits cohorts, which are nonrandomized. JUDGMENT: about 5 hours a week, alone (summarized)

Sources: [1] L000565 computed from your screening decisions, run 2026-10-25; [2] L000571 PubMed, run 2026-10-25; [3] L000572 PubMed, run 2026-10-25; [4] L000573 PubMed, run 2026-10-25; [5] L000574 PubMed, run 2026-10-25; [6] L000576 computed from the saved searches, run 2026-10-25; [7] L000577 PubMed, run 2026-10-25; [8] L000578 PubMed, run 2026-10-25; [9] L000575 PubMed, run 2026-10-25; [10] L000579 PubMed, run 2026-10-25; [11] L000569 computed from the saved records, run 2026-10-25; [12] L000566 computed from the saved records, run 2026-10-25

Next: I draft your methods and check them against your time.

Your turn, Uday: these shape the methods and your time. 1. Type: a, b, c or d? 2. Pooling: a (say how many), b or c? 3. Who checks: a, b, c or d?
For example: "1 c, 2 a: at least 3, 3 b".

Opened beside the chat: stir-l4-workspace/data/derived/step10_guidance.csv

UDAY (SIMULATED USER) session 3, message 43

pause

STIR-L4 ASSISTANT session 3, message 44

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Paused at your request. Saved this session's summary and a packed copy of your plan. Nothing you have approved is affected.

Session summary (saved and open beside the chat; say "show all" for the certified lines in full):

```
# Session summary (S3, 2026-10-25)
```

```
Step 10 of 12: Methodology | Certified 7 of 8 | Waiting on you: 3 (Uday) | Focus: which kind of weight is
lost: fat, muscle or body water, within…

Topic: the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug
is given for, any country; Asian people as a subgroup
Focus: which kind of weight is lost: fat, muscle or body water, within all adults without diabetes,
whatever the drug is given for, any country; Asian people as a subgroup

Settled this session:
- Research Question: certified by Uday; reason: this is the question (full line in the saved summary)
- Paper as an Object: certified by Uday; reason: covers my outcomes (full line in the saved summary)
- Literature as a Tensor: certified by Uday; reason: properties are fine (full line in the saved summary)
- Registering Evidence Dynamically: certified by Uday; reason: atlas is fine (full line in the saved
summary)
- Search Strategy: certified by Uday; reason: finds both known studies (full line in the saved summary)

Reopened:
- nothing

Still open: Methodology

Next questions:
- Uday (open now): 1. type of review: a systematic, b scoping, c both, d other; 2. pooling: a meta-
analysis for each part of the weight with enough studies (say how many), b no meta-analysis, results
tabled and described part by part, c other; 3. who checks screening, reading and judging: a you alone,
with my suggestion beside each record; b a, and I look again at every record you exclude; c a second
person checks a share; d other
- Then: I draft your methods and check them against your time, for you to certify
- Screening, in batches of 10 beside the methods: 2,539 records left (1,999 PubMed, 540
ClinicalTrials.gov); the full texts of the 4 included and 12 unsure records to read
- Stopping rule: of the 1,287 records the body-weight phrases added, 0 screened so far
- Next session: the OpenAlex search, in titles and abstracts with the body-weight phrases, and its known-
study check, once its free daily allowance has reset; reading past its first 100 results needs a free
access code
- Uday, next session: search the WHO trial portal by hand with the drug names; I will give the steps

Notes: Session 2 left no summary; this session began from the saved plan and record. Every saved search
result was checked against the record (133 checked). One set of PubMed records first showed a mismatch
only because its 11 saved parts were read out of order; read in order, all 11 parts and its 2,021 records
match. Not run yet: OpenAlex (refused today: its free daily allowance was used up) and the WHO trial
portal (Uday, by hand). Changed this session: the saved search results, screening decisions and counts,
the working helpers, the features to count, the record of every step, the step calculations, your plan,
the extraction form, the evidence map.

Searches this session: 27
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	3 of 3 certified	—
8. Search Strategy	search strings and stopping rule	certified by Uday	—
9. Scoping and pilot screening	first included studies	26 screening decisions	—
10. Methodology	methods for the protocol	now	Uday
11. Protocol	protocol and registration	open	—

Step	Settles	Status	Waiting on
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

When convenient: if you like, type a contact email for PubMed searches (PubMed uses it only to reach us if our searches cause a problem); I keep it private.

To save: keep this copy of your plan and record, `stir-l4-workspace_2026-10-25_S3.zip`, beside your planning folder; in a new chat, give it to me with the prompt and helper kit.

Next: when you return, I read the saved plan and record, show where we are and continue the methods.

Your turn, Uday: nothing more today. When you return, your methods rest on the three open answers in the summary: 1. Type of review: a, b, c or d? 2. Pooling: a (say how many), b or c? 3. Who checks: a, b, c or d? For example: "1 c, 2 a: at least 3, 3 b".

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_S3_summary.md

Part 2. Session 4

Session 4. The methods are settled and certified; a first protocol is drafted as a table and certified after the methods are reopened; registration is recorded as not submitted; the test script's commitment closes the steps.

Messages 1 to 4 of each session (the tool test and the calibration quiz, scored 16 of 16) are not shown.

UDAY (SIMULATED USER) session 4, message 5

passed 16/16. continue

STIR-L4 ASSISTANT session 4, message 6

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Read your saved plan, record and last summary. OpenAlex (a free index of research papers) refused your planned search again, as last session, so it waits. Nothing you have approved is affected. Say if anything looks wrong.

JUDGMENT (Uday, summarized)	
Topic	the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup
Focus	which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

Step	**Settles**	**Status**	**Waiting on**
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	3 of 3 certified	—
8. Search Strategy	search strings and stopping rule	certified by Uday	—
9. Scoping and pilot screening	first included studies	26 screening decisions	—
10. Methodology	methods for the protocol	now	Uday
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.
The line at the top of each reply shows the step, the decisions you have approved (out of 8), the answers or approvals the current step is waiting for, and your research focus.
Quick commands, in any wording: "where are we" · "next" (never accepts or certifies) · "why" (how this answer affects your other decisions) · "show <item>" · "skip" (the question stays open) · "back" (reopens only if the team agrees) · "pause" (summary and stop).

Open since last session (none pre-selected)	Option	What bears on it
1. Type of review	a systematic review: what the studies found, and how far each can be trusted	—
	b scoping review: which studies exist, and where they are missing	FACT: a 2018 guide suits scoping reviews “where the purpose of the review is to identify knowledge gaps, scope a body of literature” [1]
	c both: b for everything, then a for the parts with enough studies	FACT: a 2026 guide describes “a complementary scoping review run alongside an SR to contextualize sparse evidence” (SR, systematic review) [1]
	d other	say what
	bears on every option	JUDGMENT (your certified purpose, summarized): research gaps; finding the right question; a systematic review on this topic, its results to decide whether a full review is worth it. DRAFT: with b, check that this purpose still holds
2. Pooling results	a meta-analysis for each part of the weight with enough studies (say how many); the rest described	FACT: with very few studies, “heterogeneity cannot be reliably estimated if only very few studies are available” (2018) [1]; “it might be the best solution to refrain from a quantitative meta-analysis” (2022) [1]
	b no meta-analysis: results tabled and described part by part, reported by the guideline for synthesis without meta-analysis (SWiM)	FACT: SWiM is for reviews “in which alternative synthesis methods to meta-analysis of effect estimates are used” [2]
	c other	say what
	bears on every option	FACT: of the 26 records you screened: 4 include, 12 unsure (to read in full), 10 exclude [3]. By abstract, your 4 included studies name fat in 2, lean mass in 1, body water in 2, strength or function in 1 (named, not yet measured) [4]. JUDGMENT: your case lists as uncertain whether one answer fits every reason for treatment
3. Who checks screening, reading and judging	a you alone, with my suggestion beside each record, as now; the protocol says so	FACT: a 2020 randomized trial: “single-reviewer abstract screening missed 13% of relevant studies … dual-reviewer abstract screening missed 3% of relevant studies” [1]
	b a, and I check again every record you exclude and flag any that may fit; you decide each	FACT: a 2026 study plan on AI tools as second screeners: “yet empirical evidence comparing their performance is limited” [1]
	c a second person checks a share of your decisions (say who and how many)	FACT: a 2019 methods review, studies missed by one screener: “3% for the 6 experienced reviewers … 13% for the 3 reviewers with less experience” [1]
	d other	say what
	bears on every option	FACT: 2,539 records left to screen, after duplicates (PubMed 1,999, ClinicalTrials.gov 540) [5]; a 2026 analysis: “nonrandomized studies of interventions (NRSI) constituted the highest proportion of false exclusions” [1]; your question admits cohort studies, which are nonrandomized. JUDGMENT: about 5 hours a week, alone (summarized)

Sources: [1] L000575 PubMed, run 2026-10-25; [2] L000579 PubMed, run 2026-10-25; [3] L000565 computed from your screening decisions, run 2026-10-25; [4] L000569 computed from the saved records, run 2026-10-25; [5] L000566 computed from the saved records, run 2026-10-25

Next: I draft your methods against your time.

Your turn, Uday: these shape your methods and hours. 1. Type: a, b, c or d? 2. Pooling: a (say how many), b or c? 3. Who checks: a, b, c or d?
For example: "1 c, 2 a: at least 3, 3 b".

R **Session 4 resumes at Methodology.**

UDAY (SIMULATED USER) session 4, message 7

2 a: meta-analysis only if 3 or more similar trials, else narrative

STIR-L4 ASSISTANT session 4, message 8

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your pooling choice (summarized): meta-analysis per outcome only with 3 or more similar trials, else narrative. This may change your approved feasibility, as methods take your hours; I check it in the methods draft.

FACT: by abstract, no outcome reaches 3 among your 4 included studies [1]; adding your unsure ones, 13 studies name fat 5, body water 6, strength or function 3, lean mass 2 [1].
DRAFT: so for now all would be narrative; full texts of the unsure ones decide. Similarity (measuring method, design) is not yet judged.

The 13 studies are beside the chat; say "show all" to see them here.

Sources: [1] L000597 computed from your screening decisions and the saved records, run 2026-10-25

Next: Then I draft your methods against your 5 hours a week.

Your turn, Uday: these complete your methods. 1. Type of review (asked again): a, b, c or d? 2. Who checks (asked again): a, b, c or d? 3. Your rule names trials; your question also admits cohort studies. Count toward 3: a trials only, b both, c other?
For example: "1 c, 2 b, 3 a".

Opened beside the chat: stir-l4-workspace/data/derived/step10_pooling_rule.csv

F **Methods feed back to feasibility.** Each methods choice is checked against the hours available (flow 23), as the model requires.

UDAY (SIMULATED USER) session 4, message 9

solo, single screener, recheck 10% later. note it as a limitation

STIR-L4 ASSISTANT session 4, message 10

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your choice (summarized): you alone as the single screener, a 10% re-check later, and single screening named as a limitation. This may change your approved feasibility, as the re-check adds hours, and through it your other approved decisions.

FACT: your searches found 2,565 records so far, after duplicates; a tenth is 257 [1]. OpenAlex and the WHO trial portal may add more.

DRAFT: methods wording, for you to edit

One reviewer (Uday) screens titles and abstracts, reads full texts and judges each study alone. Later, he screens a random tenth of his decisions again (my suggestion: drawn by code, without seeing his first decision) and reports how often both passes agree (my suggestion). Limitation: a single screener; in a 2020 trial, single-reviewer abstract screening missed more relevant studies than dual-reviewer screening [2].

DRAFT: the passages shown earlier compare one screener with two; none tests a re-check by the same person, so how much yours would catch is unknown.

Sources: [1] L000604 computed from the saved records, run 2026-10-25; [2] L000575 PubMed, run 2026-10-25

Next: With your answers, I look for evidence on re-checking one's own screening and draft your methods against your 5 hours a week.

Your turn, Uday: these complete your methods. 1. Type of review (still open): a systematic, b scoping, c both, d other? 2. Count toward your 3 for pooling: a trials only, b trials and cohort studies, c other? 3. Which tenth to re-check: a all your decisions, b your exclusions only, c other?
For example: "1 c, 2 a, 3 b".

UDAY (SIMULATED USER) session 4, message 11

1 c, 2 a, 3 a

STIR-L4 ASSISTANT session 4, message 12

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your three choices (in the draft). This may change your approved feasibility (methods take hours) and so your other approved decisions.

FACT: PubMed, no filter: 88 [4], 78 [8], 163 [14] records.
Open for your focus: any part reaching 3 similar trials.

Tags: [tiab] title or abstract; [ti] title; * any ending.

Search 1: your re-check of your own screening (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("systematic review*"[tiab] OR "scoping review*"[tiab] OR "evidence synthes*"[tiab]) AND ("abstract screening"[tiab] OR "citation screening"[tiab] OR "study selection"[tiab] OR "screening decision*"[tiab] OR "full-text screening"[tiab] OR "title screening"[tiab] OR "eligibility assessment*"[tiab]) AND ("intra-rater"[tiab] OR intrarater[tiab] OR "intra-reviewer"[tiab] OR intrareviewer[tiab] OR "intra-observer"[tiab] OR intraobserver[tiab] OR "test-retest"[tiab] OR rescreen*[tiab] OR "re-screening"[tiab] OR "re-screened"[tiab] OR "same reviewer"[tiab] OR "same reviewers"[tiab] OR "self-consistency"[tiab])
```

Block	Covers	From	How it searches	Records
1	reviews	methods: your re-check of your own screening	title/abstract words, with truncation	516,997 [1]
2	screening records for a review	methods: your re-check of your own screening	title/abstract words, with truncation	33,272 [2]
3	the same person screening twice	methods: your re-check of your own screening	title/abstract words, with truncation	67,160 [3]
All	combined	—	blocks joined as shown	**88** [4]

Search 2: the hours each review task takes (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("systematic review*"[ti] OR "scoping review*"[ti] OR "evidence synthes*"[ti] OR "meta-analys*"[ti]) AND (screening[ti] OR "data extraction"[ti] OR "study selection"[ti] OR conduct*[ti] OR production[ti] OR
```

```
producing[ti] OR complete[ti] OR "risk of bias"[ti]) AND (time[ti] OR hours[ti] OR workload[ti] OR
"resource use"[ti] OR resources[ti] OR effort[ti] OR "person-hours"[ti] OR minutes[ti] OR duration[ti])
```

Block	Covers	From	How it searches	Records
1	reviews	methods: the hours each review task takes	words in the title, with truncation	519,891 [5]
2	review tasks	methods: the hours each review task takes	words in the title, with truncation	701,556 [6]
3	time or workload	methods: the hours each review task takes	words in the title	473,265 [7]
All	combined	—	blocks joined as shown	**78** [8]

Search 3: guidance for your kind of question: trials and cohort studies, a scoping part a… (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("risk of bias"[ti] AND (randomised[ti] OR randomized[ti]) AND tool[ti]) OR ("risk of bias"[ti] AND (non-
randomised[ti] OR non-randomized[ti] OR nonrandomized[ti] OR observational[ti] OR exposures[ti] OR
"follow-up studies"[ti]) AND tool[ti]) OR (GRADE[ti] AND ("quality of evidence"[ti] OR "certainty of
evidence"[ti] OR "certainty of the evidence"[ti] OR "rating quality"[ti] OR "rating the quality"[ti])) OR
(PRISMA[ti] AND "scoping reviews"[ti]) OR ("PRISMA 2020"[ti])
```

Block	Covers	From	How it searches	Records
1	risk-of-bias tools for randomized trials	methods: guidance for your kind of question: trials and cohort studies, a scoping part and a systematic part	words in the title	31 [9]
2	risk-of-bias tools for non-randomized studies	methods: guidance for your kind of question: trials and cohort studies, a scoping part and a systematic part	words in the title	14 [10]
3	grading how far the evidence can be trusted (GRADE)	methods: guidance for your kind of question: trials and cohort studies, a scoping part and a systematic part	words in the title	77 [11]
4	reporting a scoping review	methods: guidance for your kind of question: trials and cohort studies, a scoping part and a systematic part	words in the title	4 [12]
5	reporting a systematic review	methods: guidance for your kind of question: trials and cohort studies, a scoping part and a systematic part	words in the title	46 [13]
All	combined	—	blocks joined as shown	**163** [14]

Not searched: OpenAlex (refused today). PubMed ignored "rating the quality" [15].

FACT, from [16]	Title	Year	Says	Bears on
1	Neuroimmune and Neurovascular Remodeling after Perinatal Hypoxia:…	2026	“a randomly selected 10% subset re-screened after a 2-week interval to enhance internal consistency”	re-check
2	Evaluating the accuracy of full text and serial screening by one…	2026	“Compared with two reviewers, the sensitivity of single reviewer full-text screening ranged from 62% to 88%”	limitation
3	The revised Cochrane risk of bias tool for randomized trials (RoB 2)…	2020	“IRR was slight for overall judgment … Mean time to apply the tool was 28 minutes (standard deviation 13.4) per study outcome”	judging; hours
4	Risk-of-bias assessment using Cochrane's revised tool for randomized…	2023	“Staff resource averaged 358 minutes per study (SD 183)”	hours
5	RoB 2: a revised tool for assessing risk of bias in randomised trials	2019	“RoB 2: a revised tool for assessing risk of bias in randomised trials”	judging trials
6	ROBINS-I: a tool for assessing risk of bias in non-randomised studies…	2016	“a new tool for evaluating risk of bias … from studies that did not use randomisation”	judging cohort studies

FACT, from [16]	Title	Year	Says	Bears on
7	Core GRADE 4: rating certainty of evidence-risk of bias, publication…	2025	“randomised controlled trials begin as high certainty evidence and non-randomised studies of interventions (NRSI) as low certainty”	trust
8	Pooling of cohort studies and RCTs affects GRADE certainty of…	2022	“whether or not to pool these bodies-of-evidence requires proper guidance”	pooling trials only
9	PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and…	2018	“The final checklist contains 20 essential reporting items and 2 optional items”	reporting the scoping part
10	The PRISMA 2020 statement: an updated guideline for reporting…	2021	“the PRISMA 2020 27-item checklist”	reporting the systematic part

Say "show all" to see the other 319 here.

DRAFT: your methods, for you to edit (also beside the chat)	From
Type. A scoping review maps every study that meets your question: which parts of the weight were measured, how, in whom, and where trials are missing. A systematic review then judges and sums up the parts with enough studies (open: question 1).	you (summarized)
Sources. As you approved: PubMed, ClinicalTrials.gov, OpenAlex (still to run) and the WHO trial portal (you, by hand); Embase and CINAHL named as not searched.	your approved search
Selecting studies. You alone screen titles and abstracts and read full texts; you decide each record, with my suggested decision beside it (my suggestion: as now). Later you screen a random tenth of all your decisions again (257 so far [19]), drawn by code, without seeing your first decision, about 2 weeks later, and report how often the two passes agree (my suggestions; the 2-week gap as one 2026 review did, #1). My reading of the 88 abstracts: none measures what such a re-check catches.	you; [16]
Collecting data. You alone fill the extraction form you approved, with my suggested values and their exact passages beside each (my suggestion). The form needs no change for these methods.	your approved form
Judging each study (systematic part only). Risk of bias for each outcome: trials with RoB 2, the revised tool for randomized trials (#5); cohort studies with ROBINS-I, its counterpart for non-randomized studies (#6) (my suggestion).	[16]
Pooling. For each part of the weight, a meta-analysis only if 3 or more similar randomized trials report it; otherwise results tabled and described part by part, reported with SWiM [20]. How "similar" is judged, and how to pool: settled if a part reaches 3 (my suggestion). Cohort studies do not count toward the 3; how they are used: question 2.	you; [20]
FACT: counting randomized trials only (by title, abstract or record type), your 3 included trials name fat 1, body water 1, function 1; with your unsure ones, 12 trials: fat 4, body water 5, function 3, lean 1 [18].	[18]
Trust in each result. Certainty for each part with GRADE, a standard way to rate how far evidence can be trusted, where trials start high and cohort studies low (#7) (my suggestion).	[16]
Asian people. Results for Asian participants described apart.	your approved question
Reporting. Scoping part with PRISMA-ScR, systematic part with PRISMA 2020: reporting checklists for each kind (#9, #10) (my suggestion).	[16]
Limitations. One person screens, reads and judges. One reviewer's full-text screening missed relevant studies in a 2026 study (#2); raters agreed only slightly on overall risk of bias in 2020 (#3).	you; [16]

DRAFT: time these methods take one person (my estimate [19])	Hours
Screening the 2,539 records left	10.1
Re-checking 257 of your decisions	1.0
Reading the 16 full texts so far	0.7
Total	11.8: about 2.4 weeks at your 5 hours a week

DRAFT: time these methods take one person (my estimate [19])	Hours
Then, per study in the systematic part	about 36 minutes to extract, plus 28 per outcome to judge bias
FACT behind it: in a 2021 case study, two beginners' screening times added up to 4:40 hours for 586 titles and abstracts and 3:29 for 40 full texts; extracting 16 articles took 9:42 [17]; judging bias took 28 minutes per outcome (#3) and, for two reviewers, 358 per study (#4) [16]	
Not counted: records OpenAlex and the WHO portal may add, later full texts, synthesis, writing	
So the methods fit your hours, and your approved decisions still stand unless you disagree. Your own pace may differ: this rests on few studies.	

Sources: [1] L000612 PubMed, run 2026-10-25; [2] L000613 PubMed, run 2026-10-25; [3] L000614 PubMed, run 2026-10-25; [4] L000615 PubMed, run 2026-10-25; [5] L000616 PubMed, run 2026-10-25; [6] L000617 PubMed, run 2026-10-25; [7] L000618 PubMed, run 2026-10-25; [8] L000619 PubMed, run 2026-10-25; [9] L000620 PubMed, run 2026-10-25; [10] L000621 PubMed, run 2026-10-25; [11] L000622 PubMed, run 2026-10-25; [12] L000623 PubMed, run 2026-10-25; [13] L000624 PubMed, run 2026-10-25; [14] L000625 PubMed, run 2026-10-25; [15] L000626 computed from the saved PubMed responses, run 2026-10-25; [16] L000627 PubMed, run 2026-10-25; [17] L000628 PubMed Central, run 2026-10-25; [18] L000629 computed from your screening decisions and the saved records, run 2026-10-25; [19] L000630 computed from the saved texts and records, run 2026-10-25; [20] L000579 PubMed, run 2026-10-25

Next: your methods, to certify.

Your turn, Uday: to settle your methods. 1. Systematic part covers parts with: a any trial, b 3 trials, c other? 2. Cohort studies: a judged for bias, b only mapped, c other? 3. Draft: edits or "accept"?
Example: "1 a, 2 b, 3 accept".

Opened beside the chat: stir-l4-workspace/protocol/methods_draft_v1.md

UDAY (SIMULATED USER) session 4, message 13

1 b, 2 a, 3 accept

STIR-L4 ASSISTANT session 4, message 14

Step 10 of 12: Methodology | **Certified 7 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your choices (summarized) and your acceptance of the draft; both are in version 2 below. This may change your approved feasibility, as judging cohort studies takes hours, and through it your other approved decisions.

FACT: PubMed, no filter: 379 records on the time to judge one study with ROBINS-I [3].
Open for your focus: any part reaching 3 similar trials.

Search 1: the time it takes to judge a cohort study with ROBINS-I (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("ROBINS-I"[tiab] OR "ROBINS I"[tiab] OR "risk of bias in non-randomized studies of interventions"[tiab] OR "risk of bias in non-randomised studies of interventions"[tiab] OR "risk of bias in nonrandomized studies of interventions"[tiab]) AND (minutes[tiab] OR hours[tiab] OR "time to apply"[tiab] OR "time to complete"[tiab] OR "time taken"[tiab] OR "time required"[tiab] OR "time needed"[tiab] OR "time-consuming"[tiab] OR workload[tiab] OR burden[tiab])
```

Block	Covers	From	How it searches	Records
1	ROBINS-I, the risk-of-bias tool for non-randomized studies	methods: the time it takes to judge a cohort study with ROBINS-I	title/abstract words	4,966 [1]

Block	Covers	From	How it searches	Records
2	time or effort to apply it	methods: the time it takes to judge a cohort study with ROBINS-I	title/abstract words	1,385,918 [2]
All	combined	—	blocks joined as shown	**379** [3]

Not searched: OpenAlex (refused today). PubMed ignored "time to apply" [4].

FACT, from [5]	Title	Year	Says	Bears on
1	Risk of bias in nonrandomized studies of interventions showed low…	2019	“Mean time to apply the tool was 27.8 minutes (SD 12.6) per study … We found ROBINS-I difficult and demanding”	hours; judging alone
2	Methodologically rigorous risk of bias tools for nonrandomized…	2020	“time taken to apply the tool and reach a consensus … For ROBINS-I … evaluator burden was 48.45 min”	hours (two reviewers)
3	The ROBINS-I and the NOS had similar reliability but differed in…	2021	“The time in assessing a single study by ROBINS-I varied from 7 hours initially to 3 hours”	hours

Say "show all" to see the other 376 here.

DRAFT: what your answers change (version 2 is beside the chat; say "show all" to see it here)	Before	Now
Systematic part covers	the parts with enough studies (open)	only each part of the weight that reaches 3 or more similar randomized trials
Cohort studies	how they are used: open	in the systematic part, judged for bias with ROBINS-I and described; still not counted toward the 3
Time	screening, re-check and full texts so far: 11.8 hours [7]	plus judging cohort studies: FACT: 1 of your 13 included or unsure studies is not called randomized by title, abstract or record type [6]; at the paces above, 0.46 to 7 hours [6]. Still fits your 5 hours a week
Your approved purpose: a systematic review on this topic	holds	holds if a part reaches 3 after full texts; by abstract, none among your included trials does yet [8]. If none does, the review stays scoping only. Your purpose stands unless you say otherwise

Sources: [1] L000640 PubMed, run 2026-10-25; [2] L000641 PubMed, run 2026-10-25; [3] L000642 PubMed, run 2026-10-25; [4] L000643 computed from the saved PubMed response, run 2026-10-25; [5] L000644 PubMed, run 2026-10-25; [6] L000646 computed from your screening decisions and the saved records, run 2026-10-25; [7] L000630 computed from the saved texts and records, run 2026-10-25; [8] L000629 computed from your screening decisions and the saved records, run 2026-10-25

Next: once you certify, I assemble your protocol for you to edit.

```
CERTIFY Methodology | Uday (summarized): type — a scoping review of every study that meets the
question, then a systematic review of each part of the weight that reaches 3 or more similar
randomized trials; sources — PubMed, ClinicalTrials.gov, OpenAlex and the WHO trial portal, as in
the approved search, with Embase and CINAHL named as not searched; selecting studies — Uday alone
screens titles and abstracts and reads full texts, with the assistant's suggested decision beside
each record, and about 2 weeks later screens again a random tenth of all his decisions, drawn by
code and without seeing his first decision, reporting how often the two passes agree; collecting
data — Uday alone fills the approved extraction form, with the assistant's suggested values and
their passages beside each; judging each study, systematic part only — risk of bias for each
outcome, trials with RoB 2 and cohort studies with ROBINS-I, cohort studies judged and described
there; pooling — a meta-analysis for a part only if 3 or more similar randomized trials report it,
cohort studies not counted, otherwise results tabled and described part by part and reported with
SWiM; how 'similar' is judged and how to pool settled if a part reaches 3; trust — GRADE for each
part; Asian people — results described apart; reporting — PRISMA-ScR for the scoping part and
```

```
PRISMA 2020 for the systematic part; limitations — one person screens, reads and judges | reason:
___ | Uday, solo
```

Your turn, Uday: this settles your methods, the last decision before the protocol. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "change: re-check after 1 week".

Opened beside the chat: stir-l4-workspace/protocol/methods_draft_v2.md

UDAY (SIMULATED USER) session 4, message 15

certify, fits a solo review

STIR-L4 ASSISTANT session 4, message 16

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your certification (below). It may change your approved feasibility (methods take hours) and so your other decisions; my estimate fits your 5 weekly hours, so all stand unless you disagree.

```
CERTIFY Methodology | Uday (summarized): type — a scoping review of every study that meets the
question, then a systematic review of each part of the weight that reaches 3 or more similar
randomized trials; sources — PubMed, ClinicalTrials.gov, OpenAlex and the WHO trial portal, as in
the approved search, with Embase and CINAHL named as not searched; selecting studies — Uday alone
screens titles and abstracts and reads full texts, with the assistant's suggested decision beside
each record, and about 2 weeks later screens again a random tenth of all his decisions, drawn by
code and without seeing his first decision, reporting how often the two passes agree; collecting
data — Uday alone fills the approved extraction form, with the assistant's suggested values and
their passages beside each; judging each study, systematic part only — risk of bias for each
outcome, trials with RoB 2 and cohort studies with ROBINS-I, cohort studies judged and described
there; pooling — a meta-analysis for a part only if 3 or more similar randomized trials report it,
cohort studies not counted, otherwise results tabled and described part by part and reported with
SWiM; how 'similar' is judged and how to pool settled if a part reaches 3; trust — GRADE for each
part; Asian people — results described apart; reporting — PRISMA-ScR for the scoping part and
PRISMA 2020 for the systematic part; limitations — one person screens, reads and judges | reason:
fits a solo review | Uday, solo
```

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 done · 10 done · 11 now · 12 open

FACT: PubMed, no filter: 396 records [5]; PRISMA-P, the protocol checklist, has 17 items, 26 with sub-items [9]. Open: any part reaching 3 similar trials.

Search 1: guidance for your protocol: what it must report, how to state the AI's help, wh… (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("PRISMA-P"[ti] OR ("preferred reporting items"[ti] AND protocols[ti])) OR (("artificial intelligence"
[ti] OR AI[ti] OR "large language model*"[ti] OR LLM*[ti] OR generative[ti] OR ChatGPT[ti]) AND
("evidence synthesis"[ti] OR "evidence syntheses"[ti] OR "systematic review*"[ti]) AND (reporting[ti] OR
responsible[ti] OR "position statement"[ti] OR guidance[ti] OR recommendation*[ti] OR disclos*[ti] OR
transparen*[ti] OR checklist[ti])) OR ("scoping review*"[ti] AND (registration[ti] OR registering[ti] OR
registry[ti] OR registries[ti] OR PROSPERO[ti] OR "Open Science Framework"[ti] OR OSF[ti])) OR
(PROSPERO[ti])
```

Block	Covers	From	How it searches	Records
1	reporting a review protocol (PRISMA-P)	protocol: guidance for your protocol: what it must report, how to state the AI's help, where to register	words in the title	14 [1]
2	saying how AI helped a review	protocol: guidance for your protocol: what it must report, how to state the AI's help, where to register	words in the title, with truncation	75 [2]
3	registering a scoping review	protocol: guidance for your protocol: what it must report, how to state the AI's help, where to register	words in the title, with truncation	111 [3]

Block	Covers	From	How it searches	Records
4	PROSPERO, the register of planned reviews	protocol: guidance for your protocol: what it must report, how to state the AI's help, where to register	words in the title	198 [4]
All	combined	—	blocks joined as shown	**396** [5]

Not searched: OpenAlex (refused today); PROSPERO's own website.

FACT, from [6], [7], [8]	Title	Year	Says	Text	Bears on
1	Preferred reporting items for systematic review and…	2015	“PRISMA-P consists of a 17-item checklist intended to facilitate the preparation and reporting of a robust protocol”	abstract [6]	what your protocol covers
2	Preferred reporting items for systematic review and…	2015	“we encourage reviewers preparing any type of review protocol to make use of PRISMA-P as applicable”	full text [7]	your scoping part
3	An assessment of the extent to which the contents of PROSPERO…	2020	“much of the information recommended in PRISMA-P was not stated in PROSPERO registrations”	abstract [6]	a full protocol beside the register entry
4	Position Statement on Artificial Intelligence (AI) Use in…	2025	“Any use of AI or automation that makes or suggests judgements should be fully and transparently reported”	abstract [6]	the AI statement
5	Position Statement on Artificial Intelligence (AI) Use in…	2025	“Decisions to use AI should be considered and reported as part of protocol development.”	full text [8]	the AI statement, now
6	Position Statement on Artificial Intelligence (AI) Use in…	2025	“The name(s) of the AI system(s), tool(s) or platform(s), version(s), and date(s) used.”	full text [8]	question 2
7	Bibliometric, methodological and reporting characteristics of…	2026	“Sharing of prompts and human-validation procedures was insufficient”	abstract [6]	sharing the prompt
8	The nuts and bolts of PROSPERO: an international prospective…	2012	“inclusion in PROSPERO is restricted to systematic reviews of the effects of interventions … for which there is a health-related outcome”	abstract [6]	question 1
9	Contemporary practices in scoping review protocol…	2026	“Most protocols were registered on the Open Science Framework (OSF) (n = 478)”	abstract [6]	question 1

Say "show all" to see the other 390 here.

DRAFT: your protocol, version 1, in the order of the PRISMA-P items [9] (also beside the chat)	From
1a Title. Which kind of weight is lost with SGLT2 inhibitors in adults without diabetes: fat, muscle or body water? A scoping review, with a systematic review of the parts with enough trials: protocol	your focus and methods; wording my suggestion
1b Update. Not an update: a new review	your approved need
2 Registration. Register and number, once you submit (question 1)	you
3a Author. Uday; affiliation: to give. You add your e-mail and postal address when you submit; I keep no e-mail address in your files	you; my suggestion
3b Contributions. Uday plans, screens, extracts, judges and writes, and is the guarantor; the AI assistant is not an author	my suggestion

DRAFT: your protocol, version 1, in the order of the PRISMA-P items [9] (also beside the chat)	From
4 Amendments. A change after registration is reopened only if you agree, certified again with its reason, dated and sent to the register	how your plan works
5 Support. Funding, sponsor and their role: to give ("none" if none)	you
6 Rationale. Your approved case for a new review, its grounds and uncertainties, including the registers of planned reviews not yet searched	your approved need
7 Objectives. Your approved objective and question	your approved objective and question
8 Eligibility. Your approved criteria; no language or start-date limit; any publication status, including results posted in trial registers (my suggestion)	your approved question and search; my suggestion
9, 10 Sources and search. Your approved search, with PubMed search 4 in full; no contact with study authors and no reference-list checks (my suggestion)	your approved search; my suggestion
11a Records and data. Every search result, decision and extracted value kept in your planning folder: saved results with check codes, a never-edited record of every step, dated versions; licensed full texts not shared	how your plan works
11b, 11c Selecting and collecting. As in your approved methods; no contact with investigators for missing data (my suggestion)	your approved methods; my suggestion
12 Data items. Your approved extraction form; a value a study does not report is recorded as not reported, never guessed	your approved form
13 Outcomes. Main: whole-body fat, lean or fat-free mass, muscle itself, body water, each apart. Additional: strength or physical performance, total weight change (my suggestion: your question adds strength with "also"). Why: your focus asks which kind of weight is lost	your question; my suggestion
14 Risk of bias. As in your approved methods; the judgments feed the GRADE rating of each part	your approved methods
15 Synthesis. As in your approved methods, Asian participants described apart; no other subgroup or sensitivity analyses; if a part reaches 3 similar trials, its pooling method is registered as an amendment before any pooling (my suggestions)	your approved methods; my suggestion
16 Meta-bias. Each trial's registered outcomes compared with those it reports; no statistical test for publication bias, as few trials are expected: by abstract, no part reaches 3 yet [11] (my suggestions)	my suggestion; [11]
17 Confidence. GRADE for each part, as approved	your approved methods
AI statement. We will use [question 2], from 2026-10-25, with the STIR-L4 planning prompt version 1.2 and its helper kit 1.1.4, to run and log searches, count records by code, suggest screening decisions and extracted values with their exact passages, and draft wording. Uday makes and certifies every decision. Every literature fact comes from a logged search, re-counted by code; quoted passages are checked by code. Limitations: suggestions are not checked by a second person, and evidence on AI tools as second screeners is limited [10]. Prompt, kit, record and code shared with the review (my suggestion, #7). Interests in these tools: to give	#4–#6; [10]; my suggestion

Sources: [1] L000657 PubMed, run 2026-10-25; [2] L000658 PubMed, run 2026-10-25; [3] L000659 PubMed, run 2026-10-25; [4] L000660 PubMed, run 2026-10-25; [5] L000661 PubMed, run 2026-10-25; [6] L000663 PubMed, run 2026-10-25; [7] L000664 PubMed Central, run 2026-10-25; [8] L000665 PubMed Central, run 2026-10-25; [9] L000667 computed from the saved full text, run 2026-10-25; [10] L000575 PubMed, run 2026-10-25; [11] L000629 computed from your screening decisions and the saved records, run 2026-10-25

Next: the protocol, to certify.

Your turn, Uday: to finish the protocol. 1. Register: a PROSPERO, b OSF Registries (another such register), c both, d other? 2. Your chat's AI model and version? 3. Draft: edits (affiliation, funding) or "accept"?
Example: "1 b, 2 <name shown>, 3 accept".

Opened beside the chat: stir-l4-workspace/protocol/protocol_draft_v1.md

> C **Methodology certified: 8 of 8.** Every decision of the model is now certified.

UDAY (SIMULATED USER) session 4, message 17

add my name as author, no funding

STIR-L4 ASSISTANT session 4, message 18

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your edits (summarized): you as the author, and no funding; both are in version 2 of your protocol, beside the chat. Nothing you have approved is affected: your plan already had you working alone, about 5 hours a week, with no paid databases.

FACT: the protocol checklist (PRISMA-P) asks to “Provide name, institutional affiliation, and e-mail address of all protocol authors” [1].

DRAFT: what changes in version 2 (say "show all" to see it all here)	Version 1	Version 2
3a Author	Uday; affiliation: to give	Author: Uday, as you asked. Your name in full, as it should appear, and your affiliation: to give. You add e-mail and postal address when you submit
5 Support	to give ("none" if none)	No funding (you); so no funder or sponsor, and no role for one in the protocol (my suggestion)

Sources: [1] L000667 computed from the saved full text, run 2026-10-25

Next: with your answers, I complete version 3 and show you the line to certify your protocol.

Your turn, Uday: these complete your protocol. 1. Register (asked again): a PROSPERO, b OSF Registries, c both, d other? 2. The AI model and version your chat shows (asked again; "not available" if you cannot find it)? 3. Your name in full and your affiliation, as the protocol should show them?
For example: "1 b, 2 <name shown>, 3 <full name>, <university>".

Opened beside the chat: stir-l4-workspace/protocol/protocol_draft_v2.md

> C **The protocol is drafted and edited.** PRISMA-P order, a statement of how the AI helped, author, funding, registry and the author's interest in the prompt and kit: four drafts in the chat. This first protocol is a table; Session 5 rewrites it in full in the standard template.

UDAY (SIMULATED USER) session 4, message 19

1 b, 2 not available, 3 Uday only, leave affiliation blank

STIR-L4 ASSISTANT session 4, message 20

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your answers (summarized): OSF Registries; the AI model not available; your name as Uday only, affiliation left blank. All are in version 3 of your protocol, beside the chat. Registering adds a task to your hours, so it may touch your approved feasibility and, through it, your other decisions; I see nothing to change, so all stand unless you disagree.

FACT: the AI guidance shown before also lists, among what to report, “Any financial and non-financial interests the evidence synthesists have in the AI system or tool” [1].

DRAFT: what changes in version 3 (say "show all" to see it all here)	Version 2	Version 3
2 Registration	where: open	OSF Registries (you); its number added after you submit
3a Author	Uday; full name and affiliation: to give	Uday, the name as you gave it; affiliation left blank, as you asked, though the checklist asks for one [2]; you add e-mail and postal address when you submit
AI statement: model	to give	an AI chat assistant whose model and version were not available to you (you); settings such as temperature: not available (my suggestion, as for the model)
AI statement: limitations	no second person checks my suggestions	also: the model and version the guidance asks for could not be given (my suggestion)
AI statement: interests	to give	to give: your answer below

Sources: [1] L000665 PubMed Central, run 2026-10-25; [2] L000667 computed from the saved full text, run 2026-10-25

Next: with your answer, I show you the line to certify your protocol; then you submit it to OSF Registries.

Your turn, Uday: the last gap in your protocol, which the guidance above asks to declare. Any interests in these AI tools, financial or not (for example, a role in making them): a none, b say which?
For example: "a" or "b: <which tool, and how>".

Opened beside the chat: stir-l4-workspace/protocol/protocol_draft_v3.md

UDAY (SIMULATED USER) session 4, message 21

b: i wrote the stir-l4 prompt and kit

STIR-L4 ASSISTANT session 4, message 22

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your answer (summarized): you wrote the STIR-L4 prompt and its helper kit. It is now in version 4 of your protocol, beside the chat. This may change the motivations you approved, because they do not name this role. Through your methods it may also reach your feasibility and the decisions that depend on it. It adds no hours, so all stand unless you disagree.

DRAFT: what changes in version 4 (say "show all" to see it all here)	Version 3	Version 4
AI statement: interests	to give	you wrote the STIR-L4 planning prompt and its helper kit used here (you, summarized); no other interest declared (my suggestion)
AI statement: limitations	no second person checks my suggestions; model and version not given	also: the author of the prompt and kit uses them and judges their suggestions alone (my suggestion)

FACT: the guidance shown before asks for financial and non-financial interests [1].
DRAFT: your answer names a role but does not say whether money is involved. If you have a financial interest in these tools, say so under "change".

Sources: [1] L000665 PubMed Central, run 2026-10-25

Next: once you certify, you submit the protocol to OSF Registries yourself, and I record its number.

```
CERTIFY Methodology | protocol, version 4, in the order of the PRISMA-P checklist: title — Which
kind of weight is lost with SGLT2 inhibitors in adults without diabetes: fat, muscle or body water?
A scoping review, with a systematic review of the parts with enough trials: protocol; a new review,
not an update; registration — OSF Registries, its number added after submission; author — Uday,
affiliation left blank, e-mail and postal address added at submission; contributions — Uday plans,
screens, extracts, judges and writes and is the guarantor, the AI assistant is not an author;
amendments — a change after registration only if Uday agrees, certified again with its reason,
dated and sent to the register; support — no funding, no sponsor, no role for one; rationale,
objectives, eligibility, sources, search, selecting studies, collecting data, data items, judging
each study, synthesis and trust — as certified; added to them — any publication status, including
results posted in trial registers; no contact with study authors or investigators and no reference-
list checks; main outcomes whole-body fat, lean or fat-free mass, muscle itself and body water,
each apart, additional outcomes strength or physical performance and total weight change; no other
subgroup or sensitivity analyses; a pooling method registered as an amendment before any pooling;
each trial's registered outcomes compared with those it reports, with no statistical test for
publication bias; records — every search result, decision and extracted value kept with check codes
in a never-edited record, licensed full texts not shared; AI statement — an AI chat assistant,
model, version and settings not available, used from 2026-10-25 with the STIR-L4 planning prompt
version 1.2 and helper kit 1.1.4 to run and log searches, count records by code, suggest screening
decisions and extracted values with their passages and draft wording; Uday makes and certifies
every decision; prompt, kit, record and code shared with the review; limitations — no second person
checks the suggestions, the model and version could not be given, the author of the prompt and kit
uses them and judges their suggestions alone; interests — Uday wrote the STIR-L4 planning prompt
and helper kit; no other interest declared | reason: ___ | Uday, solo
```

Your turn, Uday: this settles your protocol, the last step before you register it. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..." or "change: remove the added limitation".

Opened beside the chat: stir-l4-workspace/protocol/protocol_draft_v4.md

UDAY (SIMULATED USER) session 4, message 23

certify, protocol is right

STIR-L4 ASSISTANT session 4, message 24

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Kept your certify and your reason as you gave them, but I could not record them as an approval: your methods were approved before the protocol, and an approved decision changes only if you agree to reopen it. Nothing you have approved is affected.

DRAFT: what your protocol adds to your approved methods	
Studies	any publication status, including results posted in trial registers
Searching	no contact with study authors; no reference-list checks
Outcomes	main and additional outcomes, each kept apart
Analyses	no other subgroup or sensitivity analyses; a pooling method registered before any pooling
Reporting bias	each trial's registered outcomes compared with those it reports; no statistical test
Also	how records are kept, the statement on the AI's help, author, amendments, no funding, your interest

If you reopen your methods, your other seven approved decisions count as provisional until your certification is recorded, because each connects to the methods through feasibility. They stand unless you disagree.

Your protocol, version 4, stays beside the chat; say "show all" to see it here.

Next: with a, I reopen your methods and record your certification; then you submit the protocol to OSF Registries yourself and tell me its number.

Your turn, Uday: your protocol can be approved only as part of your methods, before you register it. Reopen your methods? a yes, and record your certify above with its reason; b no: your methods stay as approved and the protocol's additions stay unapproved; c other (say what).
For example: "a" or "b".

Opened beside the chat: stir-l4-workspace/protocol/protocol_draft_v4.md

V **The kit refuses an unsafe certification.** The protocol changes certified methods, so it cannot be certified until the user agrees to reopen Methodology (message 26).

UDAY (SIMULATED USER) session 4, message 25

a

STIR-L4 ASSISTANT session 4, message 26

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Reopened your methods, as you agreed, and recorded your certify and reason (below). Your protocol is now part of your approved methods, and none of your decisions is provisional any more. This may change your approved feasibility, because registering and checking each trial's registered outcomes take hours, and through feasibility your other decisions. My judgment: they add little, so all stand unless you disagree.

```
CERTIFY Methodology | protocol, version 4, in the order of the PRISMA-P checklist: title — Which
kind of weight is lost with SGLT2 inhibitors in adults without diabetes: fat, muscle or body water?
A scoping review, with a systematic review of the parts with enough trials: protocol; a new review,
not an update; registration — OSF Registries, its number added after submission; author — Uday,
affiliation left blank, e-mail and postal address added at submission; contributions — Uday plans,
screens, extracts, judges and writes and is the guarantor, the AI assistant is not an author;
amendments — a change after registration only if Uday agrees, certified again with its reason,
dated and sent to the register; support — no funding, no sponsor, no role for one; rationale,
objectives, eligibility, sources, search, selecting studies, collecting data, data items, judging
each study, synthesis and trust — as certified; added to them — any publication status, including
results posted in trial registers; no contact with study authors or investigators and no reference-
list checks; main outcomes whole-body fat, lean or fat-free mass, muscle itself and body water,
each apart, additional outcomes strength or physical performance and total weight change; no other
subgroup or sensitivity analyses; a pooling method registered as an amendment before any pooling;
each trial's registered outcomes compared with those it reports, with no statistical test for
publication bias; records — every search result, decision and extracted value kept with check codes
in a never-edited record, licensed full texts not shared; AI statement — an AI chat assistant,
model, version and settings not available, used from 2026-10-25 with the STIR-L4 planning prompt
version 1.2 and helper kit 1.1.4 to run and log searches, count records by code, suggest screening
decisions and extracted values with their passages and draft wording; Uday makes and certifies
every decision; prompt, kit, record and code shared with the review; limitations — no second person
checks the suggestions, the model and version could not be given, the author of the prompt and kit
uses them and judges their suggestions alone; interests — Uday wrote the STIR-L4 planning prompt
and helper kit; no other interest declared | reason: protocol is right | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—

Step	Settles	Status	Waiting on
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	3 of 3 certified	—
8. Search Strategy	search strings and stopping rule	certified by Uday	—
9. Scoping and pilot screening	first included studies	26 screening decisions	—
10. Methodology	methods for the protocol	certified by Uday	—
11. Protocol	protocol and registration	now	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

Your certified protocol (version 4, with a heading naming your certification) is beside the chat; say "show all" to see it here.

To save: submit the certified protocol (protocol/protocol_v1_certified.md in your planning folder) to OSF Registries yourself, adding your e-mail and postal address on its form, not here. Keep the registration number and date.

Next: with your number, I record your registration; later changes become amendments, and screening goes on until you commit to your included studies.

Your turn, Uday: registering fixes this first version of your plan. After you submit: a registered: the number or link and the date; b not yet.
For example: "a: <number>, 2026-10-26" or "b".

Opened beside the chat: stir-l4-workspace/protocol/protocol_v1_certified.md

UDAY (SIMULATED USER) session 4, message 27

not submitting in this test. record it as not submitted

STIR-L4 ASSISTANT session 4, message 28

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your protocol as not submitted to OSF Registries (a test). Nothing you have approved is affected: your protocol already adds the number after submission.

DRAFT: what this means

- No register fixes your plan yet. Until you register, an approved decision changes only if you agree and certify it again; after registering, a change becomes an amendment sent to the register.
- You can submit later and give me the number and date.
- Still to do before you commit to your included studies: OpenAlex, the WHO trial portal (you, by hand), the remaining screening and the full texts.

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 done · 10 done · 11 now · 12 open

Next: with a, your next 10 records, my suggested decision beside each; with b, the session summary.

Your turn, Uday: what next? a screening toward your included studies (Step 12), unregistered; b pause: summary and stop; c other.
For example: "a" or "b".

H **Not submitted, and recorded as such.** The protocol was not registered in this test.

UDAY (SIMULATED USER) session 4, message 29

c. commit to the studies included so far

STIR-L4 ASSISTANT session 4, message 30

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your commitment to the 4 studies you included; this ends planning. It may change your approved search, feasibility and methods, as the studies a review holds reshape each, and through them your other decisions. Nothing changes unless you agree.

FACT: final counts at your commitment [1]	Records	Studies
Found so far, after duplicates	2,565	—
Screened by you	26	23
Included: committed	4	4, 3 of them called randomized
Unsure, to read in full: not committed	12	9 other studies; 2 records are reports of included studies
Excluded	10	10
Not yet screened	2,539; 1 report of an included study among the records not yet screened	not known
Parts named by abstract, not yet measured: your 3 included trials	—	fat 1, body water 1, function 1: none reaches your 3
… adding your 9 unsure studies called randomized	—	fat 4, body water 5, function 3, lean 1

Your approved decision (JUDGMENT, Uday, summarized)	At your commitment (FACT [1])
Search: OpenAlex and the WHO trial portal searched; strings final once no record added by the last change meets your question	neither searched; 0 of the 1,287 added records screened
Methods: you screen every record found, then re-check a random tenth	26 of 2,565 screened; no re-check
Methods: a scoping map of every study that meets your question	4 studies
Methods: a systematic part for each part of the weight with 3 similar randomized trials	none among your 4

DRAFT: what committing as now means
Your review holds these 4 studies and, by your own rule, stays scoping only
Your unsure records stay unsure; only you decide them
Your protocol stays unregistered

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 done · 10 done · 11 now · 12 done

The 23 studies and your decisions are beside the chat; say "show all" to see them here.

Sources: [1] L000714 computed from your screening decisions and the saved records, run 2026-10-25

Next: with a, I draft the changes for you to certify; with b, I build each included study's record and your evidence map; with c, your next 10 records.

Your turn, Uday: your commitment conflicts with your approved search and methods, which stand unless you reopen them. a keep it and reopen both to match it; b keep it and both as approved, the conflict recorded; c take it

back and go on screening; d other.
For example: "a" or "c".

Opened beside the chat: stir-l4-workspace/data/derived/step12_commitment.csv

V **A test step: the commitment.** The prompt's last step asks the team to commit to its included studies. In this test the script had Uday commit to the 4 records he included in the pilot, which conflicts with his approved search and methods; the assistant lays out the conflict and leaves the choice to him. The commitment and the conflict stay in the log and are not part of the final report.

UDAY (SIMULATED USER) session 4, message 31

b, keep both as approved and note the conflict. then show me the final plan

STIR-L4 ASSISTANT session 4, message 32

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded your choice. Your commitment stands, and your approved search and methods stay as they are. The conflict between them is noted in your plan. Nothing you have approved is affected.

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	3 of 3 certified	—
8. Search Strategy	search strings and stopping rule	certified by Uday	—
9. Scoping and pilot screening	first included studies	26 screening decisions	—
10. Methodology	methods for the protocol	certified by Uday	—
11. Protocol	protocol and registration	now	—
12. Commitment	the included studies (end of planning)	done	—

Interests recorded: 1 of 1 member.
Step 11 stays open only for registration, which you recorded as not submitted.

Your final plan (JUDGMENT, Uday, summarized)	What you certified	Your reason
Interests and motivation	your interest: which kind of weight is lost, fat or muscle; to learn; kinds: learn the process, personal or institutional curiosity, academic activity	testing the workflow
Team's Purpose	the focus above; kinds: research gaps, finding the right question; results to decide whether a full review is worth it; maybe publish	broadened for feasibility
Need for a Review	a new review: no review examined so far answers the focus; PROSPERO, OSF Registries, Cochrane and Epistemonikos not searched	broadened for feasibility

Your final plan (JUDGMENT, Uday, summarized)	What you certified	Your reason
Review Objective	understand which kind of weight is lost (fat, muscle or body water), and find where trials are missing	broadened for feasibility
Feasibility	broaden: all adults without diabetes, any country; Asian people as a subgroup	broaden is right
Research Question	SGLT2 inhibitors versus placebo, no treatment or usual care, in adults without diabetes (prediabetes counts); fat, lean or fat-free mass, muscle itself, body water and strength, each apart; randomized trials and controlled cohort studies	this is the question
Before the search	extraction form, property registry (the study features counted) and atlas (the map of the evidence), version 1 each	covers my outcomes; properties are fine; atlas is fine
Search Strategy	PubMed and ClinicalTrials.gov, run; OpenAlex and the WHO trial portal, planned and not run; Embase and CINAHL named as not searched; no language or date limit; a stopping rule; not peer reviewed	finds both known studies
Methodology, with the protocol	your protocol, version 4, as certified today	protocol is right
Registration	OSF Registries chosen; not submitted (you: a test)	—
Recorded conflict	your commitment against your approved search and methods, as in my last reply; both stand as approved	—

FACT, from [1]: your 4 included studies	The report you included	Year	Called randomized (by title, abstract or record type)
EMPA-TROPISM trial	Randomized Trial of Empagliflozin in Nondiabetic Patients With Heart Failure and…	2021	yes
Empire Prevent Metabolic trial	Effects of empagliflozin in patients at risk of heart failure: the Empire Prevent…	2026	yes
PubMed 29235586	Exploring glycosuria as a mechanism for weight and fat mass reduction. A pilot…	2014	no
PubMed 32568464	Effects of Empagliflozin Treatment on Glycerol-Derived Hepatic Gluconeogenesis in…	2020	yes

The full wording of each decision, the studies and the final counts are beside the chat; say "show all" to see them here.

Sources: [1] L000714 computed from your screening decisions and the saved records, run 2026-10-25

Next: with a, I write your session summary, show it with the plan map, and package your planning folder for you to keep.

Your turn, Uday: planning has ended with your commitment, so this is the last check before closing. Does the final plan look right? a yes: end the session with its summary; b something to correct: say what; c other.
For example: "a" or "b: <what to correct>".

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_S4_final_plan.md

UDAY (SIMULATED USER) session 4, message 33

a. thanks, done

STIR-L4 ASSISTANT session 4, message 34

Step 11 of 12: Protocol | **Certified 8 of 8** | **Waiting on you: 0**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Ended the session, as you chose. Saved its summary and a packed copy of your planning folder. Nothing you have approved is affected.

Session summary (also beside the chat; say "show all" for the certified lines in full):

```
# Session summary (S4, 2026-10-25)

Step 11 of 12: Protocol | Certified 8 of 8 | Waiting on you: 0 | Focus: which kind of weight is lost:
fat, muscle or body water, within…

Topic: the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug
is given for, any country; Asian people as a subgroup
Focus: which kind of weight is lost: fat, muscle or body water, within all adults without diabetes,
whatever the drug is given for, any country; Asian people as a subgroup

Settled this session:
- Methodology: certified by Uday; reason: fits a solo review (full line in the saved summary)
- Methodology, with your protocol (version 4): certified by Uday; reason: protocol is right (full line in
the saved summary)

Reopened:
- Methodology: reopened with your agreement, so that your certification of the protocol could be recorded
with it; certified again the same day

Still open: nothing

Next questions:
- Registration: your certified protocol (version 4) is not submitted to OSF Registries (you: a test). If
you submit it, bring its number and date to a new session; from then on, any change is an amendment sent
to the register
- Recorded conflict: your commitment holds 4 studies, while your approved search and methods stand as
approved: OpenAlex search 3 and the WHO trial portal not run; 2,539 records not screened, among them 0 of
the 1,287 added by the last change to the strings; the 12 unsure records not read in full; no re-check of
a tenth of your decisions. To settle it: reopen the search and methods to match the commitment, or take
back the commitment and go on screening; until then both stand
- By your pooling rule, with these studies the review stays scoping only: no part of the weight reaches 3
similar randomized trials among them

Notes: This session began from the saved plan, record and the summary of session 3. Your methods were
certified, then reopened with your agreement so that your certification of the protocol (version 4) could
be recorded with them. Registration was recorded as not submitted. Your commitment to the studies you
included ended planning; its conflict with your approved search and methods is recorded, not resolved.
Every saved search result was checked against the record (161 checked). One set of PubMed records first
showed a mismatch only because its 11 saved parts were read out of order; read in order, all 11 parts and
its 2,021 records match. Not run: OpenAlex (refused again today) and the WHO trial portal (you, by hand).
The AI model and version: not available (you). Changed this session: the saved search results, screening
decisions and counts, the working helpers, the record of every step, the step calculations, your plan,
the methods and protocol drafts, the final plan and session summaries.

Searches this session: 22
```

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 done · 10 done · 11 now · 12 done (map unchanged)

When convenient: a contact email for PubMed searches, if you like (PubMed uses it only to reach us about a problem); I keep it private.

To save: keep `stir-l4-workspace_2026-10-25_S4.zip`, this copy of your plan and record; in a new chat, give it to me with the prompt and helper kit.

Next: in a new session, I read your saved plan and record and show where we are.

Your turn, Uday: nothing more today. In a new session, bring your registration number and date if you submit the protocol, or your choice on the recorded conflict.
For example: "registered: <number>, <date>" or "reopen my search and methods".

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_S4_summary.md

C **End of Session 4.** The plan is saved; Session 5 turns it into the final report.

Part 2. Session 5

Session 5. The assistant writes the protocol in full in the standard template, with a reason and a source for each choice and every gap marked; Uday asks for corrections before certifying; the protocol is certified and the final report (Part 1) is produced.

Messages 1 to 4 of each session (the tool test and the calibration quiz, scored 16 of 16) are not shown.

UDAY (SIMULATED USER) session 5, message 5

passed 16/16. continue. planning is done on my side. please write the final report with a full protocol in the standard template. the ai assistant in every session was claude opus 5.5 from anthropic (model id claude-opus-5-5), i read it from the run settings

STIR-L4 ASSISTANT session 5, message 6

Step 12 of 12: Planning complete | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Recorded the AI model you read for all five sessions, and wrote the final report with your full protocol, version 2. This may change the Methodology you approved, because its protocol says the model was not available. Your other decisions still stand: naming the model adds no work (my reading).

From your saved plan	In your words (JUDGMENT, Uday)
Topic	the role of SGLT2 inhibitors in the weight lost by all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup
Focus	which kind of weight is lost: fat, muscle or body water, within all adults without diabetes, whatever the drug is given for, any country; Asian people as a subgroup

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	certified by Uday	—
4. Review Objective	what the review should achieve	certified by Uday	—
5. Feasibility	go, narrow, broaden or defer	certified by Uday	—
6. Research Question	the question and eligibility criteria	certified by Uday	—
7. Before the search	extraction form, property registry and atlas	3 of 3 certified	—
8. Search Strategy	search strings and stopping rule	certified by Uday	—
9. Scoping and pilot screening	first included studies	26 screening decisions	—
10. Methodology	methods for the protocol	certified by Uday	Uday
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	done	—

Interests recorded: 1 of 1 member.

Final report, beside the chat	Holds
1. Your plan as certified	the topic, the focus and your 8 decisions, each with when you certified it and your reason as given
2. Your protocol, version 2	the full text, marked DRAFT until you certify it
PDF	not made: this computer has no converter; your browser can print the page to PDF

DRAFT: protocol, version 2, item by item (in PRISMA-P order, the reporting checklist for protocols)	What it says
1a Identification	Protocol of a scoping review, with a systematic review of each part of the weight that has enough trials; title as in version 1
1b Update	A new review, not an update
2 Registration	OSF Registries (a public register of planned studies). **GAP:** registration number and date. None yet: on 2026-10-25 Uday chose not to submit the protocol, as planning was a test. If he submits it, he adds the number and date here.
3a Authors and contact	Uday, sole author; affiliation left blank at your request. **GAP:** e-mail address and postal address of the corresponding author. Uday adds them on the register's form when he submits; none is kept in the planning files.
3b Contributions	You do every review task and are the guarantor; the AI is not an author
4 Amendments	Not an amendment; any later change only with your agreement, certified again and sent to the register
5a–5c Support	No funding, no sponsor, no role for one
6 Rationale	Your clinical justification and expected impact; what the earlier reviews found in planning cover, with their search end dates; your recorded uncertainties
7 Objectives	Your objectives and your question in PICO form (population, intervention, comparison, outcomes), as certified
8 Eligibility	Adults without diabetes; any SGLT2 inhibitor against placebo, no treatment or usual care; randomized trials and controlled cohort studies; no language or start-date limit
9 Sources	PubMed and ClinicalTrials.gov, searched 2026-10-25; OpenAlex (an open index of research papers) and the WHO trial portal planned; Embase and CINAHL not searched (no access). **GAP:** run dates and counts for OpenAlex and the WHO trial portal. Not run when planning ended; Uday supplies them when he runs these searches (OpenAlex reads past its first 100 results only with a free access code).
10 Search strategy	Every string in full, with platform, date, limits and count; the known-study check; no peer review; your stopping rule; how duplicates were removed. **GAP:** the exact string for the WHO trial portal, built from the drug names of the ClinicalTrials.gov search when Uday runs it by hand.
11a Data management	Saved results with check codes, a never-edited record, version control
11b Selection	You alone screen, with my suggestion and its passage beside each record; you re-screen a random tenth about 2 weeks later; your reason as you gave it, and the published evidence on screening alone
11c Data collection	You alone extract, with my suggested values and passages beside them; no data sought from investigators. **GAP:** whether the extraction form is piloted before use, and on how many studies. Not decided; Uday decides.
12 Data items	Your certified extraction form, field by field; unreported values recorded as not reported
13 Outcomes	Main: whole-body fat, lean or fat-free mass, muscle itself, body water; additional: strength or performance, total weight
14 Risk of bias	Systematic part only, per outcome: RoB 2 for trials and ROBINS-I for cohort studies (tools for judging how far a study's design may bias its result)
15a When to pool	A meta-analysis only for a part with 3 or more similar randomized trials, with published guidance on very few studies

DRAFT: protocol, version 2, item by item (in PRISMA-P order, the reporting checklist for protocols)	What it says
15b How to pool	**GAP:** summary measures, handling of data, the method of combining studies, the measure of consistency, and how 'similar' trials are judged. Not decided; if a part reaches 3 similar randomized trials, Uday settles them and registers them as an amendment before any pooling (certified rule; registering them as an amendment was my suggestion).
15c Other analyses	Asian participants described apart; no other subgroup or sensitivity analyses
15d Without pooling	Results tabled part by part and reported with SWiM (a guideline for reporting results without meta-analysis)
16 Meta-bias	Registered outcomes compared with reported ones; no statistical test for publication bias
17 Confidence	GRADE (a system for rating how much to trust the evidence) for each part
Reporting, limitations	PRISMA-ScR and PRISMA 2020 (reporting guidelines for scoping and systematic reviews); one person screens, reads and judges
What planning found	The facts behind each decision, each with its source in the record
How STIR-L4 and AI were used	New: the model you read, the prompt and kit versions, the dates and what the AI and you each did
Gaps	The 6 gaps above, listed together
References	33, each read from a saved record, with its PubMed number or DOI (a permanent link to a paper)

DRAFT: what changes in content	Version 1 (certified)	Version 2
AI model	not available	Anthropic Claude Opus 5.5 (model id claude-opus-5-5), as you read it from the run settings, sessions 1 to 5; settings such as temperature not available
Prompt and kit	prompt 1.2, kit 1.1.4	prompt 1.1 (session 1), 1.2 (sessions 2–4), 1.3 (session 5); kit 1.1.4 (sessions 1–4), 1.1.7 (session 5); the times of each session
Limitations	the model and version could not be given	removed
Everything else	a table of drafts	your certified decisions in full sentences; searches reported as PRISMA-S (the checklist for reporting searches) asks; a numbered reference for each methods choice; what planning found; how STIR-L4 and AI were used; gaps

Say "show <item>" or "show all" to see any part here.

Next: once you certify, I record version 2 as your protocol, rebuild the report from it, and close the session with its summary and a packaged copy of your planning folder.

```
CERTIFY Methodology | protocol, version 2: version 1 as certified, written in full sentences item by item in the order of the PRISMA-P 2015 checklist, with the searches reported as PRISMA-S asks, 33 numbered references and three closing sections (what planning found; how STIR-L4 and AI were used; gaps); new in content — the AI model as Uday read it from the run settings (Anthropic Claude Opus 5.5, model id claude-opus-5-5, sessions 1 to 5; settings not available), the prompt and helper kit versions and dates of each session, and the limitation that the model could not be given removed; gaps — registration number and date (not submitted, a test); e-mail and postal address; OpenAlex and WHO trial portal run dates and counts; the WHO trial portal string; piloting the extraction form; summary measures and pooling method | reason: ___ | Uday, solo
```

Your turn, Uday: version 2 changes the protocol you approved, so recording it needs your agreement to reopen your methods, then your certification. Please read the line above. If it is right, reply "reopen, certify" and say why in your own words; otherwise say what to change. Also tell me if anything in the map looks wrong.
For example: "reopen, certify, because …" or "keep version 1".

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_final_report.html

> **The protocol in the standard template.** The assistant writes the protocol in full, item by item in PRISMA-P order, with the searches reported as PRISMA-S asks, a reason and a retrieved source for each choice, every gap marked, and a section on how STIR-L4 and the AI were used. The helper kit refuses to build the report if the protocol cites a paper that no saved search returned.

UDAY (SIMULATED USER) session 5, message 7

before i certify, three edits: 1. ref 18 is a review of perinatal hypoxia, drop it; cite a methods source or nothing. 2. in item 6 also say what the reviews in people with type 2 diabetes found on fat vs lean or muscle, that is the nearest evidence. 3. in what planning found keep the pilot screening only as a test of the criteria; take out the commitment, the conflict, the line in item 16 about included studies and the session 5 re-check, those belong in the log

> **The user reviews before certifying.** Uday asks for three edits: drop an off-topic reference, add the nearest evidence (reviews in people with type 2 diabetes), and leave the commitment and the conflict to the log.

STIR-L4 ASSISTANT session 5, message 8

Step 12 of 12: Planning complete | **Certified 8 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Made your three edits (protocol beside the chat). This still changes the Methodology you approved; nothing else is affected.

FACT: Searching for a methods source for your re-check found 17 records [4], [5]: 4 methods papers, none on re-screening one's own records, by abstract [6]; OpenAlex: allowance used up.

How to read the strings: `[tiab]` words in the title or abstract; `[ti]` words in the title; `*` any word ending (truncation); `"` an exact phrase; `AND` every block must match; `OR` any one term in the block may match; `(` parentheses group the terms of a block.

Search 1: a methods source for re-screening your own decisions (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: none)

```
("abstract screening"[tiab] OR "citation screening"[tiab] OR "study selection"[tiab] OR "screening decision*"[tiab] OR "full-text screening"[tiab] OR "title screening"[tiab] OR "eligibility assessment*"[tiab]) AND ("intra-rater"[tiab] OR intrarater[tiab] OR "intra-reviewer"[tiab] OR intrareviewer[tiab] OR "intra-observer"[tiab] OR intraobserver[tiab] OR "test-retest"[tiab] OR rescreen*[tiab] OR "re-screening"[tiab] OR "re-screened"[tiab] OR "same reviewer"[tiab] OR "same reviewers"[tiab] OR "self-consistency"[tiab]) AND (screening[ti] OR selection[ti] OR screener*[ti] OR reviewer*[ti] OR "single-reviewer"[ti])
```

Block	Covers	From	How it searches	Records
1	screening records for a review	methods: your re-check	title/abstract words, with truncation	33,272 [1]
2	the same person screening twice	methods: your re-check	title/abstract words, with truncation	67,160 [2]
3	methods papers: screening, selection or reviewers named in …	methods: your re-check	words in the title, with truncation	334,369 [3]
All	combined	—	blocks joined as shown	**17** [4]

Methods paper (FACT [6])	Year	PMID	Studies (my reading, DRAFT)
Evaluating the accuracy of full text and serial screening by one vers…	2026	42805513	one versus two reviewers, full texts

Methods paper (FACT [6])	Year	PMID	Studies (my reading, DRAFT)
Validating Loon Lens 1.0 for Autonomous Abstract Screening and Confid…	2025	40998117	an AI tool screening abstracts
Single screening versus conventional double screening for study selec…	2019	31253092	one versus two screeners
Expediting citation screening using PICo-based title-only screening f…	2017	29178925	screening by title only

Say "show all" to see the other 13 here.

New in item 6, DRAFT: the nearest evidence, from the 6 reviews in type 2 diabetes that name lean mass or muscle (FACT [7], [8], abstracts only; full texts unread)	What it pooled	Fat	Lean mass or muscle	Search end
Joongpan 2026 [7]	a meta-analysis of 8 randomized trials with 541 participants (its title names older adults)	fat mass fell (standardized mean difference −0.53)	muscle mass fell less (standardized mean difference −0.35), which the authors call substantially smaller than the fat loss	not stated in the abstract (abstract only: unknown)
Li 2026 [7]	a network meta-analysis of 33 randomized trials with 5,399 participants	fat mass fell consistently (for example dapagliflozin −1.75 kg)	lean mass fell modestly with empagliflozin (−1.22 kg), ipragliflozin (−0.99 kg) and canagliflozin (−0.80 kg) but not with dapagliflozin or tofogliflozin, and no SGLT2 inhibitor significantly reduced skeletal muscle mass compared with placebo; strength and physical performance were rarely reported	not stated in the abstract (abstract only: unknown)
Yamamoto 2026 [7]	a meta-analysis of 5 randomized trials of at least 12 months with 695 participants; its abstract names type 2 diabetes only as where these drugs are used	fat mass fell (−1.26 kg)	fat-free mass did not differ significantly (−0.53 kg; 95% confidence interval −1.62 to 0.56)	April 22, 2024
Xia 2023 [7]	a meta-analysis of 19 randomized trials with 1,482 participants	fat is not reported in the abstract	lean body mass fell by 0.66 and skeletal muscle mass by 0.35 (units not given); the quality of the evidence was low	1 May 2023
Zhang 2023 [7]	a meta-analysis of 25 randomized trials with 2,286 participants	fat mass fell (−1.49)	lean mass (−0.80), skeletal muscle mass (−0.38) and skeletal muscle index (−0.12) fell as well; body water fell (−0.96); units are not given in the abstract	February 2023
Pan 2022 [8]	a meta-analysis of 18 studies with 1,430 participants	fat mass fell (−1.16 kg)	lean mass changed by −0.76 kg (95% confidence interval −1.53 to 0.01) and skeletal muscle mass fell (−1.01 kg)	not stated in the abstract (abstract only: unknown)

Your edit	In the draft now
1. Reference 18	Dropped; item 11b cites nothing for the re-check
2. Item 6	Adds the table above, as plain sentences with the 6 references

Your edit	In the draft now
3. What planning found	Only: "To test the eligibility criteria, Uday screened 26 records by title and abstract: 4 met the criteria, 12 were unclear and 10 did not" [9]; commitment, conflict, session 5 re-check and item 16's line removed
Also, by your reason (my suggestion)	The section on how AI was used drops the commitment and the conflict too; say "put back" to keep them
References	38: one dropped, the 6 reviews added; gaps unchanged (6)

Sources: [1] L000762 PubMed, run 2026-10-25; [2] L000763 PubMed, run 2026-10-25; [3] L000764 PubMed, run 2026-10-25; [4] L000765 PubMed, run 2026-10-25; [5] L000768 PubMed, run 2026-10-25; [6] L000769 PubMed, run 2026-10-25; [7] L000084 PubMed, run 2026-10-25; [8] L000083 OpenAlex, run 2026-10-25; [9] L000772 your screening decisions, counted 2026-10-25

Next: I then record version 2, rebuild the report and close the session.

```
CERTIFY Methodology | protocol, version 2, with Uday's three edits: version 1 as certified, written in full sentences item by item in the order of the PRISMA-P 2015 checklist, with the searches reported as PRISMA-S asks, 38 numbered references and three closing sections (what planning found; how STIR-L4 and AI were used; gaps); new in content — the AI model as Uday read it from the run settings (Anthropic Claude Opus 5.5, model id claude-opus-5-5, sessions 1 to 5; settings not available), the prompt and helper kit versions and dates of each session, the limitation that the model could not be given removed, and in the rationale what 6 reviews in people with type 2 diabetes found on fat and on lean mass or muscle, the nearest evidence; no source cited for the re-check of a tenth, as no methods source was found; what planning found keeps the pilot screening only as a test of the criteria, with the commitment, the conflict, the session 5 re-check and the line in item 16 on the included studies left to the planning record; gaps — registration number and date (not submitted, a test); e-mail and postal address; OpenAlex and WHO trial portal run dates and counts; the WHO trial portal string; piloting the extraction form; summary measures and pooling method | reason: ___ | Uday, solo
```

Your turn, Uday: this reopens your methods only if you agree. Please read the line above. If it is right, reply "reopen, certify" and say why in your own words; otherwise say what to change.
For example: "reopen, certify, because …".

Opened beside the chat: stir-l4-workspace/protocol/protocol_v2_draft_r2.md

V **Corrections, each backed by a search.** The assistant searches for a methods source for the re-check, finds none and cites none; it chooses the diabetes reviews by a stated rule and checks every number against the saved abstracts.

UDAY (SIMULATED USER) session 5, message 9

reopen, certify, follows the standard template

STIR-L4 ASSISTANT session 5, message 10

Step 12 of 12: Planning complete | **Certified 8 of 8** | **Waiting on you: 0**

Focus: which kind of weight is lost: fat, muscle or body water, within…

Done: Reopened your methods, as you agreed, and recorded your certification:

```
CERTIFY Methodology | protocol, version 2, with Uday's three edits: version 1 as certified, written in full sentences item by item in the order of the PRISMA-P 2015 checklist, with the searches reported as PRISMA-S asks, 38 numbered references and three closing sections (what planning found; how STIR-L4 and AI were used; gaps); new in content — the AI model as Uday read it from the run settings (Anthropic Claude Opus 5.5, model id claude-opus-5-5, sessions 1 to 5; settings not available), the prompt and helper kit versions and dates of each session, the limitation that the model could not be given removed, and in the rationale what 6 reviews in people with type 2 diabetes found on fat and on lean mass or muscle, the nearest evidence; no source cited for the re-check of a tenth, as no methods source was found; what planning found keeps the pilot screening only as a test of the criteria, with the commitment, the conflict, the session 5 re-check and the line in item 16 on the included studies left to the planning record; gaps — registration number and
```

```
date (not submitted, a test); e-mail and postal address; OpenAlex and WHO trial portal run dates
and counts; the WHO trial portal string; piloting the extraction form; summary measures and pooling
method | reason: follows the standard template | Uday, solo
```

Version 2 is now your protocol; version 1 is kept. Your other decisions still stand: version 2 keeps the methods of version 1, so it asks no more work (my reading).

The final report beside the chat now holds version 2 as certified, with its 6 gaps; say "show <item>" to see any part here.

Session 5 summary (saved; say "show all" to see it here):

From your record	
Settled	Methodology: your protocol, version 2, with your three edits; certified by you
Reopened	Methodology, with your agreement, so that version 2 could replace version 1
Still open	nothing to certify; the 6 gaps, for you when you submit or run the two searches left
Registration	not submitted to OSF Registries (a test)
Still on record	the conflict between your commitment and your approved search and methods, unchanged; kept out of the protocol, as you asked
Searches this session	5, each shown when it ran

Steps: 1 done · 2 done · 3 done · 4 done · 5 done · 6 done · 7 done · 8 done · 10 done

To save: keep `stir-l4-workspace_2026-10-25_S5.zip`, your plan and record; in a new chat, give it to me with the prompt and helper kit.

Next: in a new session, I read your saved plan and record and show where we are.

Your turn, Uday: nothing more is needed today. If you submit the protocol, bring its number and date to a new session, so your plan records it.
For example: "registered: <number>, <date>" or "done".

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_final_report.html

> C **Protocol certified; the final report is produced.** The plan as certified and this protocol form the final report in Part 1. (The one-line steps view here leaves out steps 9, 11 and 12, and the kit would have called an unsubmitted protocol registered: both fixed before release.)